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A multiscale study on multivalence targeting

Zhendong Xie

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Tesi doctoral

Un estudio multiescala sobre la focalización multivalencia

UNIVERSITAT DE BARCELONA

Autor/a: Zhendong Xie

Director/a: Giuseppe Battaglia

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Autor/a: Zhendong Xie

Director/a: Giuseppe Battaglia

Tutor/a: Josep Samitier Martí

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A multiscale study on multivalence targeting

Zhendong Xie

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Nanoscience, Faculty of Physics, University of Barcelona

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Resum

El disseny de fàrmacs de precisió utilitza informació biològica detallada, com ara perfils genètics, fenotips cel·lulars i patrons d'expressió molecular, per desenvolupar terapèutics altament específics, efectius i segurs. En aquest treball, demostrem el potencial de la nanomedicina en el desenvolupament de fàrmacs de precisió. Establim un conjunt de principis de disseny per guiar la selectivitat de les nanopartícules, fonamentats en la termodinàmica, la cinètica i la farmacocinètica.

La nanomedicina pot aprofitar l'efecte multivalent per augmentar l'afinitat d'unió. Per comprendre millor les interaccions nanopartícula–cèl·lula, vam desenvolupar un model termodinàmic anomenat teoria de la diana fenotípica (PAT). Generalment, un nombre més gran de lligands de baixa afinitat pot augmentar de manera efectiva la selectivitat d'unió.

Vam aplicar PAT per estudiar nanopartícules de poli(2-(metacriloiiloxi)etil fosforilcolina)-poli(2-(diisopropilamino)etil metacrilat) (PMPC-PDPA) amb diverses topologies. El model prediu la fracció d'unió i la constant d'associació per a diferents arquitectures de nanopartícules. Basant-nos en PAT, vam analitzar el paràmetre de selectivitat de nanopartícules esfèriques en funció del radi de la partícula i del grau de polimerització d'una sola cadena de PMPC. Atès que el PMPC s'uneix a tres receptors diferents, també vam explorar com els canvis en la densitat de receptors afecten la fracció d'unió. Aquests resultats proporcionen un marc quantitatiu per optimitzar el disseny de nanopartícules per a una diana selectiva.

Vam ampliar el PAT per incloure l'orientació a receptors grans, com el receptor relacionat amb les lipoproteïnes de baixa densitat 1 (LRP1) i el receptor de la manosa (MRC1/CD206). Quan es dirigeixen aquests receptors grans, cal tenir en compte tant la penalització estèrica osmòtica com la penalització estèrica per compressió. A més, sorgeix una penalització estèrica a partir de lligands no units. Quan s'utilitza un nombre excessiu de lligands funcionals, aquesta penalització augmenta, provocant una reducció de la força d'unió, que finalment pot resultar negligible o inexistent. Aquest efecte estèric està influït tant pel nombre de lligands funcionals com per un paràmetre d'interferència. Per tenir en compte aquests factors, vam redefinir el paràmetre de selectivitat com a funció del nombre de lligands i no del nombre de receptors. A continuació, vam analitzar aquest paràmetre revisat en relació amb l'energia d'unió, el paràmetre d'interferència, la penalització estèrica per lligand no unit i el nombre de llocs d'unió disponibles.

Després de desenvolupar el PAT, un model termodinàmic, el vam ampliar per descriure la cinètica d'unió de les interaccions nanopartícula-cèl·lula. Vam investigar la cinètica d'unió de nanopartícules de PMPC amb diverses topologies, ajustant dades experimentals amb un model matemàtic. En integrar PAT amb un marc cinètic, vam aclarir els paràmetres clau que governen les interaccions multivalents, incloent-hi la velocitat d'associació, la taxa de dissociació i la velocitat d'endocitosi. A més, vam explorar el trànsit intracel·lular de nanopartícules de PMPC amb diferents topologies per obtenir una visió més profunda del seu destí cel·lular. També vam aplicar el model per ajustar la cinètica d'unió entre nanopartícules decorades amb mUNO, funcionalitzades amb diferents densitats de lligand, i macròfags derivats de THP-1.

Per ampliar l'aplicabilitat del nostre model a escenaris biològics més complexos, vam desenvolupar un marc avançat que té en compte la cinètica d'unió, l'endocitosi, l'exocitosi, el reciclatge de receptors i la síntesi de receptors. Addicionalment, vam introduir un model de dinàmica caòtica per simular el comportament d'oscil·lació dels receptors observat en certes condicions experimentals. Preveiem que aquest model basat en el caos es pugui refinar i aplicar millor per interpretar experiments de captació que impliquin regulació dinàmica dels receptors.

Per quantificar l'eficiència terapèutica, vam desenvolupar un model farmacocinètic basat en la fisiologia (PBPK) per descriure l'absorció, distribució, metabolisme i excreció (ADME) de les nanopartícules. Vam aplicar aquest model a dos escenaris diferents. El primer implicava interaccions no específiques amb nanopartícules de PMET (poli(metionina)), mentre que el segon es basava en la teoria de la diana fenotípica (PAT) aplicada a nanopartícules de PMPC. Per a les nanopartícules de PMET, vam quantificar els paràmetres farmacocinètics clau i vam analitzar la seva influència en la selectivitat de la diana. En el cas de les nanopartícules de PMPC, vam proposar una estratègia per ajustar els paràmetres de disseny de les nanopartícules per millorar la selectivitat de la diana, assegurant que les nanopartícules interactuïn preferentment amb tipus cel·lulars específics.
El nostre objectiu és utilitzar mètodes in silico, incloent-hi models termodinàmics, cinètics i farmacocinè

Abstract

Precision drug design utilizes detailed biological information, such as genetic profiles, cellular phenotypes, and molecular expression patterns, to develop therapeutics that are highly specific, effective, and safe. In this work, we demonstrate the potential of nanomedicine in the development of precision drugs. We establish a set of design principles to guide nanoparticle selectivity, grounded in thermodynamics, kinetics, and pharmacokinetics.

Nanomedicine can leverage the multivalent effect to enhance binding avidity. To better understand nanoparticle–cell interactions, we developed a thermodynamic model called the phenotypic targeting theory (PAT). Generally, a larger number of low-affinity ligands can effectively increase binding selectivity.

We applied PAT to study poly(2-(methacryloyloxy)ethyl phosphorylcholine)-poly(2-(diisopropylamino)ethyl methacrylate) (PMPC-PDPA) nanoparticles with various topologies. The model predicts the binding fraction and association constant for different nanoparticle architectures. Based on PAT, we analyzed the selectivity parameter of spherical nanoparticles as a function of particle radius and the degree of polymerization of a single PMPC chain. Since PMPC targets three distinct receptors, we also explored how changes in receptor density affect the binding fraction. These results provide a quantitative framework for optimizing nanoparticle design for selective targeting.

We extended the PAT to include the targeting of large receptors, such as the low-density lipoprotein receptor-related protein 1 (LRP1) and the mannose receptor (MRC1/CD206). When targeting such large receptors, both osmotic steric penalty and compression steric penalty must be taken into account. Additionally, a steric penalty arises from unbound ligands. When an excessive number of functional ligands is used, this steric penalty increases, leading to a reduction in binding strength, ultimately resulting in negligible or no binding. This steric effect is influenced by both the number of functional ligands and an interference parameter. To account for these factors, we redefined the selectivity parameter as a function of the number of ligands rather than the number of receptors. We then analyzed this revised selectivity parameter in relation to binding energy, the interference parameter, the steric penalty per unbound ligand, and the number of available binding sites.

After developing the PAT, a thermodynamic model, we extended it to describe the binding kinetics of nanoparticle-cell interactions. We investigated the binding kinetics of PMPC nanoparticles with various topologies, fitting experimental data using a mathematical model. By integrating PAT with a kinetic framework, we clarified key parameters governing multivalent interactions, including the on-rate, dissociation rate, and endocytosis rate. In addition, we explored the intracellular trafficking of PMPC nanoparticles with different topologies to gain deeper insight into their cellular fate. We also applied the model to fit the binding kinetics between mUNO-decorated nanoparticles, functionalized with varying ligand densities, and THP-1-derived macrophages.

To expand the applicability of our model to more complex biological scenarios, we developed an advanced framework that accounts for binding kinetics, endocytosis, exocytosis, receptor recycling, and receptor synthesis. Additionally, we introduced a chaotic dynamics model to simulate receptor oscillation behavior observed in certain experimental conditions. We anticipate that this chaos-based model can be further refined and applied to better interpret uptake experiments involving dynamic receptor regulation.

To quantify therapeutic efficiency, we developed a physiologically based pharmacokinetic (PBPK) model to describe the absorption, distribution, metabolism, and excretion (ADME) of nanoparticles. We applied this model to two distinct scenarios. The first involved non-specific

interactions using PMET (poly(methionine)) nanoparticles, while the second was based on the Phenotypic Targeting Theory (PAT) applied to PMPC nanoparticles. For the PMET nanoparticles, we quantified key pharmacokinetic parameters and analyzed their influence on targeting selectivity. In the case of PMPC nanoparticles, we proposed a strategy to tune nanoparticle design parameters in order to enhance targeting selectivity, ensuring that the nanoparticles preferentially interact with specific cell types.

We aim to use in silico methods, including thermodynamic, kinetic, and pharmacokinetic modeling, to assist researchers in designing more selective nanomedicines and gaining deeper insights into nanoparticle-cell interactions.

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PUBLICATIONS AND CONFERENCES

LIST OF PUBLICATIONS

1. Ding H, Su L, Xie Z, et al. Morphological insights in oxidative sensitive nanocarrier pharmacokinetics, targeting, and photodynamic therapy[J]. Journal of Materials Chemistry B, 2025, 13(12): 3852-3863.
2. Chen J, Xie Z, Sapienza R, et al. Supramolecular Design of Polymeric Random Lasers[J]. Advanced Optical Materials, 2025: 2500541.

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Poster Presentations

1. 2022, 15th Symposium of IBEC (Institute for Bioengineering of Catalonia): The physiologically based pharmacokinetic (PBPK) modelling of superselective nanomedicine.
2. 2023, XVII International Congress of the Spanish Biophysical Society: Modelling the process of phenotypic targeting in vivo conditions using Physiologically-based Pharmacokinetics.
3. 2023, 16th Symposium of IBEC (Institute for Bioengineering of Catalonia): Integrating phenotypic targeting in physiologically-based pharmacokinetics modeling.
4. 2024, 1st edition of the Barcelona Blood-Brain-Barrier Conference: Multiscale physiologically-based pharmacokinetics(PBPK) modeling for enhanced brain targeting.
5. 2024, 17th Symposium of IBEC (Institute for Bioengineering of Catalonia): Multiscale physiologically-based pharmacokinetics modeling.

Oral Presentations

1. 2024, International Conference on Nanomedicine and Nanobiotechnology 2024: Integrating phenotypic targeting in physiologically-based pharmacokinetics modeling.

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Chapter 1: Introduction

1.1 Seeking precision drugs

1.1.1 Challenges in reducing the side effects.

Specificity and selectivity are two desired features for the drugs. Side effects of small-molecule drugs commonly stem from their poor specificity. These compounds tend to diffuse passively throughout the body, impacting not only diseased cells but also healthy tissues. This lack of targeted action frequently leads to significant off-target effects and reduced therapeutic efficiency. For example, doxorubicin intercalates into DNA and disrupts cell division, affecting not only cancer cells but also healthy tissues such as the bone marrow, gastrointestinal tract, and hair follicles, leading to side effects including myelosuppression, diarrhea, and hair loss[1]. To reduce the side effects, doxorubicin can be encapsulated within nanoparticles or liposomes, which alters its biodistribution and reduces accumulation in healthy tissues. It's known as doxorubicin liposomal (Doxil®)[2].

To enhance specificity, some drugs are designed to target specific receptors expressed on diseased cells. Ligand–receptor targeting therapies are more specific than non-targeted small molecules[3-5]. However, this approach can still cause side effects. Targeted receptors are not exclusive to diseased cells, as many are also expressed at low levels in healthy tissues. For instance, some breast cancer cells overexpress the human epidermal growth factor receptor 2 (HER2). Drugs targeting HER2 accumulate at the diseased site; however, they can cause cardiotoxicity, as HER2 is also expressed in heart muscle cells[6].

Additionally, drugs, as exogenous agents, can trigger immune responses such as immune activation, cytokine release, and allergic reactions[7,8]. Namely, TGN1412 was deemed safe and effective in preclinical studies and was subsequently tested in healthy human volunteers in a first-in-human clinical trial. In a phase I clinical trial, six human volunteers experienced life-threatening conditions involving multiorgan failure and were admitted to the intensive care unit, despite receiving only a very low dose[9]. TGN1412 could trigger a severe cytokine storm, thus resulting in life-threatening side effects.

Drug side effects are a major cause of failure in drug development and post-approval market withdrawal. Minimizing adverse effects is therefore critical to improving clinical success, patient safety, and long-term treatment viability[10]. Developing highly selective drugs enhances therapeutic efficacy by targeting disease-specific pathways while reducing off-target toxicity. Such precision increases patient outcomes and improves the overall success rate of therapies.

1.1.2 Biological barriers and disease burden

Overcoming biological barriers is essential for achieving effective therapeutic outcomes. From the moment a drug enters the body, it faces a series of challenges, ranging from molecular-level interactions to complex physiological barriers[11-13]. A thorough understanding of these barriers is crucial for designing drugs that can efficiently target, penetrate, and act within the body. In this section, we systematically introduce these obstacles, highlighting the difficulties drugs encounter at each biological level. As shown in **Fig. 1.1**, the biological barriers are illustrated progressively from the molecular level, to the cellular level, and finally to the organ level.

At the molecular level, several factors influence a drug's ability to reach its target effectively. Water solubility is a primary barrier for many drugs must dissolve in the body's aqueous environment to be absorbed and transported, yet poorly soluble compounds often suffer from

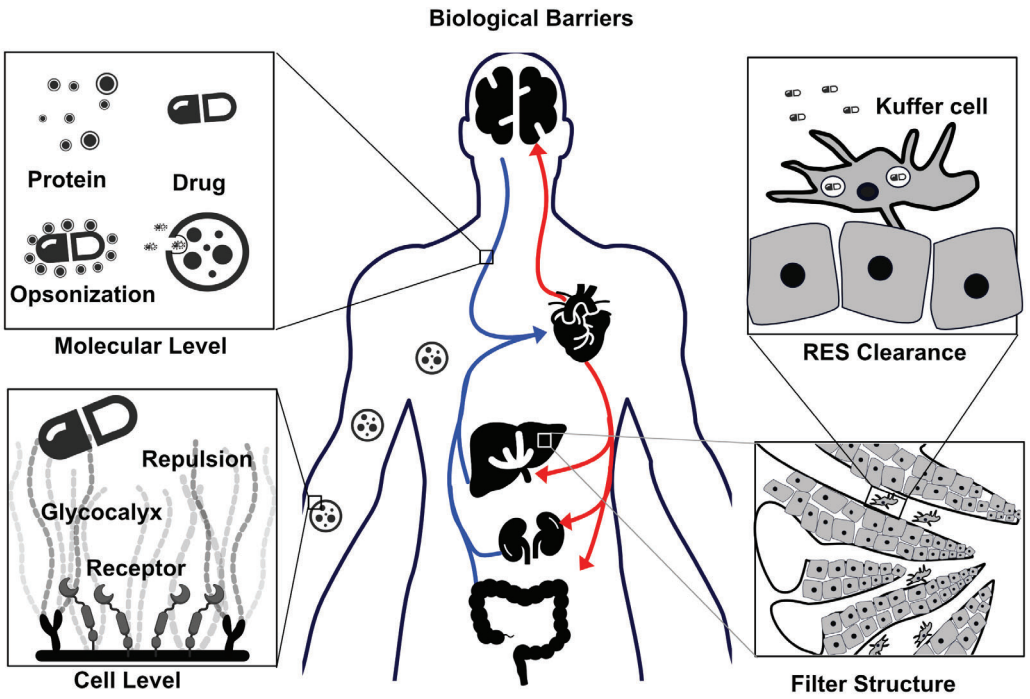

Figure 1.1 The biological barrier for the drugs. At the molecular level, certain plasma proteins interact with injected drugs or nanoparticles, forming a protein corona that tags them for recognition by macrophages, ultimately leading to their clearance from the bloodstream. At the cellular level, the glycocalyx, a dense, protective layer on the surface of many cells, can act as a barrier that repels drugs, reducing their ability to interact with target cells. On the organ level, the reticuloendothelial system (RES), particularly the liver and spleen, plays a key role in filtering drugs from circulation. Macrophages residing in the RES further contribute to this clearance by engulfing and degrading foreign substances. Together, these multi-scale biological barriers significantly influence the pharmacokinetics and biodistribution of therapeutic agents.

low bioavailability[14,15]. Additionally, once in the bloodstream, drugs may interact with plasma materials, primarily antibodies or components of the complement system, forming a protein corona[16-18]. This process is called opsonization. Once opsonized, these drugs are more readily recognized by phagocytic cells, such as macrophages and neutrophils, through their surface receptors (e.g., Fc receptors or complement receptors)[6,19-21]. Another important factor is pH variation across different body compartments[22-25]. For example, the acidic environment of the stomach or the varying pH along the gastrointestinal tract can degrade pH-sensitive drugs or alter their absorption[26].

At the cellular level, biological barriers include both extracellular and intracellular components. The cell membrane, made up of a heterogeneous mix of lipids, proteins, and carbohydrates, serves as a selective barrier that regulates the entry and exit of substances[27,28]. Passive diffusion is typically limited to small, nonpolar molecules, while most drugs require more complex mechanisms to cross the membrane. In addition to receptor targeting, the dense glycocalyx on the cell surface can repel drugs and hinder effective interactions[29,30]. Internalization may occur through pinocytosis, phagocytosis, or receptor-mediated endocytosis, generally involving membrane wrapping[31,32]. Once inside the cell, drugs may be expelled through exocytosis or transported to the lysosome, an organelle abundant in degradative enzymes and a highly acidic environment, which can inactivate many therapeutic agents and reduce their efficacy.

At the physiological level, biological barriers are influenced by both tissue architecture and cellular composition[33,34]. The structural characteristics of the vasculature govern drug distribution[35]. Different organs display varying vascular permeability, which critically affects drug extravasation. For example, in the liver and spleen, fenestrated capillaries and irregular vascular geometry slow down blood flow velocity, promoting efficient filtration of circulating substances[36,37]. These organs, which act as primary blood filters, also contain numerous phagocytic cells that constitute the reticuloendothelial system (RES). This active defense mechanism identifies and removes foreign agents, including many drugs[38,39]. In contrast, organs like the brain are protected by the highly selective blood–brain barrier, which significantly restricts drug penetration[40,41]. Once drugs diffuse into the interstitial space, the lymphatic drainage system further aids in their clearance by transporting interstitial fluid and macromolecules away from tissues, thereby reducing local drug retention[42,43].
Biological barriers play a crucial role in protecting the body from harmful external substances, but they also pose significant obstacles to drug delivery. Overcoming these barriers remains a major challenge in the development of effective therapeutics.

1.2 Nanomedicine: the next generation of therapies

1.2.1 Enhanced therapeutics through nanomedicine

Minimizing side effects reduces patient discomfort and improves quality of life. To overcome the limitations posed by passive drug distribution, delivery systems are widely utilized to enhance biodistribution. Besides, therapeutic agents can be shielded from degradation, their solubility enhanced, and their release precisely controlled[44]. This targeted delivery approach significantly enhances therapeutic efficacy while minimizing side effects, serving as a classic example of nanomedicine in action. As mentioned above, Doxil® and other drug-encapsulating formulations are examples of nanomedicine in application. Nanomedicine refers to the application of nanotechnology in the diagnosis, treatment, monitoring, and prevention of diseases, playing a key role in next-generation drugs[45]. The nanoscale offers a degree of control over drug behavior. The properties of nanomedicines are relatively easy to manipulate, allowing for more precise tuning of drug release, distribution, and therapeutic effects[46].
Nanomedicine offers a powerful strategy for overcoming these biological barriers, thanks to its small size, engineered surface properties, and stimuli-responsive behavior-all of which enhance its ability to navigate the body's complex defense systems[47]. The defining feature of nanomedicine is its nanoscale size (typically 1–100 nanometers), which enables unique interactions with biological systems and facilitates targeted drug delivery[48,49]. The enhanced permeability and retention (EPR) effect is a well-known example of the advantages offered by nanomedicine at the nanoscale[50]. The nanoscale size of nanomedicines also enables them to penetrate deeply into tissues. Through surface modifications, including targeting ligands or stealth coatings (e.g., PEGylation), nanomedicines can extend circulation time, evade immune surveillance, and enable site-specific delivery-ultimately overcoming the blood clearance and enhancing both therapeutic efficacy and safety[51]. Through surface modification, nanomedicine can also be designed to overcome the biological barriers. The poly(lactic-co-glycolic acid) (PLGA) nanoparticles can be surface-functionalized with targeting ligands such as transferrin, lactoferrin, or apolipoprotein E (ApoE) to facilitate receptor-mediated transcytosis across the BBB[52]. In **Fig. 1.2 (a)**, we show examples of the polymers used for nanoparticle surface modifaction[53].

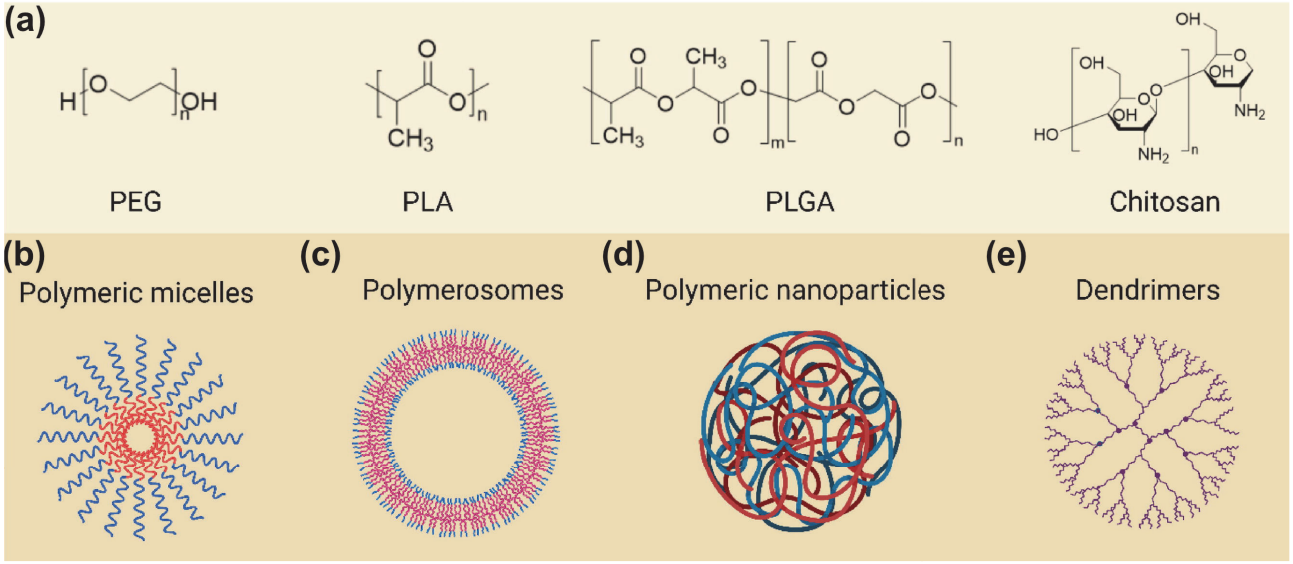

Figure 1.2 Examples of different nanodelivery systems. (a) Polymers used in nanoparticle formulation include polyethylene glycol (PEG), an FDA-approved homopolymer that enhances solubility, reduces immunogenicity, and prolongs circulation, as well as poly(lactic-co-glycolic acid) (PLGA), polylactic acid (PLA), and chitosan, all of which are biodegradable and biocompatible. **(b)** Dendrimers are highly branched, tree-like macromolecules with a well-defined, monodisperse structure, making them ideal carriers for drug delivery. **(c)** Nanoemulsions are nanoscale oil-in-water or water-in-oil dispersions stabilized by surfactants, widely used as drug delivery systems due to their high solubility, stability, and bioavailability for hydrophobic compounds. **(d)** Dendrimers are highly branched, tree-like macromolecules with a well-defined, monodisperse structure, making them ideal carriers for drug delivery.. (from **Ref. 53**)

One central component of nanomedicine is nanodelivery systems[53-55]. These systems are designed to improve the precision, efficiency, and safety of therapeutic delivery by leveraging nanoscale materials with tunable properties. Organic nanoparticles, such as amphiphilic block copolymers, liposomes and dendrimers, are widely used due to their biocompatibility and flexibility[56]. The amphiphilic block copolymers differ in size, shape, and structure, such as the polymeric micelle and polyersome. Liposomes, composed of phospholipid bilayers, can encapsulate both hydrophilic and hydrophobic drugs, enhancing stability and bioavailability[57]. Polymeric nanoparticles are tiny, engineered particles composed of biodegradable or synthetic polymers. Dendrimers are highly branched, tree-like polymers that allow precise control over size and surface functionality, making them ideal for multivalent drug attachment and controlled release[58]. Some of these organic nanoparticles are shown in **Fig.1.2(b)-(e)**. Inorganic nanoparticles, such as gold nanoparticles and silica-based carriers, offer structural stability and tunable surface properties, often useful in imaging and theranostics[59]. Together, these delivery systems provide versatile strategies to overcome biological barriers and improve therapeutic outcomes.

Despite these advancements, there remains considerable room for improvement in achieving precise targeting. Continued research is needed to optimize nanomedicine and fully realize its potential in clinical applications.

1.2.2 Design precision nanomedicine using the multivalency effect

It is not trivial to overstate the importance of two highly desirable properties in drug design: selectivity and specificity. As we already discussed, the answer is easy if the diseased cell

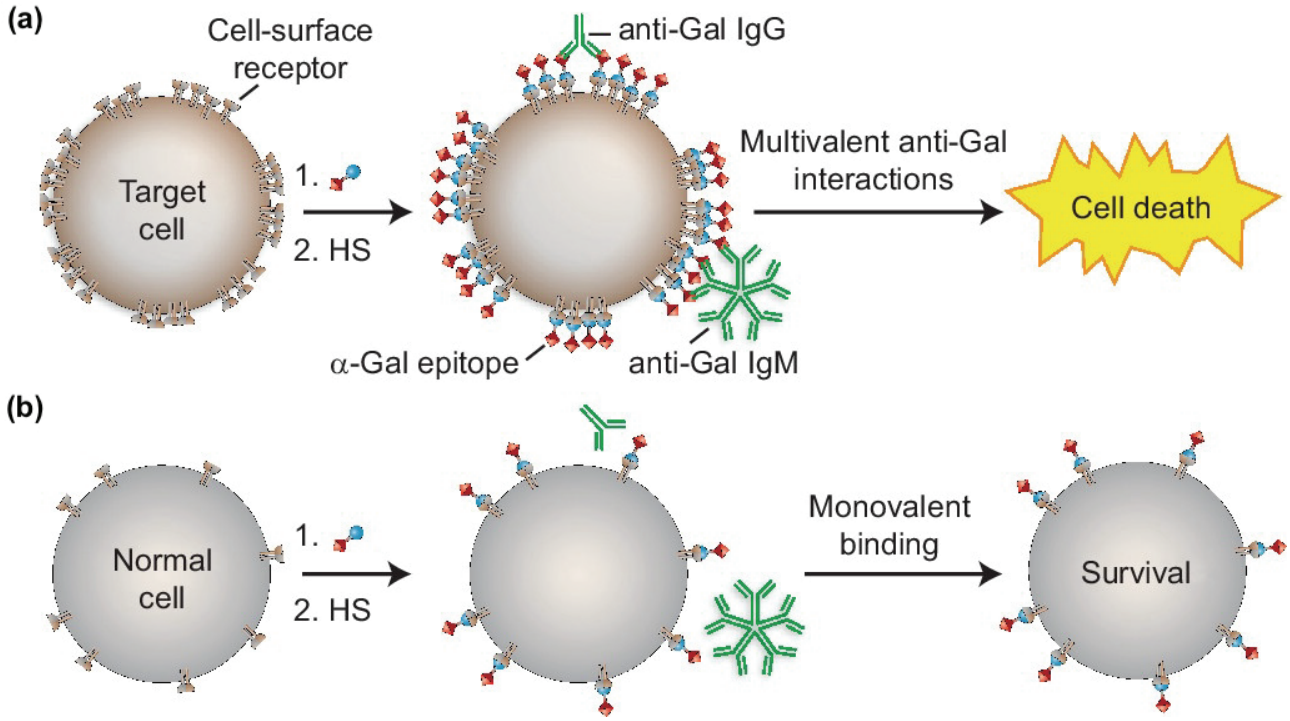

Figure 1.3 Graphical representation of cell-targeting strategy based on multivalent binding. A bifunctional conjugate is designed to bind with high affinity to a specific cell-surface receptor, such as integrin, which is highly expressed on target cells. In this system, the blue circle denotes the integrin ligand. Attached to the conjugate is a low-affinity α-Gal epitope (represented by a red diamond), which recruits naturally occurring bivalent anti-Gal IgG and decavalent anti-Gal IgM antibodies (shown in green) when a noncovalent multivalent array forms on the cell surface. **(a)** In cells expressing high levels of the target receptor, this recruitment leads to strong antibody binding and triggers complement-mediated cell death. **(b)** Conversely, cells with low receptor expression remain unaffected, as the monovalent interaction with anti-Gal is too weak to initiate an immune response. (From **Ref. 60**)

expresses a unique receptor. The problem becomes quite frustrating if the diseased cells have homogenous receptors. This is particularly relevant for tumor cells, which often express receptors that are also found on healthy cells. As a result, designing highly specific ligand–receptor interactions that selectively target tumor cells while sparing healthy cells remains a significant challenge. Thus, there is a pressing need to develop new strategies to enhance the selectivity-more specifically, to equip drugs with the ability to distinguish between cells even when they express homogenous receptors. There is some evidence that may help address this problem. Coby B. Carlson *et al.* found that using low-affinity, multivalent interactions could enable selective tumor cell targeting[60]. As illustrated in **Fig. 1.3**, a bifunctional conjugate is designed to bind with high affinity to a cell-surface receptor, such as integrin, which is highly expressed on target cells. In the schematic, the blue circle represents the integrin ligand, while the red diamond denotes the low-affinity α-Gal epitope. When the conjugate binds to cells with high receptor expression, it forms a multivalent array on the cell surface that recruits bivalent anti-Gal IgG and decavalent anti-Gal IgM antibodies (shown in green), ultimately triggering complement-mediated cell death (**Fig. 1.3(a)**). In contrast, as shown in **Fig. 1.3(b)**, cells with low levels of the target receptor do not form stable interactions with the antibodies, and therefore remain unaffected due to the weak monovalent anti-Gal binding. A higher number of binding sites increases the likelihood of binding, a phenomenon known as the multivalency effect. The multivalency effect refers to the enhanced strength and specificity of interactions that occur when multiple binding sites on one molecule simultaneously bind to multiple receptors on

another surface[61]. Such selectivity can be attained through multivalency, inspiring approaches to design drugs that leverage the multivalency effect.

Nanomedicines can effectively exploit the multivalency effect by being functionalized with multiple ligands or multiple binding sites[62,63]. However, nanomedicines still require further refinement to distinguish between cell populations, an approach popularized as the 'magic bullet' that brought fame to Paul Ehrlich[64]. The multivalency effect plays a pivotal role in the design of such precision nanomedicine. Establishing quantitative standards to guide nanomedicine formulation is a critical process, driven by the need to elucidate the underlying mechanisms of nanomedicine-cell interactions.

Certain theoretical models have been established to derive the guiding principle for designing the precision target system. As the interactions between nanoparticles and cells become better understood, the development of corresponding models has been significantly advanced. In 2003, Pavel I. Kitov and David R. Bundle proposed a quantitative model to analyze the thermodynamic parameters of multivalent interactions[61]. This model considered all bound species, described multivalent binding, and obtained the overall binding free energy as a function of the degeneracy of bound states. The model inspired the theoretical explanation of the multivalency effect, even if it's limited to the molecular level. In 2011, Francisco J. Martinez-Veracoechea and Daan Frenkel derived the thermodynamics model considering nanoparticle interaction with the cell surface and validated the multivalency effect using the Monte Carlo simulation[65]. Building on the milestone of theory, the theoretical framework for the multivalent effect has seen significant development over the past decades, with numerous outstanding studies published[66-69].

By leveraging the multivalency effect, nanomedicine, including nanoparticles, can be engineered to exhibit superselective binding to their targets, which overexpress the target receptors. Theoretical modeling allows us to predict the binding probability of nanoparticles with varying parameters. This enables the determination of optimal nanoparticle size, ligand–receptor binding affinity, and the number of functional ligands required for effective targeting[68]. By integrating these predictive tools into the design process, researchers can systematically optimize nanoparticle formulations, reduce trial-and-error in experimental studies, and accelerate the development of more selective and efficient nanomedicines.

1.3 Physical mechanism behind the superselectivity based on the multivalency effect

1.3.1 Theoretical model and general principle for superselectivity

The superselectivity model is based on thermodynamic principles, considering only the interactions between ligands and receptors[70]. At the initial explanation of the model, let's consider a question everyone has heard: Why does an apple fall to the ground? The answer is easy: the ground represents the state of low gravitational potential energy. This is the key point for understanding the ligand-receptor interaction model. Let's consider a nanoparticle with N_L functional ligands binding to N_R receptors in the binding volume. The binding energy for each bond is ϵ_B. Ultimately, the system settles into the lowest energy state, regardless of the time it takes to get there. We assume the ligand is flexible enough to reach every receptor, and the probability of each combination of the ligand-receptor binding is equal. For example, the probability of ligand L1 binding to receptor R2 is equal to that of ligand L1 and ligand L2 binding to receptor R1 and receptor R2, respectively. We also need to keep in mind that the

system tends to traverse all the possible states. Thus, we could drive the partition function of the system as:

$$q = \sum_{\lambda=1}^{min(N_L,N_R)} \binom{N_L}{\lambda}\binom{N_R}{\lambda}\lambda! e^{-\lambda\beta\epsilon_B} \qquad (1.1)$$

where λ is the number of bound ligands, $\beta = \frac{1}{kT}$ with k e being the Boltzmann constant and T the absolute temperature. To avoid misunderstanding of the lowest energy state, I would give an example of rolling the dice. 1 would be the lowest number of dice. Reaching the lowest energy state does not mean we get 1 when we roll the dice, but the mathematical expectation of rolling the dice[70]. The partition function describes the overall strength of the nanoparticle-receptor binding.

The degeneracy part (combination of ligand/receptor pairs or combinatorial part), $\Omega = \sum_{\lambda=1}^{min(N_L,N_R)} \binom{N_L}{i}\binom{N_R}{\lambda}\lambda!$, is the number of all the possible ways of forming the bound. The degeneracy depends on the binding topology. In the previous text, we mainly talked about the case when the ligands are flexible. In the aspect of ligand topology, there are mainly 3 topology types[61]. As the system forms λ bonds, the combinatorial part would be:

1. Indifferent topology, only one ligand/receptor complex forms,
   $\Omega_{\lambda=1} = N_L N_R$

2. Linear topology, where rigid ligands bind with receptors together,
   $\Omega_\lambda = (N_L - \lambda + 1)(N_R - \lambda + 1)$.

3. Radial topology, where any ligand can bind with any receptor,
   $\Omega_\lambda = \binom{N_L}{\lambda}\binom{N_R}{\lambda}\lambda!$.

Thus, we could rewrite the partition function for a more general case:

$$q = \sum_{\lambda=1} \Omega_i e^{-\lambda\beta\epsilon_B} \qquad (1.2)$$

Compared with all the topologies, it's obvious that the radial topology has the largest combinatorial value. The nanoparticle should be designed with flexible ligands to increase the binding states. Unless otherwise stated, we henceforth assume that the particles possess flexible ligands. Except for the ligand topology, the value of the partition function increases nonlinearly with the increase in the number of receptors, as shown in **Fig. 1.4(a)**. Increasing the number of receptors leads to a corresponding increase in the system's degeneracy. The number of ligands, ligand topology, and the number of receptors all contribute to the partition function through the system's degeneracy.

Under the assumption that adsorption sites on the cell surface are independent and each can bind at most one nanoparticle, the fraction of bound nanoparticles can be described using the well-established Langmuir adsorption model:

$$\theta = \left\langle \frac{zq}{1+zq} \right\rangle \qquad (1.3)$$

where z is the activity of the nanoparticle, $1 = \Omega_0 e^{-0\cdot\beta\epsilon_B}$ stands for the nonbinding situation and $\langle \ldots \rangle$ describes the Poisson distribution of receptors[67,68,71]. In a dilute solution $z \approx [P]N_A v_B$,

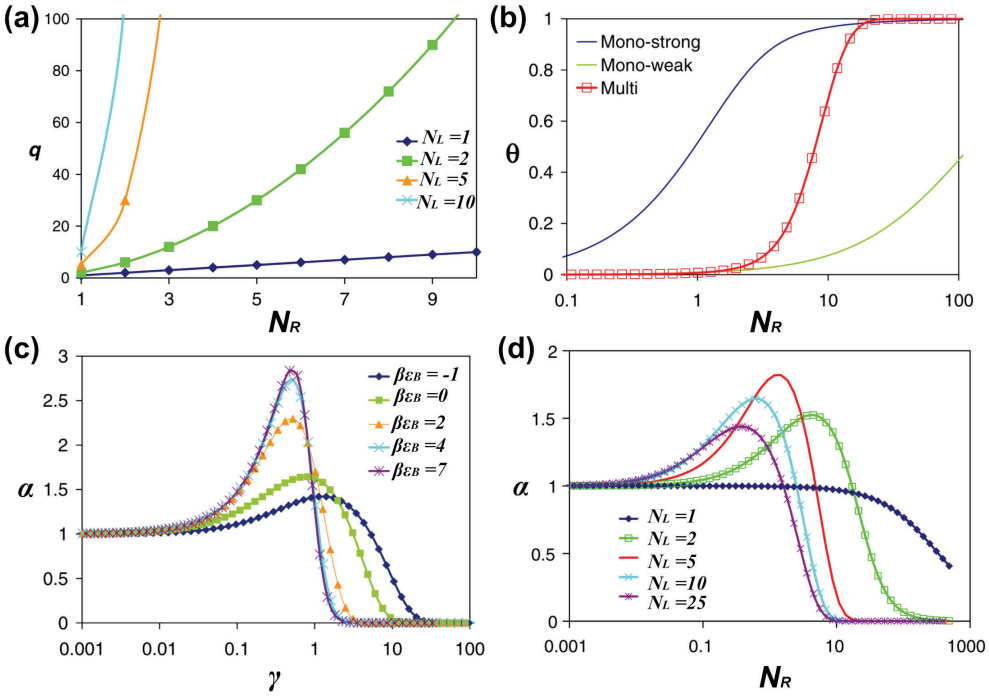

Figure 1.4 Design a nanoparticle based on the multivalency effect. (a) Partition function values plotted against receptor numbers for various ligand counts. **(b)** Binding fraction as a function of receptor number for three scenarios: monovalent strong interaction, monovalent weak interaction, and multivalent weak interaction. **(c)** Selectivity parameter plotted against γ for different binding energies. **(d)** Selectivity parameter versus receptor number for varying numbers of ligands. (From **Ref. 65**)

where $[P]$ is the nanoparticle concentration, N_A refers to the Avogadro number, and v_B presents the binding volume that nanoparticles can explore while binding with the adsorption site[65,67,68]. θ can be experimentally observed as the cell surface coverage, which offers simplicity in the binding preference. Through the equation, we could find that it is the partition function that governs the nontrivial dependence of the adsorption probability on the cell surface. The adsorption model helps explain why multivalent systems exhibit high binding strength with cells that express a high level of receptors.

The adsorption model enables the prediction of the fraction of bound nanoparticles, providing valuable insight for optimizing nanoparticle formulation. In **Fig. 1.4(b)**, Francisco J. Martinez-Veracoechea and Daan Frenkel presented the binding fraction as a function of receptor number, comparing mono-weak binding, mono-strong binding, and multi-weak binding scenarios[65]. Compared to mono-weak and mono-strong binding, multi-weak binding exhibits a sigmoidal profile, indicating a switch-like activation of binding once the receptor number exceeds a certain threshold. They defined super selectivity as the phenomenon where the fraction of bound nanoparticles increases more rapidly than linearly with the surface concentration of receptors. This thermodynamics model provides a theoretical explanation and the guiding rule for exploiting the multivalency effect to achieve superselectivity.

To quantify the selectivity and determine the condition where nanoparticles are most sensitive to the variation of receptors, Francisco J. Martinez-Veracoechea and Daan Frenkel. defined a selectivity parameter α:

$$\alpha = \frac{dln\theta}{dlnN_R} \qquad (1.4)$$

In **Fig. 1.4(c)** and **(d)**, they showed the selectivity parameter with the increase of γ ($\gamma = N_R e^{-\beta \epsilon_B}$) and the number of receptors, respectively. The system tends to have a higher selectivity when the ligand-receptor affinity is lower[68,72]. This is due to the nonlinear increase of the combination between the ligands and receptors and the lesser contribution of the binding energy. However, the binding energy can't be smaller than the unspecific interaction; otherwise, the model deviates from the specific ligand-receptor interaction, and the nanoparticles will target cells without specific targeting.

The minimal requirements for achieving superselective binding are multivalency, combinatorial entropy, and low ligand/receptor affinity[69]. Superselectivity benefits from multiple functional ligands, which can form numerous ligand–receptor complexes, thereby enabling nanoparticles to recognize cells with high receptor expression. To increase the combinatorial part, the ligand should be flexible enough to bind to every receptor. Last but not least, the ligand–receptor affinity should be carefully controlled within a specific range. It must be weak enough to ensure sensitivity to receptor expression, but not so weak that nonspecific interactions become dominant. By using the thermodynamics model, we could offer numerous prospects for the development of superselective nanoparticles.

However, the cell surface is highly complex, featuring not only receptors but also the glycocalyx, which acts as a protective barrier against nonspecific interactions. Furthermore, the interaction between nanoparticles and receptors can generate steric hindrance that prevents effective ligand binding. To enable more precise nanomedicine design and establish detailed guidelines for nanoparticle fabrication, further refinement of superselectivity theory is necessary.

1.3.2 Phenotypic association theory (PAT) based on the interaction between the nanoparticle and the cell surface

Cell phenotype refers to the observable characteristics of a cell, arising from the interaction between its genetic makeup (genotype) and environmental factors. Here, we refer to cell phenotype as the expression of receptors and glycans on the cell surface, as these are the key elements recognized by the nanoparticles[73]. Our group has developed a phenotypic association theory (PAT) that accounts for the steric effect. The steric effects arise from the spatial arrangement of the atoms and are nonbinding interactions that influence the shape (conformation) and reactivity of the ions and molecules. The steric effect plays a pivotal role in the nanoparticle/receptor interactions.

In biological systems, proteins regulate their repulsive interactions through surface charge and structural configuration. Cells are coated with long polysaccharides known as glycans, which are either chemically linked to proteins to form glycoproteins or attached to dedicated receptors. These glycans exhibit a high affinity for water and are often densely packed, causing them to extend outward and form a brush-like structure that helps protect the cell. Such a structure creates a strong repulsive steric potential protecting the cell surface from fouling[74,75].

Nanoparticles made of hydrophilic polymer brushes prevent nonspecific interactions and protein fouling, thereby ensuring long-term compatibility[76,77]. Besides, using non-immunogenic moieties featuring neutral polar or zwitterionic functional groups can help to control the nonspecific interaction, thus preventing opsonization[16,78,79]. These polymer chains will interact with the water in proximity and generate repulsive potentials less sensitive than other species. The polymers generally used are poly(ethylene oxide) (PEO), poly(2methacryloyloxyethyl phosphorylcholine) (PMPC), poly(glycerol)s, poly(amino acid)s, polysarcosine, poly(2-oxazoline)s, and poly(N-(2hydroxypropyl)methacrylamide). They could satisfy all the requirements.

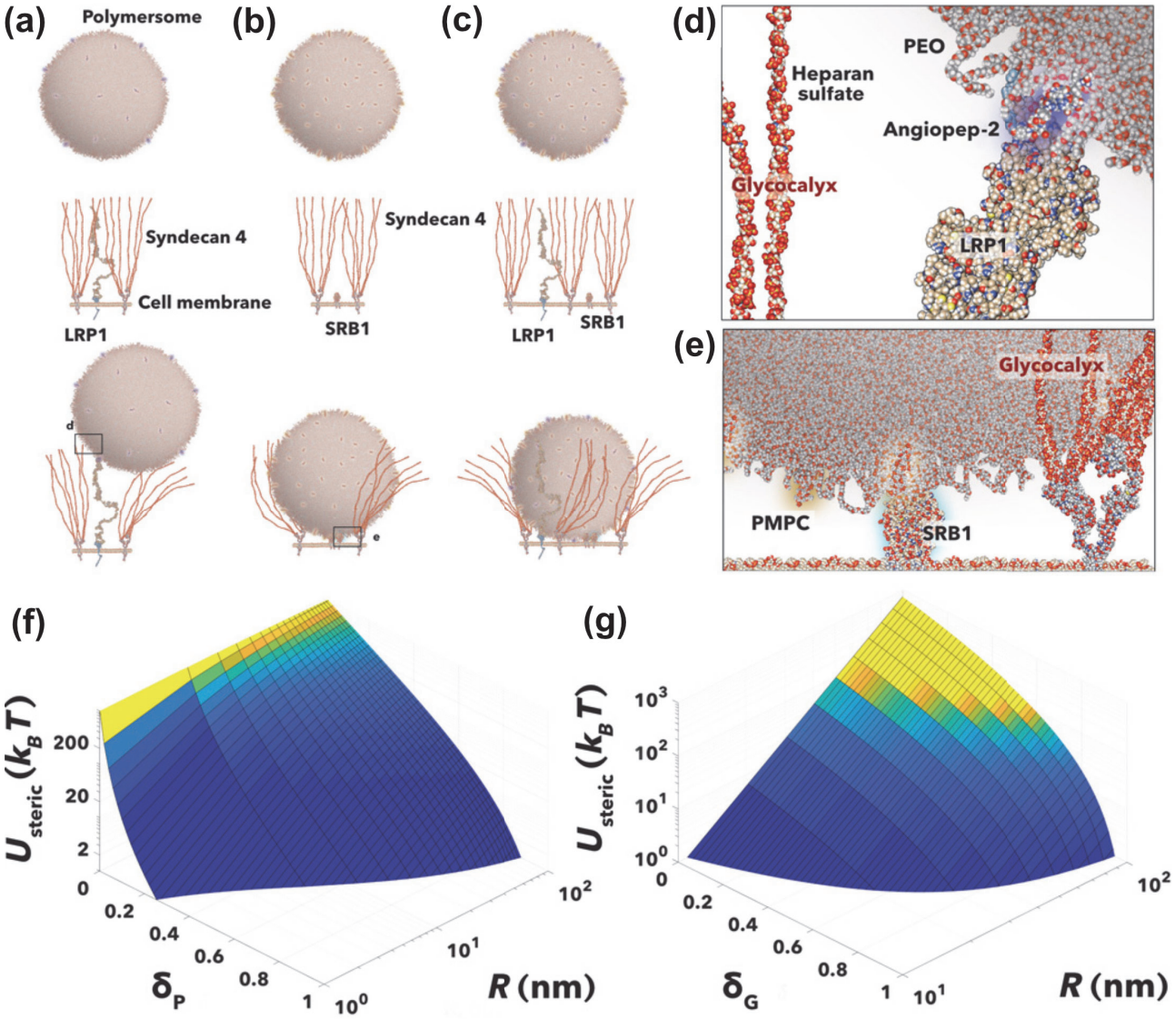

Figure 1.5 Binding scenario and the steric potential for multivalent POEGMA-PDPA polymersomes functionalized for targeted delivery. (a) Polymersomes decorated with angiopep peptides targeting LRP1 receptors. **(b)** Polymersomes functionalized with PMPC chains targeting SRB1 receptors. **(c)** Dual-ligand polymersomes simultaneously targeting both LRP1 and SRB1 receptors. **(d)** Detailed molecular interactions between angiopep and LRP1, modulated by steric hindrance from both PEO and glycocalyx brushes. **(e)** Detailed molecular interactions between PMPC and SRB1, similarly influenced by steric repulsion from the PEO and glycocalyx brushes. **(f)** Repulsive steric potentials associated with LRP1 insertion into the PEO brush, calculated as functions of polymersome radius (R) and PEO insertion depth (δ_P). **(g)** Repulsive steric potentials during polymersome insertion into the glycocalyx brush, shown as functions of radius (R) and insertion depth into the glycocalyx heparan sulfate chains (δ_G). (From **Ref. 68**)

In the scenario of a nanoparticle approaching the cell surface, the nanoparticle needs to overcome the glycan brush to target receptors buried inside the brush, as shown in **Fig. 2(a)-(d)**[68]. This process gives rise to steric potential, which originates from two main factors: osmotic pressure due to volume exclusion, and elastic resistance resulting from the compression of glycan chains. The steric repulsion depends on the relative height of the receptor concerning the glycan height, according to Halperin[80]. The relative height is defined as an interference parameter δ_G. If the nanoparticle radius is smaller than the glycan brush, the elastic component can be ignored. In nanomedicine, a particle radius of approximately 50 nm is considered

optimal for cellular uptake[81-83]. The glycans are approximately 100 nm. We can thus calculate the potential as the nanoparticle inserts into the glycan brush as a function of the interference parameter, considering only the osmotic pressure component, as follows:

$$\beta U_G = \frac{4\pi R^3 (1-\delta_G)^{9/4}}{3\sigma_G^{3/2}} \qquad (1.5)$$

where σ_G is the area per glycan on the cell surface. Since we are considering only the osmotic pressure component, the steric potential is primarily determined by the interference parameter, which does not directly depend on glycan height; in other words, it is independent of the degree of glycan polymerization.

It is important to mention that Shihu Wang *et al.* suggested that nanoparticles can be designed with greater selectivity when ligands are buried within the polymer chains, as validated by Monte Carlo simulations[84]. The paper considers nanoparticles possessing the ligand buried inside the polymer chain. After nanoparticles penetrate the glycan brush and reach the vicinity of the receptor, the receptor must then overcome the steric potential imposed by the polymer brush to bind with ligands inside. The steric potential operates through the same mechanism as when nanoparticles are inserted into the glycan brush. If the receptor is small and does not compress the polymer chains, the interference steric potential from the polymer brush would be:

$$\beta U_P = \frac{V_R (1-\delta_P)^{9/4}}{\sigma_P^{3/2}} \qquad (1.6)$$

where δ_P is the relative length between the ligand and receptor, V_R refers to the volume of the receptor, and σ_P indicates the area per polymer chain. Using Zhulina's model, it can be derived as:

$$\sigma_P = \sigma_0 \left(1 + \frac{\delta_P h_P}{R}\right)^{\gamma - 1} \qquad (1.7)$$

with σ_0 being the area per chain on a plane surface and γ being a geometrical parameter to represent the packing of the chains on the surface[85]. For $\frac{h_0}{R} \leq (\sqrt{3}-1)$, $\gamma = \left(\frac{h_0}{R}+1\right)^2$ unless $\gamma = 3$. h_0 is the polymer length on a planar surface. According to Zhualina *et al.*, the brush height on the surface can be written as:

$$h_P = R\left[\left(1 + \frac{(\gamma+2)N_P}{3R}\left(\frac{v a_0^2}{3\sigma_0}\right)^{\frac{1}{3}}\right)^{\frac{3}{\gamma+2}} - 1\right] \qquad (1.8)$$

where a_0 stands for a single repeat unit length of the polymer brush and N_P refers to the degree of polymer chain polymerization.

Here, we present our group's previous work to elucidate the steric potential. In **Fig.1.5**, we show a binding scenario and the steric potential for multivalent poly[oligo(ethylene glycol) methyl methacrylate]-poly(2-(diisopropylamino)ethyl methacrylate) (P[(OEG)10MA]20-PDPA100) polymersomes for selective targeting. Panels **(a)** to **(c)** show schematic representations of polymersomes functionalized with angiopep peptides targeting low-density lipoprotein receptor-related protein1 (LRP1) receptors **(a)**, PMPC chains targeting scavenger receptor class B member 1 (SRB1) receptors **(b)**, and a combination of both ligands for dual-receptor targeting

(c). Panels **(d)** and **(e)** delve into the molecular interactions between each ligand–receptor pair, emphasizing how these interactions are modulated by steric hindrance from the polymersome's PEO brush and the cell's glycocalyx. Panels **(f)** and **(g)** further quantify these effects by showing the repulsive steric potentials encountered during LRP1 insertion into the PEO brush and polymersome insertion into the glycocalyx brush, as functions of the polymersome radius and correlated interference parameter. Together, these diagrams highlight how ligand configuration and physical barriers shape the efficiency and selectivity of nanoparticle-cell interactions.

Having all this steric potential, we could thus modify the partition function. We can write the binding energy between a single ligand and a receptor as: $e_B = \epsilon_B + U_P$. The partition function between the nanoparticle and the receptors would be:

$$Q = \sum_{i=1} \Omega_i e^{-i\beta(\epsilon_B + U_P)} \qquad (1.9)$$

Considering the steric from the glycans, the final partition function would be:

$$\Xi = e^{-\beta U_G} \sum_{i=1} \Omega_i e^{-i\beta(\epsilon_B + U_P)} \qquad (1.10)$$

The binding fraction of the partition function would be:

$$\theta = \frac{a\Xi}{1 + a\Xi} \qquad (1.11)$$

where $a = [P]N_A v_B$ is the activity of the nanoparticle. $v_B = \pi/3((R+d)^3 - R^3)$ for specific nanoparticles with $d = \delta_P h_P$ being the ligand length.

As shown in the final partition function, the partition function represents the attractive interactions, while the steric potential accounts for the repulsive contributions. Suppose nanoparticles are designed to target a specific cell phenotype. In that case, the binding scenarios can be categorized into four types: low receptor expression with low glycan expression, low receptor expression with high glycan expression, high receptor expression with low glycan expression, and high receptor expression with high glycan expression.

1.3.3 Optimization for the PAT to discriminate cell types

Obviously, it's not difficult to design nanoparticles that selectively target cells with high receptor and low glycan expression based on the multivalency effect. This is ideal for targeting tumor cells, as they generally express higher levels of receptors. For more precision targeting, a problem arises when nanoparticles are designed to target cells with a low receptor density. Some diseases are characterized by downregulated receptor expression, such as heart failure and Parkinson's disease[86,87]. One approach to addressing this issue is to design nanoparticles with range selectivity, enabling binding only within a specific range of receptor densities[67]. This can be achieved by introducing a steric factor associated with the receptors. To tackle more complex targeting tasks, we propose strategies for optimizing selectivity informed by PAT.

The use of "protective" receptors on the nanoparticle surface, as proposed by Stefano Angioletti-Uberti, can enhance binding selectivity[66]. Increasing the number of ligands typically results in high avidity for cells. However, redundant ligands must bind exclusively to specific target receptors, as interactions with non-target receptors can result in nonspecific binding and potentially harmful effects. As illustrated in **Fig. 1.6**, we present the binding behavior of a multivalent nanoparticle to a cell surface expressing various receptors. Typically, the nanoparticle does not bind to cells with low expression of the target receptor, as shown **in Fig. 1.6(a)**. Binding occurs when the cell expresses a high level of the target receptors (**Fig. 1.6(b)**). In a more realistic setting, ligands can interact with both target and non-target receptors, although they bind the latter with low affinity. As also shown in **Fig. 1.4(a)**, the partition function increases with the number of ligands. Even when the binding energy is relatively low, the

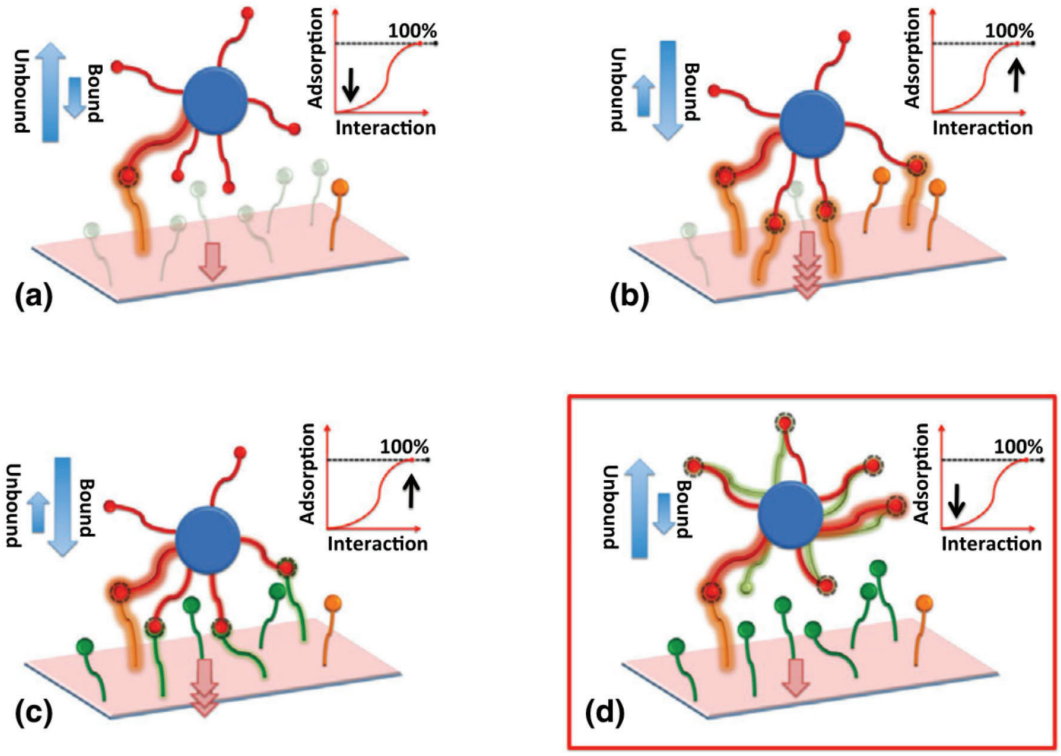

Figure 1.6 Binding of a multivalent nanoparticle (blue) to the cell surface (pink) expressing various receptors. **(a)** When the expression level of the target receptor is low, no significant binding occurs. **(b)** Once receptor expression exceeds a certain threshold, the ligands bind selectively, resulting in nanoparticle adsorption. **(c)** In a more realistic scenario, ligands can interact with both targeted and untargeted receptors, binding the latter with low affinity. Nevertheless, the accumulation of multiple weak interactions can still lead to nonspecific adsorption, even in the absence of target receptor overexpression. **(d)** This challenge can be addressed by incorporating "protective" receptors (light green) directly onto the nanoparticle surface. If these protective receptors form stronger interactions than the untargeted receptors, selective binding can be effectively recovered. (From **Ref. 66**)

increased number of binding configurations (degeneracy) leads to higher entropy, which in turn promotes interaction with unintended receptors, thereby reducing binding selectivity. To mitigate this effect, "protective" receptors can be introduced on the nanoparticle surface. In the absence of target receptors, ligands form intraparticle bonds with these protective receptors, preventing them from interacting with untargeted receptors. When ligands encounter target receptors with higher affinity than the protective ones, they dissociate from the protective receptors and bind to the target receptors instead (**Fig. 1.6(d)**). In this way, selective binding can be effectively restored.

Pattern recognition based on the multivalence effect can also lead to selectivity[88,89]. Multivalent pattern recognition is a mechanism by which particles or molecules equipped with multiple ligands selectively bind to target surfaces based on the spatial organization and density of receptors. Hale Bila et al. observed super-selective binding in hexavalent architectures, indicating that nanoscale control of geometric patterns can be used to discriminate between receptor densities in a highly selective manner[89]. By adjusting the number, spacing, and geometric arrangement of ligands, researchers can fine-tune the binding behavior, ensuring that the nanoparticle binds only when the receptor density exceeds a specific threshold. This strategy provides enhanced precision in controlling nanoparticle selectivity.

Receptor clustering and uniformity could enhance the local receptor density, thereby facilitating nanoparticle binding[90,91]. Zhaoping Xie et al. demonstrated that changes in inter-receptor attraction smaller than the thermal energy can selectively switch receptor clustering and activation on or off in an almost stepwise manner[91]. If the local receptor density is below the threshold required for effective binding, inter-receptor interactions suppress binding. When the local density exceeds this threshold, these interactions enable binding. Such receptor clustering–mediated enhancement of binding is often reported in the immune system. This inspires the design of ligand-decorated nanoparticles or multivalent architectures that follow similar principles, binding only when receptors are sufficiently clustered to enable highly specific targeting.

The number of ligands and the receptor density must be optimized within a specific range to ensure effective binding. Meng Liu *et al.* demonstrated that increasing both receptor density and ligand number can introduce significant steric hindrance, which restricts binding to a narrow window of receptor expression levels and ligand number[67]. Binding occurred only when both receptor density and ligand number fell within this optimal range; outside of it, no adsorption on the cell surface was observed. This phenomenon, known as range selectivity, places constraints on the optimal ligand number to achieve maximum binding efficiency in multivalent constructs. The range selectivity in the receptor density enhances the nanoparticle's ability to discriminate cells with different receptor expression.

Multiplexing in super-selectivity refers to the strategy of decorating a single nanoparticle or surface with multiple types of ligands, each designed to bind selectively to a different receptor type[68,73]. Multiple types of ligand–receptor interactions may occur, and it is essential to account for all relevant interaction types. This allows the system to recognize and bind to complex cellular signatures, not just a single receptor. Many cell types share individual markers, but differ in combinations and densities of receptors. Multiplexing can leverage these subtle differences to enhance the ability of nanoparticles to distinguish between different cell populations. As shown in PAT, the nanoparticles are very sensitive to receptor density. Targeting an additional receptor type may result in a synergistic effect, thereby lowering the threshold of receptor expression required for effective binding. If the target cells express low levels of the primary receptor, a feasible strategy is to identify a synergistic receptor to enhance binding efficiency.

In summary, we believe that a deeper understanding of nanoparticle–receptor interactions could contribute to the development of a true 'magic bullet' for treating various diseases.

1.4 Multivalence uptake kinetics in nanoparticle delivery systems

1.4.1 Nanomedicine binding kinetics

As demonstrated in the previous section, we addressed how to design nanomedicine with selective targeting. However, the thermodynamic model overlooks the factor of time. How long does it take for the designed nanomedicine–receptor interactions to reach equilibrium? It could take a day, a week, or even longer-an outcome that is far from ideal for practical applications. In addition to selectivity and specificity, the kinetics of nanomedicines is also an important factors that significantly influence their therapeutic performance[92]. Nanomedicines have complex interactions with biological systems, making accurate kinetic profiling essential for rational design and successful clinical translation. The ultimate goal of understanding binding kinetics is to design nanomedicines that act rapidly on target cells while remaining slow-acting or inactive toward non-target cells. Such fast-acting nanomedicines could also enhance therapeutic efficacy while minimizing off-target accumulation and associated side effects.

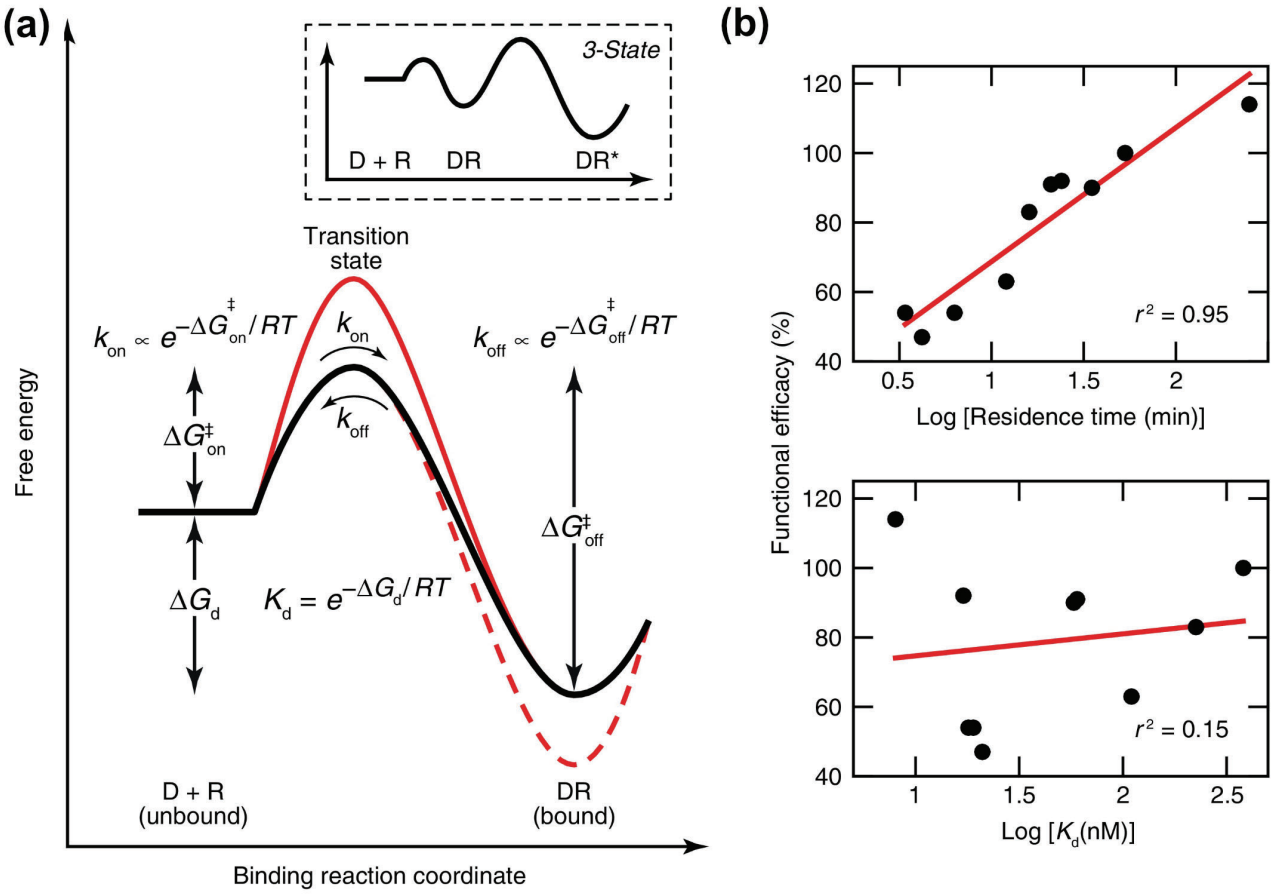

Figure 1.7 The affinity and binding rates of a drug are determined by the free energy landscape underlying the binding interaction. (a) A simple free energy profile (black) of a drug (D) binding to a receptor (R) to form a drug–receptor complex (DR). The free energy difference between the bound and unbound states, ΔG_d, determines the binding affinity. **(b)** Drug residence time, rather than affinity, predicts the functional efficacy of a series of agonists targeting the A_{2A} adenosine receptor. Residence time shows a strong correlation with functional efficacy, whereas binding affinity exhibits little correlation. (From **Ref. 94**)

There is growing recognition that the rates at which drugs associate with and dissociate from their targets directly impact both efficacy and safety[93,94]. However, the binding kinetics between the drugs and the targets are poorly understood. The equilibrium dissociation constant k_d, or its proxies such as the drug concentration producing half-maximal inhibition or effect (IC_{50} or EC_{50}), provides a basis for estimating appropriate dosing and is often used as a surrogate for *in vivo* efficacy[94]. The equilibrium dissociation constant reflects the drug–target affinity between a drug and its target, and it is relatively well understood. The drug–target affinity can often be achieved by evaluating the equilibrium constant. In contrast, binding kinetics are typically assessed through real-time measurements of the on-rate (k_{on}) and dissociation rates (k_{off}).

For drugs targeting a specific receptor and using a chemical approach, the binding kinetics can be expressed as:

$$D + R \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} DR \qquad (1.12)$$

where D represents the drug, R refers to the receptors, DR stands the drugs/receptor complex[94]. Many factors affect the binding kinetics, such as molecular size, conformation fluctuations, electrostatic interactions, and hydrophobic effects.

A drug-receptor reaction typically involves overcoming an energy barrier (ΔG_{on}) known as the activation energy, as shown in **Fig. 1.7(a)**. This energy input is required to reach the transition state, where reactant bonds are destabilized and new bonds begin to form. The on-rate is related to the energy barrier. Once this barrier is surpassed, the system proceeds to a lower-energy product state, releasing energy in the process. This release can occur as heat, light, or other forms of energy, depending on the nature of the reaction. For a reaction to proceed spontaneously, it must be thermodynamically favorable-that is, the change in Gibbs free energy (ΔG_d) must be negative. This ensures that the products are more stable than the reactants, making the overall process energetically downhill. Therefore, both the kinetics (energy barrier) and thermodynamics are essential in determining whether and how a reaction occurs. The drug's efficacy is more closely associated with the dissociation rate rather than the dissociation constant, as illustrated in **Fig. 1.7(b)**. For cells exhibiting a low dissociation rate, the drug is less likely to detach from the receptor, resulting in higher functional efficiency.

1.4.2 The uptake kinetics

In addition to binding kinetics, the rate at which nanoparticles are internalized by target cells is a critical factor influencing drug efficacy[92]. Many drugs, especially modern therapeutics such as RNA-, DNA-, and protein-based agents, target intracellular sites[95-97]. Without cellular internalization, these drugs cannot reach their intended site of action. A drug may bind efficiently to its receptor on the cell surface, but without effective internalization, its therapeutic effect can be limited. Thus, the internalization should be included as a pivotal character for designing drugs.

The receptor-mediated endocytosis of a cell has a preference for the size of the nanoparticle, significantly influencing the uptake efficiency. Numerous studies have demonstrated that nanoparticles with diameters ranging from 30 to 100 nm are most efficiently internalized via receptor-mediated endocytosis[81-83,98-100]. This size range is optimal for facilitating effective membrane wrapping and vesicle formation without imposing excessive mechanical resistance. In contrast, very small nanoparticles may fail to induce sufficient membrane curvature or receptor clustering necessary to initiate endocytosis. On the other hand, larger nanoparticles demand greater membrane deformation energy, which may surpass the energy supplied by receptor–ligand interactions, thereby resulting in reduced uptake efficiency or diversion to alternative pathways such as macropinocytosis or phagocytosis.

Nanoparticle shape plays a critical role in determining the rate and mechanism of cellular endocytosis[101-103]. Spherical nanoparticles are typically internalized efficiently via receptor-mediated endocytosis due to their uniform curvature, which facilitates consistent membrane wrapping. In contrast, rod-like or elongated nanoparticles often exhibit slower initial uptake but can achieve higher overall internalization, attributed to their increased surface area for receptor interactions. Disc-shaped nanoparticles generally show lower internalization efficiency, as their flat geometry makes membrane wrapping energetically less favorable. Cubic and other faceted shapes, with sharp edges and corners, can hinder membrane wrapping and introduce mechanical resistance. Consequently, their cellular uptake tends to be less efficient or may proceed through non-classical endocytic pathways.

It's also found that the adhesion (association constant or avidity) between the nanoparticle and cell controls endocytosis[104-106]. Raluca Groza *et. al* explored the role of membrane adhesion energy in endocytosis by employing receptors with affinities spanning seven orders of magnitude. Their study demonstrated that once a critical threshold in adhesion energy is surpassed, enabling membrane deformation, endocytosis proceeds reliably, regardless of

further increases in receptor-ligand affinity[106]. Our group demonstrated that in brain endothelial cells, endocytosis predominantly occurs at high adhesion energy levels, whereas transcytosis is facilitated only within a specific intermediate adhesion energy window[104]. Therefore, in the design of nanomedicines, it is essential to predict adhesion energy accurately, highlighting the critical importance of our thermodynamic model in guiding effective formulation strategies.

Several interrelated parameters, such as nanoparticle size and shape, simultaneously influence nanoparticle avidity (as described by the PAT), binding affinity, and internalization rate. Therefore, the predesign process must take a comprehensive approach to ensure optimal binding efficiency. Uptake kinetics models provide a valuable framework for quantifying the rate of nanoparticle internalization, offering a rational means to optimize design parameters. Although much remains to be understood about how each parameter quantitatively affects uptake kinetics, we emphasize that integrating thermodynamic and kinetic models is essential. This combined approach will significantly advance the development of precision nanomedicines.

1.5 Nanoparticle pharmacokinetics

1.5.1 The journey of a nanoparticle in the body

We have already explored the interactions of nanoparticles at the molecular and cellular levels. The next crucial step is to establish a more comprehensive and realistic understanding of their behavior within the whole body to validate their therapeutic potential. Preclinical pharmacokinetics (PK) testing is essential before advancing drugs to clinical trials. The rational design of precise nanomedicines relies on a comprehensive understanding of the absorption, distribution, metabolism, and excretion (ADME) processes that determine their biological fate. Unlike small-molecule drugs, nanomedicines exhibit distinct ADME characteristics due to their unique chemical and physical properties[107-109]. In this section, we present the journey of a nanoparticle through the body to illustrate these ADME principles.

Absorption refers to the process by which nanoparticles transition from the external site of administration into the internal biological environment. The mechanisms and efficiency of absorption vary significantly depending on the route of administration. Common delivery routes include oral, intravenous (IV), nasal, pulmonary, transdermal, and subcutaneous pathways. Each route presents distinct biological barriers and physiological conditions that influence both the efficiency and kinetics of nanoparticle uptake. The choice of administration route plays a pivotal role in determining the extent and localization of nanoparticle absorption. For instance, a study by Marco E. Zamora *et al.* demonstrated that changes in the administration route led to significant differences in both absorption profiles and tissue-specific targeting, notably between the lungs and brain[110].

After absorption, nanoparticles enter systemic circulation and are transported through the bloodstream, enabling them to reach and interact with various organs and tissues throughout the body. Numerous studies have demonstrated that nanoparticles can circulate systemically and access nearly all major organs via blood flow. Upon reaching these organs, nanoparticles may localize within the extracellular matrix, adhere to immune cells such as macrophages, bind to the vascular endothelium, or be actively internalized by resident cells through endocytic pathways[111]. These interactions are influenced by several key factors, including the nanoparticles' size, shape, surface chemistry, and the characteristics of the local microenvironment[112]. By precisely controlling these parameters, researchers can strategically

regulate nanoparticle biodistribution and achieve tissue-specific diffusion, which is essential for the development of effective precision nanomedicine.

Metabolism of nanoparticles refers to the biological processes that modify their physicochemical properties once inside the body. These metabolic transformations largely depend on the composition, structure, and surface characteristics of the nanoparticles[113,114]. Inorganic nanoparticles, such as gold or silica-based particles, tend to be chemically stable and are often resistant to metabolic breakdown, resulting in prolonged retention in tissues[115,116]. In contrast, polymeric nanoparticles are typically engineered to be biodegradable or biocompatible[117,118]. For example, PLGA nanoparticles are engineered to degrade into non-toxic byproducts, such as lactic acid and glycolic acid, which are easily metabolized by the body. Polystyrene (PS) nanoparticles, on the other hand, are generally considered biocompatible but not inherently biodegradable, and are often used in research as model systems rather than for therapeutic delivery[119].

Excretion refers to the elimination of nanoparticles from the body, primarily through feces and urine[120,121]. The liver and kidneys are the key organs responsible for this process. Nanoparticles smaller than approximately 5.5 nanometers can undergo efficient and rapid renal clearance via glomerular filtration, making the kidneys the primary route of excretion for these small particles[122]. In contrast, larger nanoparticles are cleared more slowly and are predominantly eliminated through the hepatobiliary pathway[123]. In this case, the liver processes nanoparticles by transporting them across hepatocytes and excreting them into bile, which eventually reaches the intestines[124]. Although hepatocytes exhibit some phagocytic capacity, they are generally less efficient at capturing nanoparticles compared to Kupffer cells, the specialized macrophages of the liver. The Kupffer cells belong to the RES, and after nanoparticles are internalized and degraded by the RES, their components are excreted through the renal system (urine) or the hepatobiliary system (feces), depending on solubility and breakdown products[125].

1.5.2 Physiologically based pharmacokinetic (PBPK) modeling

In addition to analysing the ADME data from *in vivo* experiments, a robust analytical framework is needed to validate these findings and guide the rational design of future nanoparticles. Physiologically based pharmacokinetic (PBPK) modeling offers a powerful solution. PBPK models simulate the distribution and behavior of substances in the body by dividing it into interconnected compartments, each representing specific organs or tissues, based on physiological and anatomical data[126]. These compartments are linked according to biological fluid flow, such as blood circulation. A typical PBPK model structure is illustrated in **Fig. 1.8**[127]. One of its major advantages is the ability to extrapolate results across species, enabling the translation of preclinical (e.g., animal) data to human contexts, which is essential for clinical development and dose optimization.

When constructing a PBPK model, it is essential to include key physiological compartments that represent major organs. Commonly included organs in a PBPK model are the liver, kidneys, spleen, lungs, brain, and heart[128]. Among these, the heart plays a critical role due to its central function in systemic circulation. Therefore, both in modeling and in experimental validation, special attention must be paid to the heart to ensure that no harmful or clinically significant effects are overlooked. After establishing the structural framework of the PBPK model, a corresponding set of mathematical equations must be developed to describe the kinetics of nanoparticle transport and distribution across compartments. Some parameters within these equations may be unknown or specific to a given biological system, and thus require estimation

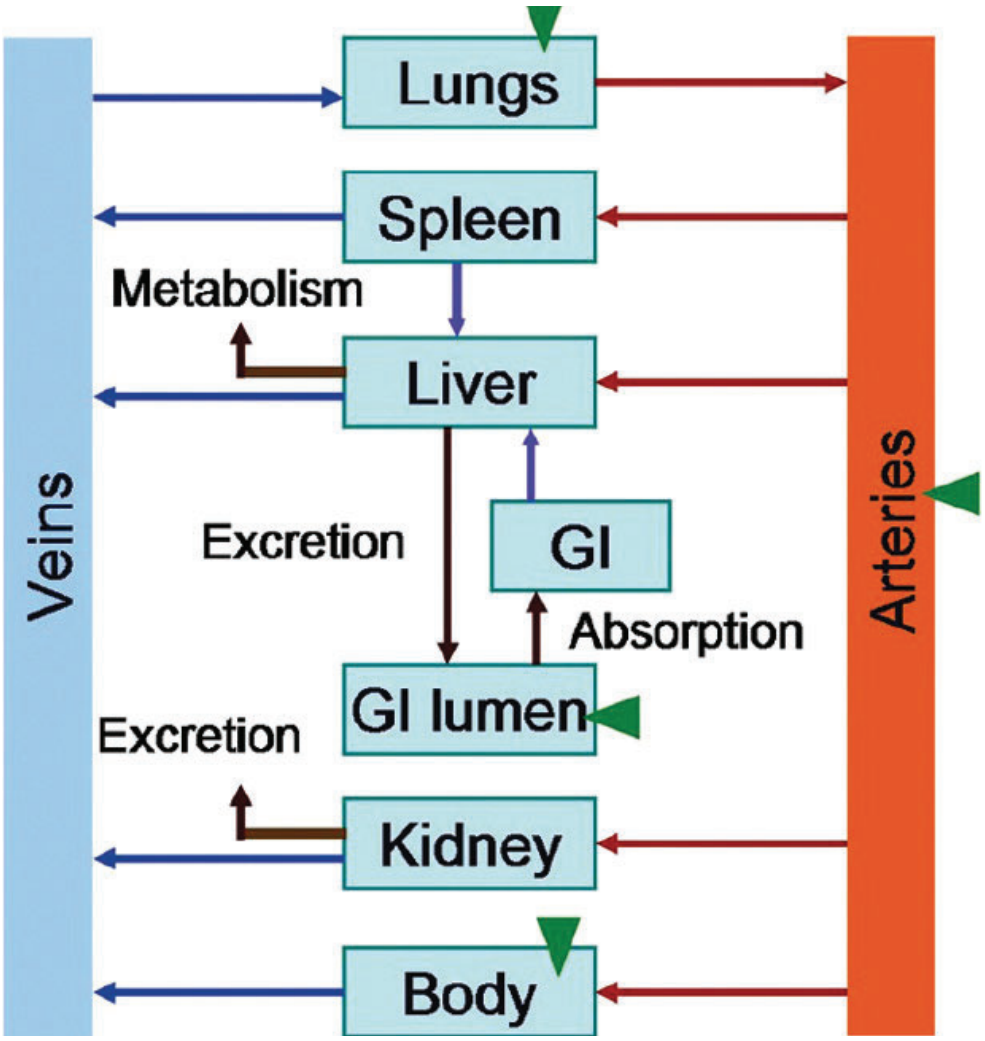

Figure 1.8. A typical blood flow-limited PBPK model structure. (From Ref. 127)

or calibration using experimental data. Accurate parameterization is essential to ensure the model's predictive capability and biological relevance.

The key parameters related to fluid transport in a blood-circulation-based PBPK model are the influx rate (k_{in}) and efflux rate (k_{out}). As illustrated in **Fig. 1.8**, the influx rate represents the rate at which nanoparticles enter an organ, while the efflux rate describes the transport of nanoparticles from the interstitial space back into the bloodstream. Blood serves as the primary medium for nanoparticle distribution throughout the body. The concentration of nanoparticles in the blood can be expressed as:

$$V_P \frac{\partial \rho_P}{\partial t} = \sum (-V_P k_{P-i} \rho_P + V_i k_{i-P} \rho_i) \qquad (1.13)$$

where V_P and ρ_P refer to the blood volume and nanoparticle concentration in the blood, respectively. k_{P-i} stands for the influx rate from blood to the ith organ and k_{i-P} represents the efflux rate from the ith organ to blood. As for the nanoparticle concentration in a general organ, it can be expressed as:

$$V_i \frac{\partial \rho_i}{\partial t} = V_P k_{P-i} \rho_P - V_i k_{i-P} \rho_i \qquad (1.14)$$

For nanoparticles in the excretion organ, it can be expressed as:

$$V_i \frac{\partial \rho_i}{\partial t} = V_P k_{P-i} \rho_P - V_i (k_{i-P} + k_{ex}) \rho_i \qquad (1.15)$$

with k_{ex} being the excretion rate. The nanoparticle will be sequestered in the organ, thereby serving as a reservoir for sustained nanoparticle circulation in the bloodstream.
Nanoparticle interactions with cells depend on their physicochemical properties. For instance, the sediment rate is relevant for metallic nanoparticles due to their relatively high mass density compared to plasma, resulting in gravitational forces that exceed buoyant forces[126]. In contrast, for polymeric nanoparticles, the difference between gravitational and buoyant forces is minimal and typically negligible. It is also commonly acknowledged that macrophages play a key role in the uptake of nanoparticles[120]. The amount of material internalized by macrophages can be expressed as:

$$\frac{\partial \rho_\phi}{\partial t} = k_{mac}\rho_i - k_{cl}\rho_\phi \qquad (1.16)$$

with ρ_ϕ being the nanoparticle concentration in the macrophages, k_{mac} being the macrophage rate, and k_{cl} being the nanoparticle clearance rate.
From the perspective of designing precision nanomedicine, additional factors and optimization strategies must be considered to more accurately mimic the in vivo behavior of nanoparticles. The physicochemical properties of nanoparticles, which govern their specific interactions with biological systems, are critical design parameters. For instance, transcytosis through endothelial cells has been identified as a key transport mechanism and has been extensively reported in the literature[129]. Incorporating such biological processes into nanoparticle design and modeling will enhance the accuracy and relevance of predictive models, ultimately supporting more effective therapeutic outcomes.

1.6 Aim and objective

We aim to harness the multivalency effect to enhance the therapeutic performance of nanomedicines. A thorough understanding of the physical mechanisms underlying multivalent interactions is essential for the rational design of precision nanomedicine. These mechanisms can vary depending on the specific binding scenario. To account for this variability, we develop PAT based on thermodynamic models tailored to each scenario. The PAT theory incorporates both cell phenotype and nanoparticle properties to predict the binding fraction based on these parameters. We could drive guidelines and pre-design the nanoparticle properties to enhance the binding efficiency.
Since the precision nanomedicine discussed in this thesis relies on multivalent interactions and the PAT overlooks the time, it is also crucial to characterize the multivalent kinetics of these interactions with cells. Therefore, we integrate the thermodynamic model with a kinetic framework to quantify these relationships and elucidate the underlying physical mechanisms. The nanoparticle properties influence the binding kinetics, but little is known about how these properties influence the kinetics. In addition to binding kinetics, uptake kinetics and intracellular trafficking are also critical for evaluating the therapeutic efficacy of nanomedicines. A mathematical model to integrate all these kinetics and quantify these processes would help us to pave the way to design more effective nanomedicines.
Furthermore, we extend our analysis from the cellular level to the whole-body level by constructing a PBPK model to validate the predictions against in vivo experimental data. Traditional PBPK models treat organs and cells as compartments and use ordinary differential equations (ODEs) to simulate the ADME processes of nanoparticles. However, these models often lack a detailed representation of nanoparticle-cell interactions. In this work, we aim to integrate PAT and uptake kinetics to further elucidate nanoparticle behavior within the body. This combined approach enables the development of rational strategies to improve the design

and efficiency of precision nanomedicines. To facilitate broader application, we have developed a Python package that allows users to design their PATs and fit correlated multivalent kinetics and pharmacokinetics. This package predicts the binding fraction based on input parameters such as nanoparticle properties and cell type.

In summary, the aims of this thesis are:

1. To advance the theoretical framework of PAT for multivalent nanoparticle interactions.
2. To quantify multivalent binding kinetics through integration with PAT theory.
3. To develop a PBPK model for validating experimental results and enabling the extrapolation of treatment strategies of multivalent systems.
4. To build a Python package to enable users to design their own model.

Chapter 2: Designing nanoparticles based on PAT

2.1 Phosphorylcholine-driven cellular phenotypic targeting

2.1.1 Intrinsic properties of the PMPC-PDPA nanoparticles

PMPC-PDPA is an amphiphilic diblock copolymer composed of a hydrophilic PMPC segment and a pH-sensitive hydrophobic PDPA segment[130]. The PMPC block mimics the phosphorylcholine headgroups found in natural cell membranes, providing excellent biocompatibility, anti-fouling properties, and stealth behavior in biological environments[131,132]. In contrast, the PDPA block is responsive to pH changes due to its tertiary amine groups, which become protonated under mildly acidic conditions (pKa ~6.2), transforming the block from hydrophobic to hydrophilic[133-135]. This amphiphilicity enables the copolymer to self-assemble into well-defined polymersomes in aqueous environments, with a hydrophilic PMPC corona and a hydrophobic PDPA membrane. The chemical structure of the PMPC chain and PMPC polymersome is shown in **Fig. 2.1(a)**.

This targeting is based on the interaction between the phosphorylcholine (PC) groups present in chains of the PMPC Psome (POs) and the cell receptors: SRB1, member 3 (CD36), and the transmembrane protein CD81, which belongs to the tetraspanin family, as shown in our group's previous work[73]. Tetraspanins are known as described as receptors for cholesterol, which play diverse roles in the immune system and cancer. SRB1 and CD36 can induce the internalization of the PMPC. The all-atom modelling of the receptors is shown in **Fig. 2.1(b)**.

PMPC nanoparticles exhibit a high colloidal stability and tunable size and topology, which can be precisely controlled by adjusting the block ratio and assembly conditions[130]. Among these factors, the degree of polymerization of a single PC chain, the size, and shape of the polymersomes are key elements influencing their targeting performance. In **Fig. 2.1(c)-(e)**, we show the transmission electron microscope (TEM) images of different nanoparticle topologies. The disks exhibit small, spherical, flattened, and non-vesicular structures with a hydrophobic core. The radius is about 10-15 nm (**Fig. 2.1(c)**). The spherical is large, spherical, and vesicular with an aqueous core. The radius of spherical POs is about 40-60 nm (**Fig. 2.1(d)**). Tubes present elongated tubular and vesicular structures with a length of ~150–300 nm, each capped with a hemisphere of diameter ~30–50 nm (**Fig. 2.1(e)**).

To fabricate spherical PMPC nanoparticles, we gradually increased the pH of an acidic polymer solution beyond the pKa of the PDPA block[130]. This induced hydrophobicity in the PDPA segment, triggering the self-assembly of polymer unimers into membrane-enclosed vesicular structures, effectively shielding the hydrophobic domains from the aqueous environment. A similar bottom-up strategy was used to generate disk-like nanoparticles; however, in this case, the pH increase was performed rapidly at 60 °C, promoting the formation of kinetically trapped, non-equilibrium structures with a hydrophobic core. In contrast, tubular nanoparticles were produced using a top-down approach. A preformed film of the copolymer was hydrated over the time-period of two weeks, allowing for the gradual formation of tubular structures. These were then isolated from residual assemblies (e.g., spherical polymersomes) via density gradient purification, as previously described.

Remarkably, viruses exhibit a wide range of morphologies, such as the tubular shape of the Ebola virus and the spherical structure of the human immunodeficiency virus (HIV)[136,137]. These distinct shapes play a crucial role in how viruses interact with host cells, particularly influencing the rate and pathway of endocytosis, which will be discussed in the next section. Understanding

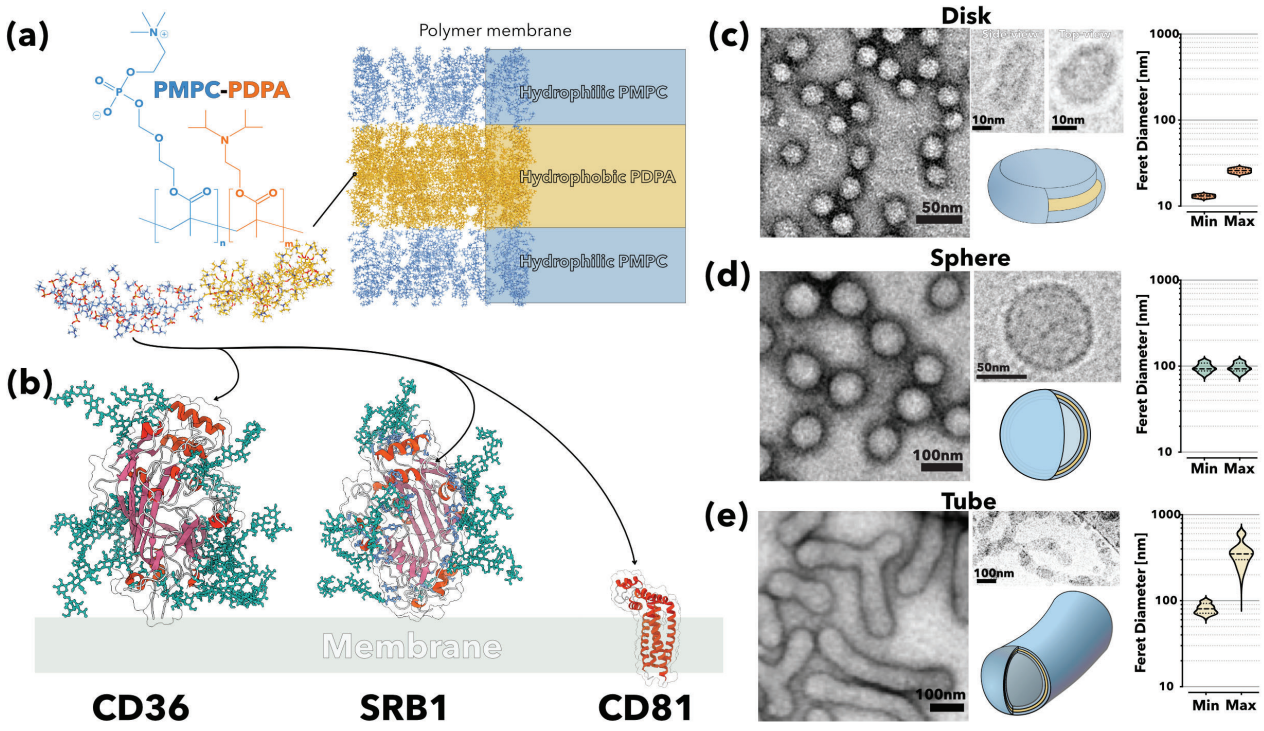

Figure 2.1. Tailoring the morphology of soft nanoparticles. (a). Schematic illustration of nanoparticle/cell surface binding. The PMPC ligand primarily targets three receptors: CD36, SRB1, and CD81. **(b)-(d).** Transmission electron microscopy (TEM) images of **(b)** spherical disks of a hydrophobic core, **(c)** spherical vesicles *(i.e.*, Psomes), and **(d)** tubular vesicular structures formed by the spontaneous self-assembly of $PMPC_{25}$-$PDPA_{70}$. Differences in the morphology were obtained by engineering both the self-assembly procedure and purification steps. **(e).** The violin plot of polydispersity index (PDI), measured and calculated from TEM images.

this relationship highlights the importance of nanoparticle topology in disease modeling and therapeutic design.

2.1.2 Physical mechanism of phosphorylcholine-driven cellular phenotypic targeting

The application of the PAT to the binding scenario between PMPC nanoparticles and their targets can aid in the optimal design of nanoparticle properties. Since PMPC nanoparticles are composed solely of PMPC chains, the number of binding ligands on their surface is directly related to the size of the nanoparticles. Additionally, each PMPC chain contains multiple binding sites, and the number of these sites is associated with the degree of polymerization of the PC units within the PMPC chains.

Considering that a single PMPC ligand binding with the receptor is a critical event in the interaction process. The affinity of this binding can significantly influence the overall efficacy and targeting capabilities of PMPC nanoparticles. Our group has modeled the PMPC binding to SRB1, CD36, and CD81[73]. The number of binding sites, λ_ζ, could be approximated as $\lambda_\zeta \simeq 1 + b_\zeta(N_{PC} - 1)$, where b_ζ is an arbitrary constant and N_{PC} represents the degree of polymerization of the PC unit along a single PMPC chain. The binding sites are within the reach of each PC unit. We can write the partition function between a single PMPC chain and the ζ receptor as:

$$q_{PC/\zeta}^{free} = \sum_{i=1}^{\lambda_\zeta} \left[\binom{\lambda_\zeta}{i} e^{-j\beta(\epsilon_{B/\zeta} + U_{P/\zeta})} \right] \approx \left[1 + e^{-\beta(\epsilon_{B/\zeta} + U_{P/\zeta})} \right]^{\lambda_\zeta} - 1 \qquad (2.1)$$

where $\epsilon_{B/\zeta}$ is the binding energy between the PC unit and the relative receptor, and $U_{P/\zeta}$ is the repulsive penalty from the chain, which can be approximated as $\beta U_{P/\zeta} = u_\zeta(N_{PC}-1)$ with u_ζ being the steric repulsion between the non-bound PC units and the receptor. Thus, we can write the binding energy as $\beta E^{free}_{PC/\zeta} = -ln(q_{PC/\zeta})$ between the single PMPC chain and the receptor.
When the receptors bind with the PMPC chain, they must insert into the PMPC brush, thereby experiencing a steric potential, $U_{S/\zeta}(z)$. As the PMPC-PDPA chain self-assembles, the PMPC chains packed to the nanoparticle surface form an archetypal Alexander De Gennes polymer brush[138]. The height of the PMPC chain can be approximated as $H \simeq a_{PC}N_{PC}$, with a_{PC} being the PMPC monomer length. We can write the insertion steric as:

$$\beta U_{S/\zeta} = \frac{V_\zeta\left(1-\left(\frac{N_{PC}-z+1}{N_{PC}}\right)^2\right)^{9/4}}{\left[\sigma_0\left(1+\frac{a_{PC}(N_{PC}-z)}{R}\right)^{\gamma-1}\right]^{3/2}} \qquad (2.2)$$

where $z \in [1,\cdots,N_{PC}]$ is an integer representing the depth of the insertion. σ_0 and γ represents the same nanoparticle parameters shown before[80]. For $\frac{a_{PC}N_{PC}}{R} \leq (\sqrt{3}-1)$, $\gamma = (\frac{a_{PC}N_{PC}}{R}+1)^2$ unless $\gamma = 3$. Thus, we can derive the binding energy as:

$$E_{PC/\zeta} = \sum_{z=1}^{N_{PC}} [E^{free}_{PC/\zeta}(z) + U_{S/\zeta(z)}] \qquad (2.3)$$

To account for the case when $U_{S/\zeta}(z)$ overcoming the attractive force, the binding energy $\epsilon_{PC/\zeta}$ per PMPC single chain to the receptor is defined as:

$$\epsilon_{PC/\zeta} = \begin{cases} E_{B/\zeta} & for \quad E_{PC/\zeta} \geq E_B \\ E_{PC/\zeta} & otherwise \end{cases} \qquad (2.4)$$

We can write the partition function between the POs and the receptor ζ as:

$$q_{PO/\zeta} = \sum_{\lambda=1}^{min(N_{PMPC},N_\zeta)} \binom{N_{PMPC}}{\lambda}\binom{N_\zeta}{\lambda}\lambda! e^{-\lambda\beta\epsilon_{PC/\zeta}} \qquad (2.5)$$

where $N_{PMPC} = \frac{4\pi R^2}{\sigma_0}$ is the number of the PMPC chain on the nanoparticle surface, and N_ζ is the number of ζ receptors expressed on the cell surface. The nanoparticles contain thousands of the PMPC chain, thus , $N_{PMPC} \gg N_\zeta$, we can simply the partition function as:

$$q_{PO/\zeta} = \left(1+N_{PMPC}e^{-\beta\epsilon_{PC/\zeta}}\right)^{N_\zeta} - 1 \qquad (2.6)$$

The cells have the glycocalyx protecting them from nonspecific interaction. The glycocalyx chains will repel the nanoparticles when they approach the cell surface, as shown in the equation:

$$\beta U_G = \frac{4}{3}\pi R^3 \sigma_{GAG}^{-3/2}\left(1-\frac{min(h_\zeta)}{h_G}\right)^{9/4} \qquad (2.7)$$

where h_ζ is the height of the receptor, and h_{GAG} is the height of the glycocalyx. For $h_G \gg h_\zeta$, we can do an approximation: $\beta U_G = \frac{4}{3}\pi R^3 \sigma_{GAG}^{-3/2}$. Thus, we can write the total partition function as:

$$\Xi = \prod_{\zeta=SRB1,CD36,CD81}\left[\left(1+N_{PMPC}e^{-\beta\epsilon_{PC/\zeta}}\right)^{N_\zeta}-1\right]e^{-\beta U_G} \qquad (2.8)$$

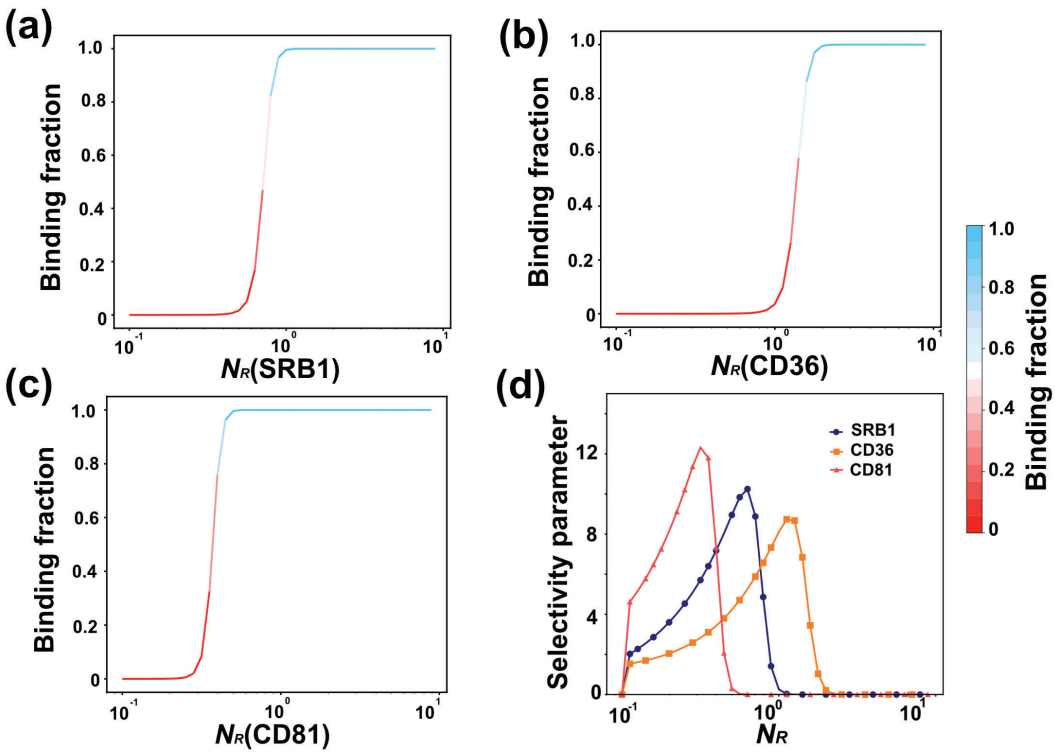

Figure 2.2. The binding fraction and the corresponding selectivity parameters. **(a)-(c)** The binding fraction as a function of receptor number for SRB1, CD36, and CD81, respectively, while keeping the other two receptors constant in each case. **(d)** The corresponding selectivity parameter.

Thus, we can derive the total binding energy of the single nanoparticle to a cell expressing SRB1, CD36, and CD81 as: $\beta E_{tot} = -ln(\Xi)$.

The activity of the nanoparticle is defined as the volume that the nanoparticle could explore, which is related to the nanoparticle topology. For the spherical nanoparticle, we can write the activity as: $a \simeq \pi/3\, N_A[P][3(R + a_{PC}N_{PC})^3 - R^3]$[67,68]. For the tubular nanoparticle, the activity is $a = \frac{L}{2}\left[(\theta_1(R+d)^2 - \sqrt{4Rd}) - (\theta_2 R^2 - \sqrt{2Rd - d^2})\right]$, where $\theta_1 = cos^{-1}\frac{R-d}{R+d}$ and $\theta_2 = cos^{-1}\frac{R-d}{R}$. The disk-like nanoparticle activity can be written as: $a = \pi R[Ra_{PC}N_{PC} + \pi[(h_D/2 + a_{PC}N_{PC})^2 - (h_D/2)^2]]$, where h_D is the height of the disk.

As in the Langmuir-Hill isotherm, the cell surface is considered an adsorption surface, and the fraction of the bound nanoparticle can be approximated as:

$$\theta = \left\langle \frac{a\Xi}{1 + a\Xi} \right\rangle \qquad (2.9)$$

2.1.3 Binding fraction prediction and target selectivity of PMPC nanoparticles

Using the PAT model, the binding fraction can be predicted based on cell type and receptor expression levels. For the PMPC nanoparticles targeting three types of receptors, we analyzed the binding fraction to each receptor type. As shown in **Fig. 2.2(a)-(c)**, the binding fraction is plotted as a function of the number of a single receptor type, while the levels of the other two receptors are constant. The inter-glycocalyx spacing is assumed to be 15 nm. The PMPC nanoparticles possess a radius of 30 nm and a polymerization of 25. The incubation concentration is 10^{-10} *M*. The results show that the binding fraction exhibits a sigmoid-like curve,

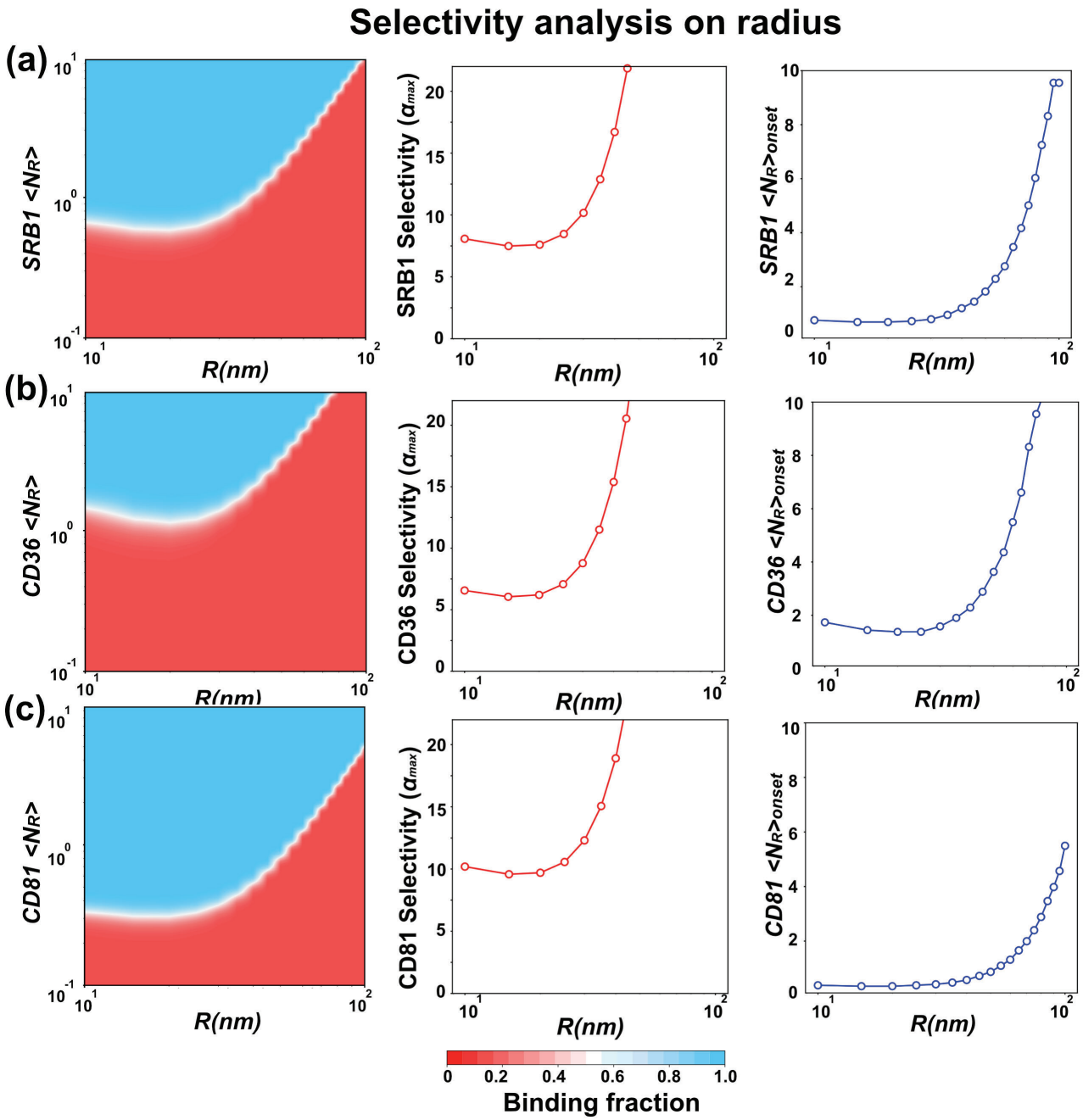

Figure 2.3. Selectivity analysis of R with various receptors. (a) Left: Binding fraction for SRB1 and *R*; Middle: Maximum selectivity parameter of SRB1 as a function of *R*; Right: Onset number of SRB1 relative to *R*. **(b)** Left: Binding fraction for CD36 and *R*; Middle: Maximum selectivity parameter of CD36 as a function of *R*; Right: Onset number of CD36 relative to *R*. **(c)** Left: Binding fraction for CD81 and *R*; Middle: Maximum selectivity parameter of CD81 as a function of *R*; Right: Onset number of CD81 relative to *R*.

reflecting a nonlinear increase as receptor number rises. Once the receptor level exceeds a certain threshold, the binding fraction increases sharply. Consequently, PMPC nanoparticles preferentially bind to cells that express the receptor above this threshold, while remaining unbound to cells with lower receptor expression. The selectivity parameter (α) is applied to quantify the PMPC nanoparticle selectivity as shown in **Fig. 2.2(d)**. Non-selective binding corresponds to $\alpha < 1$, whereas super-selective binding is characterized by $\alpha > 1$.

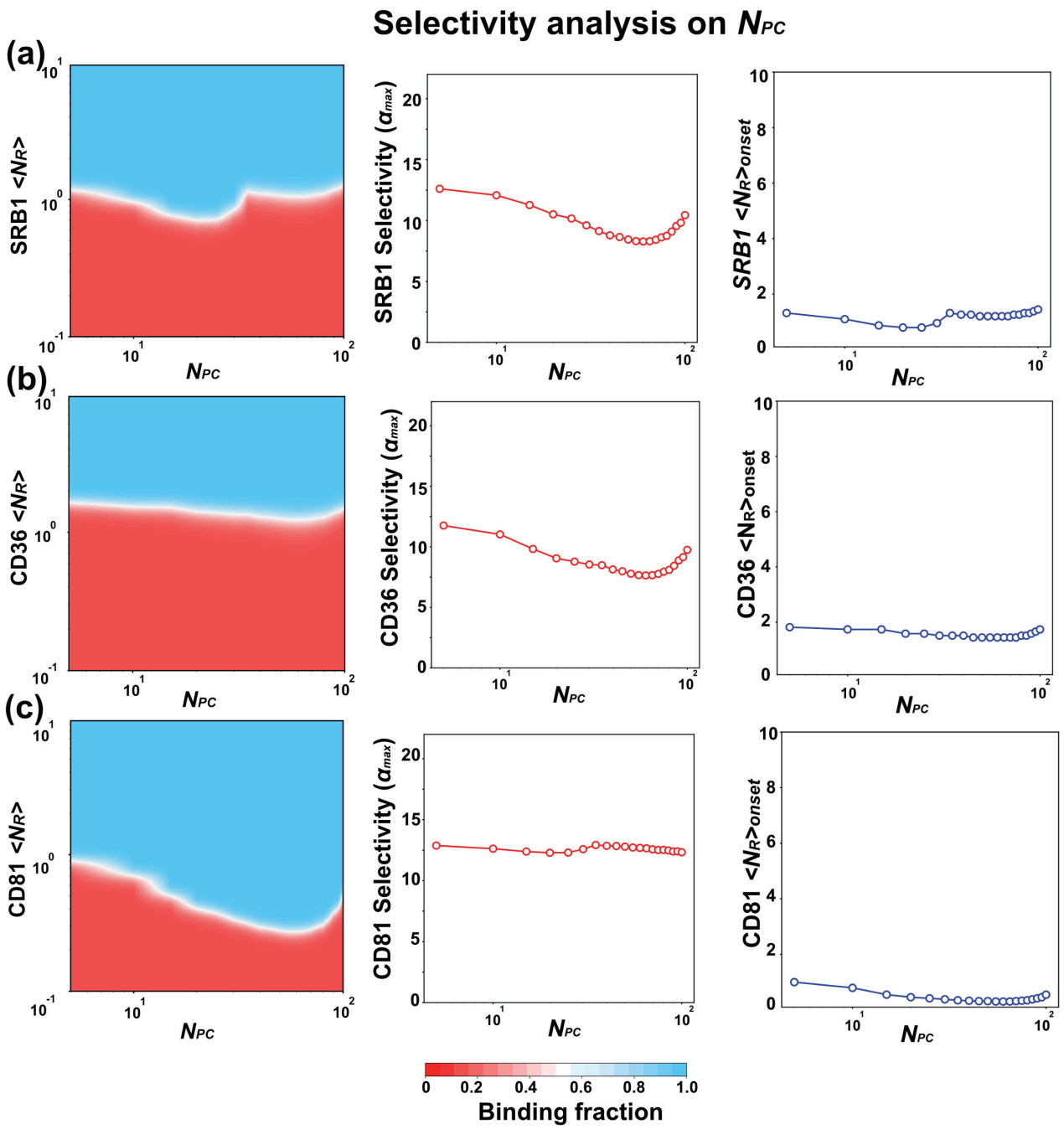

Figure 2.4 Selectivity analysis of N_{PC} with various receptors. (a) Left: Binding fraction for SRB1 and N_{PC}; Middle: Maximum selectivity parameter of SRB1 as a function of N_{PC}; Right: Onset number of SRB1 relative to N_{PC}. **(b)** Left: Binding fraction for CD36 and N_{PC}; Middle: Maximum selectivity parameter of CD36 as a function of N_{PC}; Right: Onset number of CD36 relative to N_{PC}. **(c)** Left: Binding fraction for CD81 and N_{PC}; Middle: Maximum selectivity parameter of CD81 as a function of N_{PC}; Right: Onset number of CD81 relative to N_{PC}.

Modifying key parameters of PMPC nanoparticles-such as radius, degree of polymerization, and shape-can significantly enhance their targeting efficiency. These variations provide a foundation for developing guiding principles in the optimal design of nanoparticle properties. To better understand how selectivity depends on nanoparticle parameters, we introduce two key descriptors: the maximum selectivity value and the receptor onset threshold. The receptor number at which α reaches its maximum value is defined as the onset density, N_{onset} (noting that this number represents the average per site that governs the Poisson distribution). The

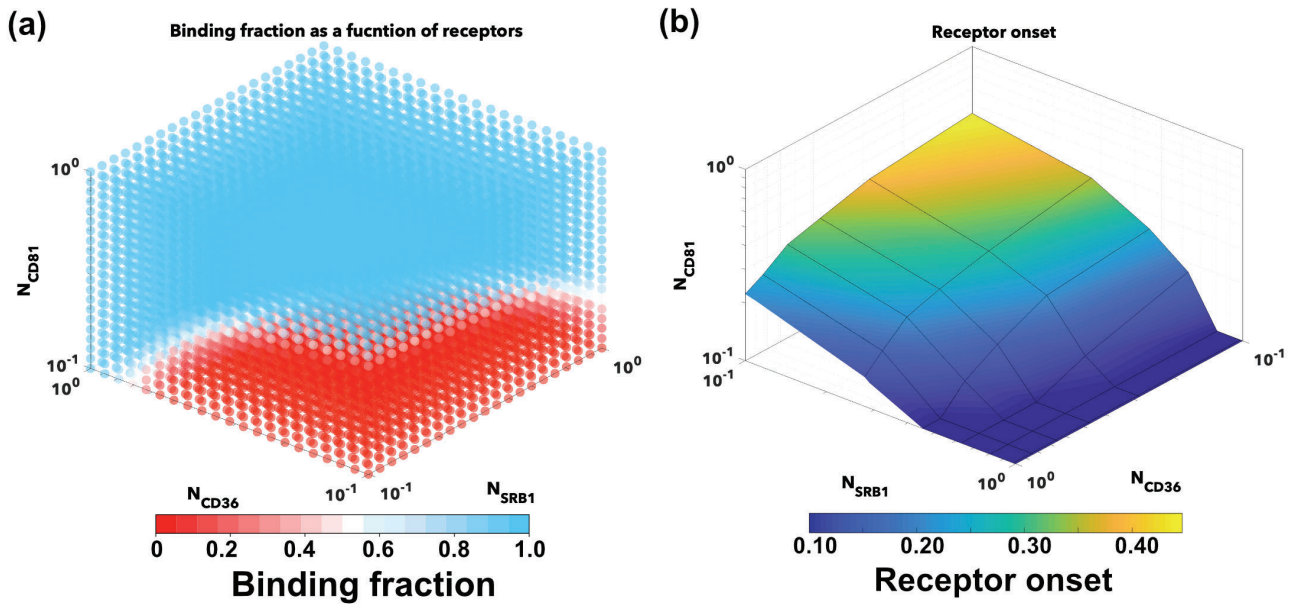

Figure 2.5. Multiplexing of PMPC Nanoparticles and Synergistic Effects. (a) A 4D plot showing the binding fraction as a function of all three targeted receptors. **(b)** The onset threshold for CD81 receptor expression as a function of the other two receptors. As the expression levels of the other two receptors increase, the onset threshold for CD81 decreases, indicating a synergistic effect that enhances binding.

corresponding maximum value, $\alpha_{onset} \equiv \alpha_{max}$, is defined as the super-selectivity parameter. **Fig. 2.3** and **Fig. 2.4** present the binding fraction as a function of a single receptor type and PMPC nanoparticle parameters (R and N_{PC}). The corresponding maximum selectivity and receptor onset threshold as functions of the nanoparticle parameters are shown in **Fig. 2.3** and **Fig. 2.4**, respectively.

Fig. 2.3 presents the analysis of the selectivity parameter as a function of nanoparticle radius for different receptor types, with all other parameters held constant—both the maximum selectivity parameter and the receptor onset value increase with increasing radius across all receptor types. The number of receptors needed for effective binding grows as the nanoparticle radius increases. As the radius of the PMPC nanoparticle increases, the number of PC chains grows, enhancing the combinatorial entropy and thus promoting binding. However, this also amplifies the repulsive steric hindrance from the glycocalyx, which opposes binding. To overcome this steric barrier, a higher receptor density is required to sufficiently enhance the ligand–receptor partition function and enable effective binding.

Fig.2.4 shows the analysis of the selectivity parameter as a function of nanoparticle polymerization for different receptor types, while keeping all other parameters constant. As the polymerization of the PMPC chain increases, the number of receptors required for effective binding first decreases and then increases. This behavior is linked to the variation in the total binding energy between a single PMPC chain and a receptor. As described in **Eqs. (2.3)** and **(2.4)**, the total binding energy depends on two competing factors: the binding free energy of the PC chain with receptors and the steric insertion potential from the PC chain. Increasing the polymerization enhances the total binding free energy due to more PC units, but also amplifies the steric insertion penalty. Initially, the gain in binding free energy outweighs the increase in steric repulsion, resulting in a net increase in total binding energy and thus a reduction in the receptor onset. However, beyond a certain chain length, the steric penalty grows more rapidly than the binding energy, causing a net decrease in total binding energy. This leads to a decrease in the receptor onset, reflecting the need for more receptors to achieve strong binding.

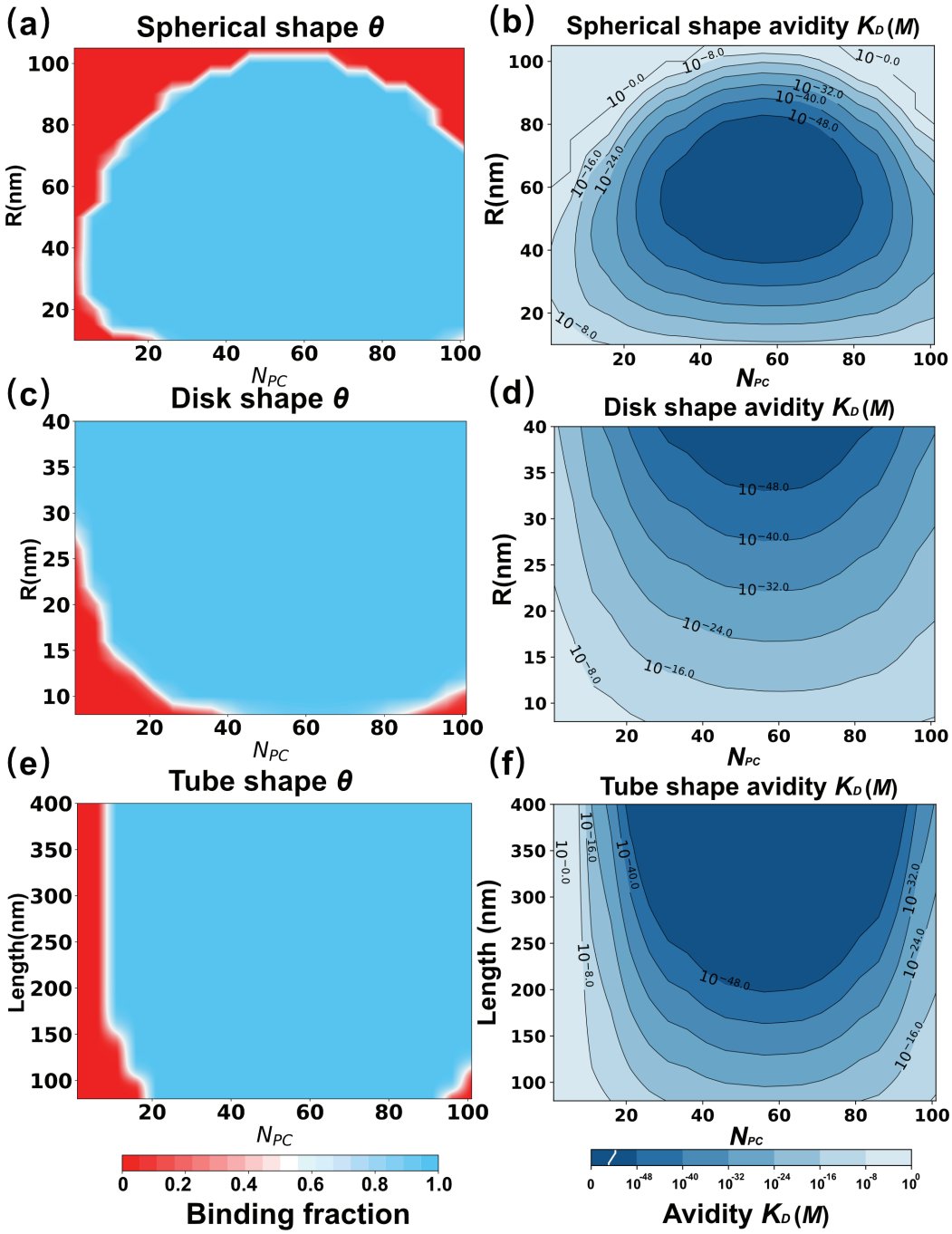

Figure 2.6 The binding fraction and avidity as a function of *R* and N_{PC} with different topologies. (a)-(b) Binding fraction and avidity prediction of spherical nanoparticles versus PMPC polymerization (N_{PC}) and radius (R). **(c)-(d)** Binding fraction and avidity prediction of disk-like nanoparticles versus PMPC polymerization (N_{PC}) and radius (R). **(e)-(f)** Binding fraction and avidity prediction of tube-like nanoparticles versus PMPC polymerization (N_{PC}) and radius (R).

As shown in the selectivity analysis, PMPC nanoparticles with a radius of 30 nm and a polymerization degree of 25 exhibit the lowest receptor onset, indicating that this configuration is optimal for achieving efficient binding. For PMPC nanoparticles targeting three distinct receptor types, it is essential to consider the combined contribution of all receptors to determine the overall binding fraction. **Fig. 2.5(a)** presents a 4D plot of the binding fraction as a function of the expression levels of all three target receptors. Once the total number of receptors exceeds a critical threshold, the nanoparticles begin to bind effectively, regardless of

the exact distribution among receptor types. **Fig. 2.5(b)** shows how the onset threshold of CD36 varies with the expression levels of the other two receptors. Notably, an increase in either SRB1 or CD81 significantly reduces the number of CD36 receptors needed to initiate binding, illustrating a clear synergistic effect among the three receptors. This synergistic interaction implies that targeting multiple receptors can amplify binding even when individual receptor levels are low, enhancing the specificity and robustness of nanoparticle-cell targeting in heterogeneous cellular environments. This principle could guide the design of multivalent systems that exploit receptor combinations for improved selectivity in complex biological settings.

Given a specific cell type, the binding fraction can be predicted based on the properties of the PMPC nanoparticle, specifically its radius and polymerization degree. **Fig. 2.6(a)** shows the binding fraction of spherical PMPC nanoparticles as a function of R and N_{PC} by incubating A549 cells at a concentration of 10^{-10} M. In most cases, the nanoparticles exhibit effective binding to A549 cells. However, the binding fraction is also influenced by the incubation concentration. To better understand this dependency, **Fig. 2.6(b)** presents the avidity as a function of R and N_{PC}. The maximum avidity occurs within a specific range of these parameters, forming a ring-like region. This pattern reflects the steric hindrance from the glycocalyx and from the PMPC chains, which together define the optimal design space for effective nanoparticle binding.

Topology is another crucial factor influencing effective nanoparticle binding. **Fig. 2.6(d)–(f)** present the binding fraction and avidity for PMPC nanoparticles with different shapes, specifically disk-like and tube-like topologies. For disk-like nanoparticles, the binding fraction increases with the radius, as shown in **Fig. 2.6(c)**. This enhancement arises from their high surface-to-volume ratio, which allows for a greater number of PC ligands to contribute to binding while experiencing relatively less steric hindrance from the glycocalyx. Tube-like PMPC nanoparticles also exhibit a higher surface-to-volume ratio compared to spherical particles and similarly benefit from an increase in size, enhancing their binding performance. In summary, the surface-to-volume ratio is a key parameter when selecting nanoparticle topology. The PAT can be employed to predict both binding fraction and avidity for nanoparticles of various shapes, guiding the rational design of PMPC nanoparticle topologies for optimal targeting performance.

2.2 Angiopep-2(AP2)-driven cellular phenotypic targeting

2.2.1 Intrinsic properties of the PEG-PCL nanoparticles

PEG-PCL (poly(ethylene glycol)-block-poly(ε-caprolactone)) nanoparticles are amphiphilic block copolymers that exhibit distinct intrinsic properties arising from their molecular architecture. The hydrophilic block is composed of PEG, chosen for its well-established protein anti-fouling capabilities, which help reduce nonspecific interactions in biological environments[139]. The hydrophobic PCL segment forms the inner core, which enables the encapsulation of hydrophobic drugs[140]. It degrades slowly in physiological environments through hydrolysis of its ester linkages, making it well-suited for long-term drug release systems. PCL also elicits minimal inflammatory responses and exhibits low toxicity, making it appropriate for both systemic and localized delivery. These nanoparticles typically self-assemble into spherical micelles or vesicle-like structures in aqueous environments, with their size depending on the PEG-to-PCL block ratio and the preparation conditions.

Angiopep-2 (AP2) is a synthetic peptide (sequence: TFFYGGSRGKRNNFKTEEY) derived from the Kunitz domain of aprotinin, designed to facilitate transcytosis across the BBB[141-143]. It binds to the LRP1, which is expressed on the surface of brain capillary endothelial cells and many

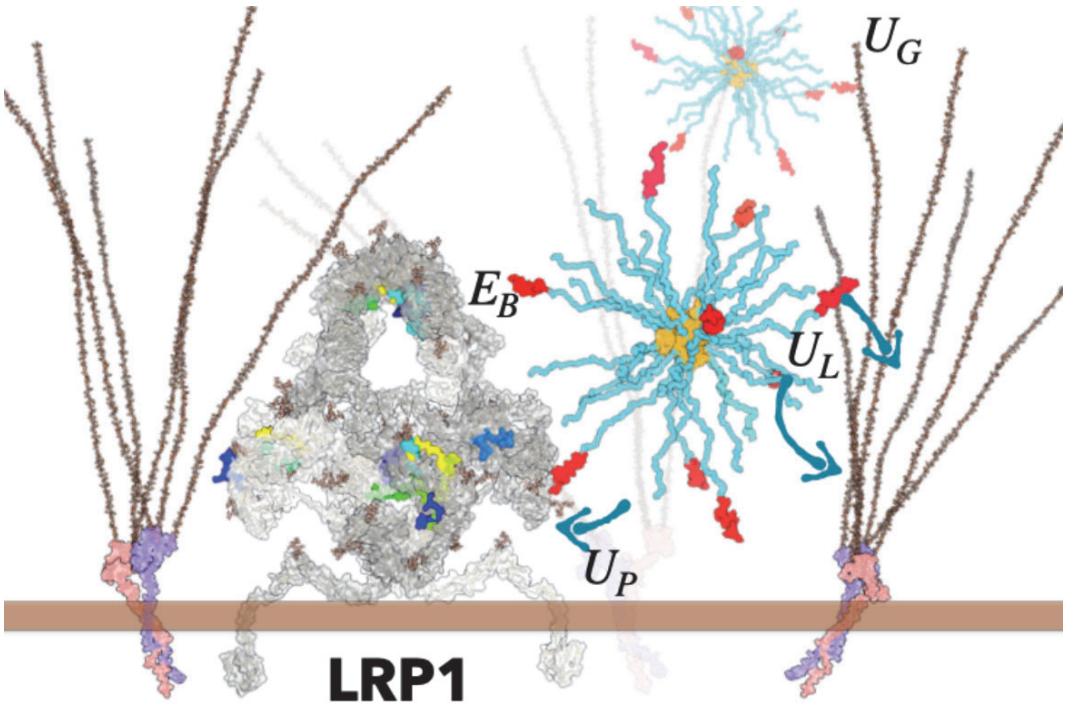

Figure 2.7 The binding schematic between the AP2-decorated PEG/PCL nanoparticle and LPR1.

tumor cells. The all-atom modeling of LRP1 receptor is shown in **Fig. 2.7**. We utilize AP2–decorated PEG-PCL nanoparticles to target and cross the BBB. The combination of PEG-PCL's favorable pharmacokinetic profile with AP2's targeting capability makes this platform highly promising for the delivery of chemotherapeutics or biologics to brain tissues, offering improved efficacy and reduced systemic toxicity.

2.2.2 Experimental validation of AP2-driven cellular phenotypic targeting

We fabricated various formulations of nanoparticles and evaluated their binding and transcytosis behaviors across three different cell types: the mouse brain endothelial cell line bEnd.3, the mouse astrocyte cell line C8-D1A, and the human glioblastoma cell line T98G. The experimental results are presented in **Fig. 2.8(a)-(c)**.

In the binding profiles, we observed a distinct on–off binding behavior as the number of surface ligands was varied. Specifically, when the interference parameter is 1 (as shown in **Fig.2.8(a)**), no significant binding occurred among all the cell types when the number of ligands was fewer than 2. Binding became apparent once the number of ligands reached 4. Interestingly, when the number of ligands exceeded 20, binding was no longer observed, a result that is counterintuitive based on traditional multivalency principles (increasing the number of ligands enhances the degeneracy of ligand–receptor pairings, thereby increasing the overall binding avidity and improving binding efficiency). As shown in **Fig. 2.8(d)**, we show the binding profile as a function of ligand number for glucose-decorated nanoparticles, targeting the same CNS cell lines. These nanoparticles primarily target glucose transporter proteins (GLUTs), a family of relatively small membrane proteins responsible for facilitating glucose transport across the cell membrane[144]. In contrast to the binding assay with AP2–decorated nanoparticles, increasing the number of glucose ligands does not significantly affect the binding efficiency. When the number of ligands exceeds a certain threshold, strong binding is consistently observed. Notably, GLUTs are significantly smaller than receptors such as LRP1. Thus, we hypothesize that there is an

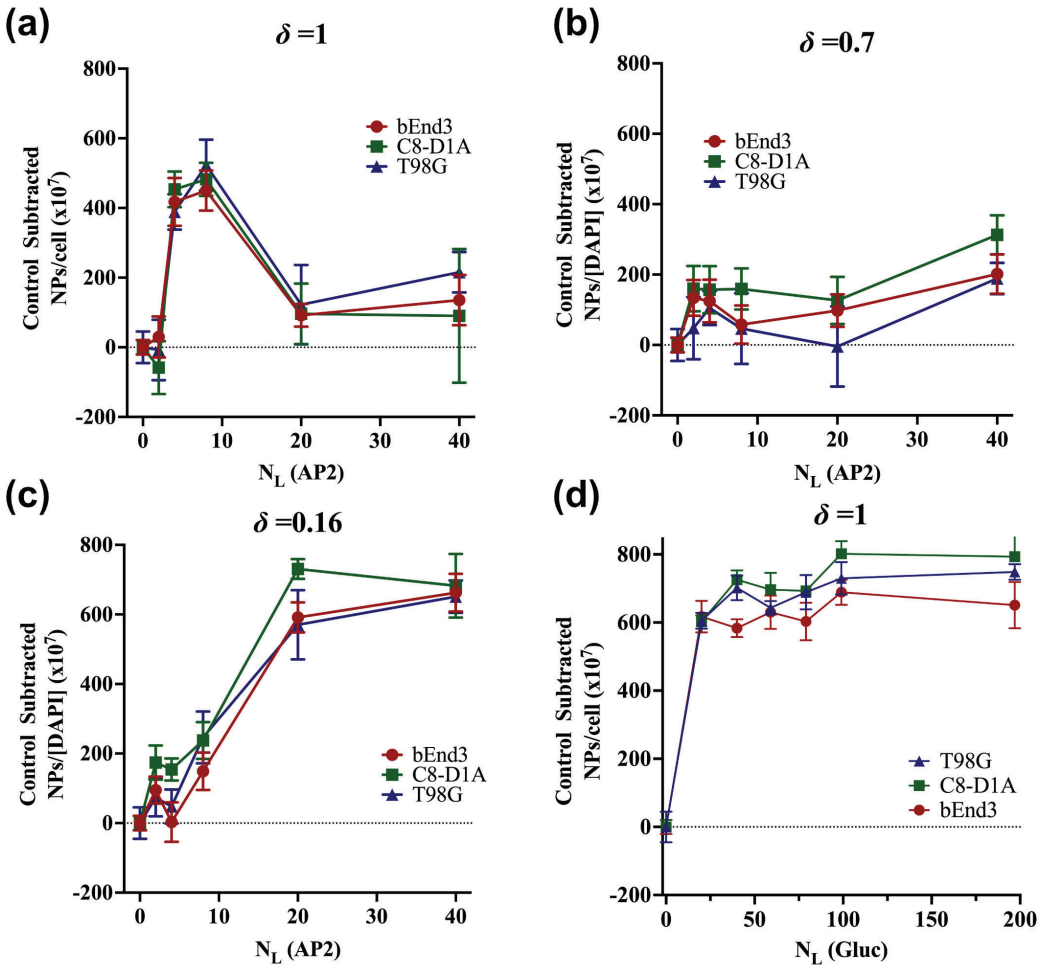

Figure 2.8 Binding profile of the AP2- and glucose-decorated nanoparticles targeting CNS cell lines. (a)-(c) Binding profiles as a function of ligand number for AP2-decorated nanoparticles targeting bEnd3, C8-D1A, and T98G cell lines, with interference parameters of 1, 0.7, and 0.16, respectively. **(d)** Binding profiles as a function of ligand number for glucose-decorated nanoparticles targeting bEnd3, C8-D1A, and T98G cell lines, assuming an interference parameter of 1.

additional steric effect from the unbinding ligands as nanoparticles are designed to target large receptors.

The steric potential from unbound ligands may arise from the size of the targeting receptor, altering the binding conformation. As illustrated in the schematic in **Fig. 2.7**, when a nanoparticle binds to large receptors on the cell surface(here, the case is the AP2-decorated PEG-PCL nanoparticle interacting with LRP1), it must adjust its position to bind with the available binding sites properly. While one side of the nanoparticle faces the target binding site, the opposite side may be exposed to non-binding components on the cell surface. Unbound ligands on this side can interact with these non-specific surface structures, introducing additional steric hindrance that impairs effective nanoparticle binding. The steric potential from unbinding ligands limits the enhancement of binding avidity by arbitrarily increasing the number of ligands, resulting in a confined range for adjusting ligand numbers.

The on–off binding behavior varies with different interference parameters. As shown in **Fig. 2.8(a)-(c)**, the onset of binding for nanoparticles occurs at different ligand numbers depending on the interference parameter. Nanoparticles with smaller interference parameters tend to exhibit binding onset at higher ligand numbers. Furthermore, binding did not turn off when the

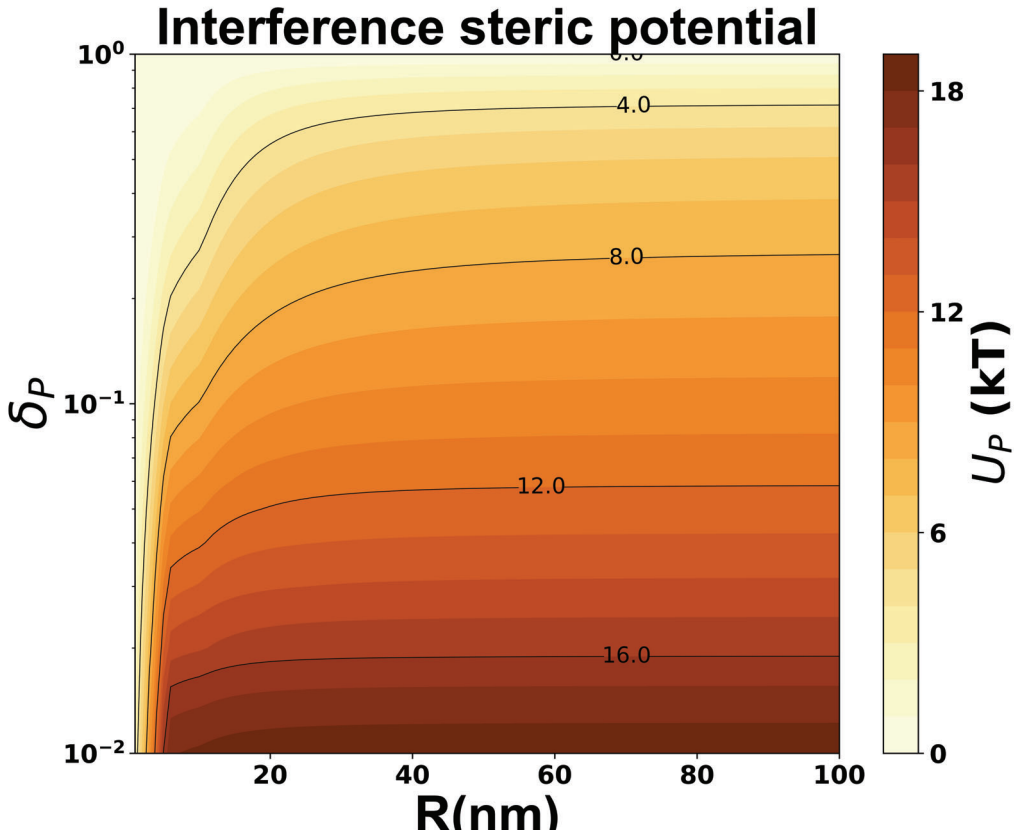

Figure 2.9 The interference steric potential between LPR1 and AP2-decorated PEG-PLA nanoparticles as a function of the interference parameter and particle radius.

ligand number exceeded 40. This observation suggests that reducing the interference parameter decreases the steric potential exerted by unbound ligands.

The steric potential from unbound ligands offers important insights for nanoparticle design. This steric hindrance is particularly prominent when nanoparticles are engineered to target large receptors. Here, we restate our hypothesis as follows: when the nanoparticle inserts into the glycocalyx, ligands on the receptor-facing side engage with binding sites on the large receptor; meanwhile, unbound ligands may interact with other components on the cell surface, generating additional steric hindrance and ultimately reducing binding efficiency. In such cases, unbound ligands are more likely to interact nonspecifically with surrounding cellular components, which impedes effective binding. However, by reducing the interference parameter, burying ligands deeper within the nanoparticle surface, these nonspecific interactions can be minimized, thereby decreasing steric penalties. This phenomenon highlights the need for a more rational adjustment of the number and presentation of functional ligands during nanoparticle design.

2.2.3 Physical mechanism of Angeopep2-driven cellular phenotypic targeting

We applied the PAT model to analyze the binding fraction to central nervous system cells across varying ligand densities. Compared to the PMPC-based system, the targeting receptor LRP1 is significantly larger than the receptors typically engaged by PMPC nanoparticles. In this case, the large receptor LRP1 provides multiple binding sites that can be accessed by a single nanoparticle, enabling the formation of a multivalent binding system. This allows ligands on one nanoparticle to simultaneously engage with several domains on a single LRP1 receptor,

significantly enhancing the overall binding strength and stability. Such multivalent interactions not only improve binding avidity but also reduce dissociation rates, making them especially beneficial for achieving strong and sustained nanoparticle attachment to the cell surface.

In this binding scenario, the compressive steric penalty becomes a critical factor, as the bulky LRP1 receptor is repelled by the protective polymer brush, limiting effective interactions with AP2 ligands. Furthermore, we identified an additional steric penalty arising from unbound ligands, which further decreases the likelihood of productive binding. As discussed in the previous section, increasing ligand density led to a reduction in binding efficiency. Notably, these steric penalties were significantly reduced when the ligands were embedded deeper within the polymer brush layer. Based on these observations, we optimized the PAT model to better account for nanoparticle binding behavior in the presence of large target receptors like LRP1.

In the case of targeting large receptors, we need to consider both the osmotic component and the compression component. The interference steric potential would be:

$$\beta U_P = \begin{cases} \frac{\pi h_L^3}{6\sigma_P^{3/2}}\left(1-\delta_P^2\right)^{\frac{9}{4}} & \textbf{if } h_L < \frac{1}{2}\sqrt{\frac{\sigma_P}{\pi}} \\ \frac{h_L N_P^2}{\delta_P^{1/4}\sigma_P}\left(\frac{a_0^2}{\sigma_P}\right)^{\frac{7}{6}}(1-\delta_P^3) & \textbf{if } \frac{1}{2}\sqrt{\frac{\sigma_P}{\pi}} \le h_L < R \\ \frac{R N_P^2}{\delta_P^{1/4}\sigma_P}\left(\frac{a_0^2}{\sigma_P}\right)^{\frac{7}{6}}(1-\delta_P^3) & \textbf{if } h_L \ge R \end{cases} \qquad (2.10)$$

where h_L is the ligand length[80]. In this binding scenario, the energetic penalty arises when the large LRP1 receptor compresses or deforms the polymer brush on the nanoparticle surface to access and bind the AP2 ligand. The size of the ligand and the interference parameter play crucial roles in determining the interference steric potential. The larger the ligands, the greater the steric penalty when they are compressed or their volume is depleted. The variation of steric potential as a function of the interference parameter and nanoparticle radius is illustrated in **Fig. 2.9**. As the interference parameter decreases, the steric interference potential increases. When the nanoparticle radius is larger than 20 nm, its influence on the steric potential is minimal. However, for radii smaller than 20 nm, the steric potential decreases with decreasing radius.

We hypothesize that this loss of binding at high ligand number is due to an additional steric potential arising from the unbound ligands, denoted as U_L, which imposes a physical barrier that hinders receptor access and thus suppresses binding. Thus, we can write the partition function as:

$$Q = \sum_{\lambda=1}^{min(N_L,N_B)} \binom{N_L}{\lambda}\binom{N_B}{\lambda}\lambda! e^{-\beta(\epsilon_B+U_P+U_L)} \qquad (2.11)$$

where N_B refers to the number of binding sites accessible to the nanoparticle, which depends on the nanoparticle's size. As discussed in the previous section, this steric is related to the number of functional ligands and the interference parameter. By fitting the binding fraction for different incubation results shown in **Fig. 2.6(a)-(c)**, we could write the steric penalty from the unbound ligands as:

$$\beta U_L = U_A(N_L - \lambda)\delta^2 \qquad (2.12)$$

where U_A the repulsion penalty between the unbinding ligands and other material on the cell surface. **Eq. (2.12)** corresponds to the observed phenomenon: the greater the number of unbound ligands, the higher the steric potential; conversely, the lower the interference

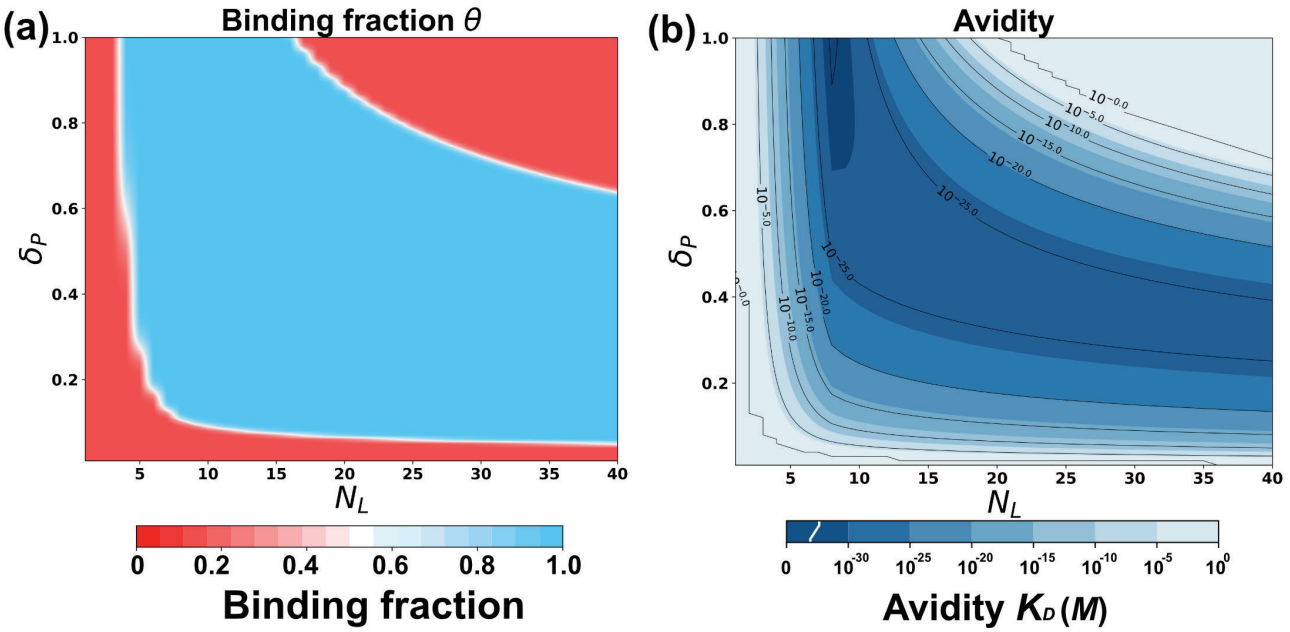

Figure 2.10 Binding fraction and avidity as functions of ligand number and the interference parameter. (a) Binding fraction as a function of ligand number and the interference parameter. **(b)** Avidity as a function of ligand number and the interference parameter.

parameter, the smaller the steric potential. Considering the steric from the glycocalyx, we can write the final particle function as:

$$\Xi = Qe^{-\beta U_G} \qquad (2.13)$$

The final binding fraction can be expressed as **Eq. (1.11)** and **Eq. (2.9)**.

2.2.3 Binding fraction and target selectivity of AP2 decorated PEG-PCL nanoparticles

Using the experimental data and PAT, we determined the repulsion penalty from the unbinding ligands. We present the total binding avidity and the binding fraction as a function of the ligand number and interference parameter in **Fig. 2.10**. There are approximately 10 AP2 binding sites on LRP1, which have an estimated size of about 40 nm. The PEG-PLA nanoparticles used in this study have a radius of 16.64 nm. The nanoparticle's size constrains the number of accessible binding sites, and this limitation is accounted for in the predictive model.

In **Fig. 2.10(a)**, we present the predicted binding fraction, which aligns well with the experimental data. The binding fraction depends on both the number of ligands and the interference parameter. When the interference parameter is large, nanoparticles with a high number of ligands exhibit a low binding fraction due to steric hindrance from unbound ligands. Reducing the interference parameter lowers the steric hindrance from unbound ligands. When the interference parameter is small, increasing the number of ligands within a certain range can enhance combinatorial entropy while minimizing steric hindrance from unbound ligands, thereby improving binding efficiency. However, if the interference parameter drops below a critical threshold, the resulting steric interference can exceed the binding energy, effectively preventing any binding events. As shown in **Fig. 2.10(b)**, the highest avidity occurs at approximately 8 AP2 ligands when the interference parameter is equal to 1. At this point, the number of ligands is relatively low, and the steric interference penalty can be considered negligible.

Since we only target a single receptor, we redefine the selectivity parameter as:

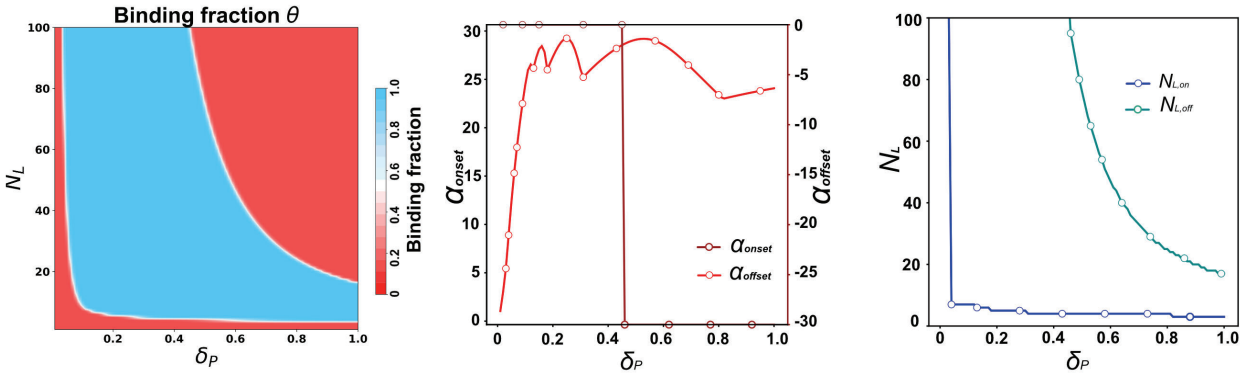

Figure 2.11 The selectivity analysis versus the interference parameter. The left panel shows the binding fraction as a function of the number of ligands and the interference parameter. The middle panel shows the selectivity parameter versus the interference parameter. The right panel shows the onset and offset ligand number value as a function of the interference parameter.

$$\alpha_{N_L} = \frac{\partial ln\,\theta}{\partial ln\,N_L}$$

As the number of ligands increases, the binding fraction may first increase and then decrease. The ligand number at which α reaches its maximum value is defined as the onset ligand, $N_{L,on}$. The decrease in the binding fraction would lead to a negative selectivity parameter. The ligand number at which α is negative and its minimum value is defined as the offset ligand, $N_{L,off}$. The corresponding offset value is defined as α_{offset}. In **Fig. 2.11**, we plot the binding fraction as a function of the number of ligands (extending the number of ligands to 100) and the interference parameter. The middle panel shows the selectivity parameter. The right panel shows the onset and offset ligand number value as a function of the interference parameter, which correlates to the binding fraction map on the left panel.

When targeting large receptors, the variation in binding behavior across different cell phenotypes is relatively small, as shown in **Fig. 2.8(a)-(c)**. In contrast, for nanoparticles targeting small receptors, only a single receptor typically falls within the binding volume, and multivalent interactions rely on multiple binding sites on that individual receptor. In this case, cell phenotype primarily influences binding through differences in glycocalyx expression, which contributes to steric hindrance. In short, there is no significant binding difference between the CNS cell types because of the AP2-decorated PEG-PLA targeting the large receptor, LRP1.

By predicting the binding fraction and analyzing selectivity, we can rationally design nanoparticles for targeted applications. When targeting large receptors, tuning both the interference parameter and the number of ligands allows for effective binding. If the goal is strong binding with high avidity, a design featuring a high interference parameter and a relatively low number of ligands is optimal. However, in cases where the ligands themselves contribute to therapeutic efficacy, it is preferable to use a low interference parameter. This enables the nanoparticle to accommodate a greater number of ligands, enhancing its therapeutic potential.

2.3 Extending PAT to model nanoparticle interactions with large receptors

2.3.1 mUNO-driven cellular phenotypic targeting and the application of PAT for targeting large receptors

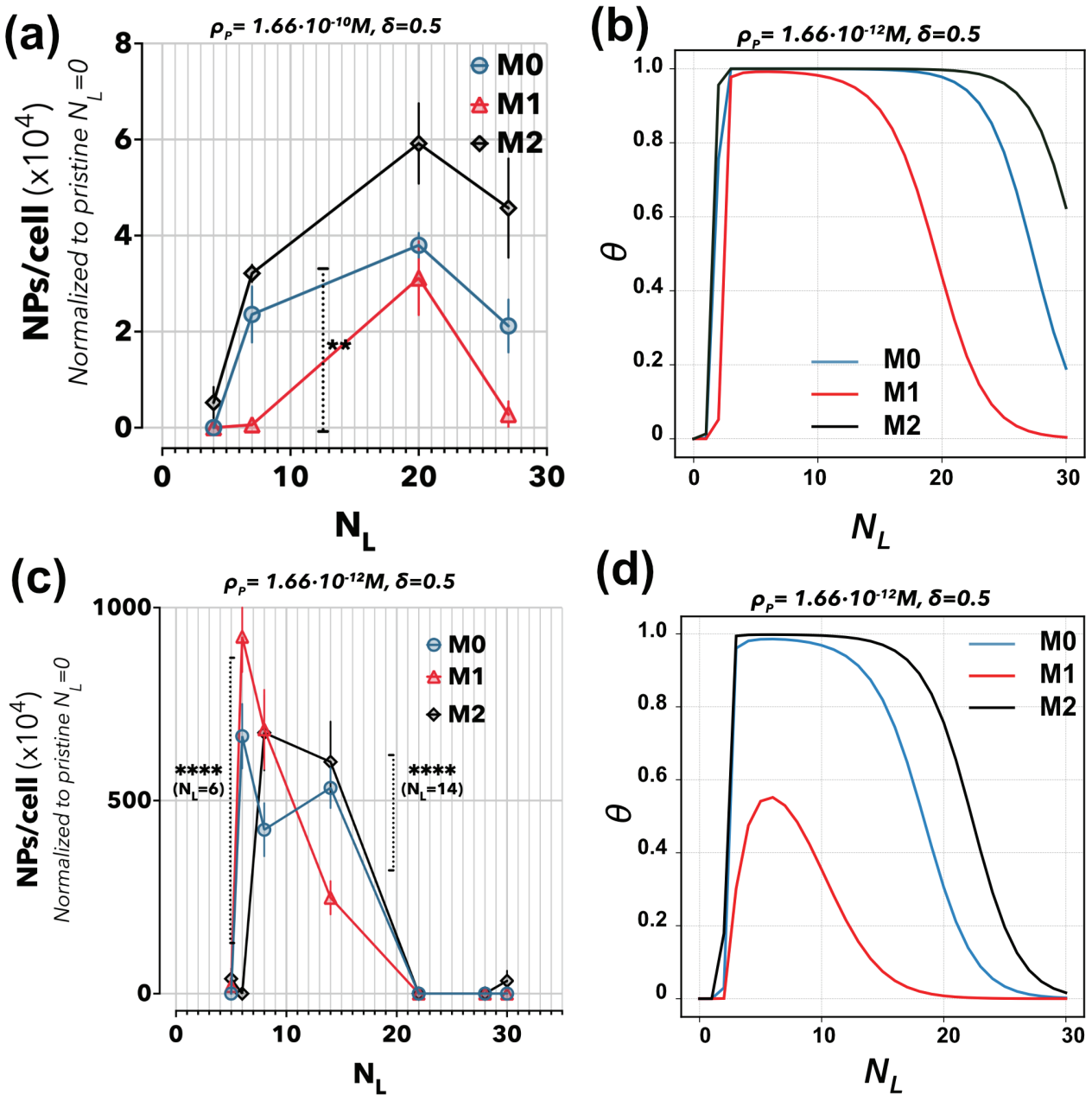

Figure 2.12. Experimental data and model predictions for the interaction between PEG-PLA nanoparticles and different cell types. (a) Experimental measurements of nanoparticle binding on various cell types after 10 minutes, using an interference parameter of 0.5. **(b)** Predicted binding fraction of PEG-PLA nanoparticles, with the same interference parameter, based on the PAT model for large receptors.

CD206, also known as the mannose receptor (MRC1), is a type I transmembrane C-type lectin receptor predominantly expressed on M2-polarized macrophages, dendritic cells, liver sinusoidal endothelial cells, and certain skin cells[145,146]. It mediates clathrin-dependent endocytosis and continuously cycles between the plasma membrane and endosomes[147]. Functionally, CD206 recognizes terminal mannose, fucose, N-acetylglucosamine, and sulfated glycans on both pathogens and endogenous glycoproteins, playing a critical role in antigen uptake, microbial clearance, tissue remodeling, and anti-inflammatory responses[148,149]. It is involved in host defense by recognizing pathogens such as Candida, Mycobacterium tuberculosis, and HIV. The soluble form of CD206 (sCD206) is being investigated as a biomarker for conditions including pneumonia, tuberculosis, and cardiovascular diseases[150]. Clinically,

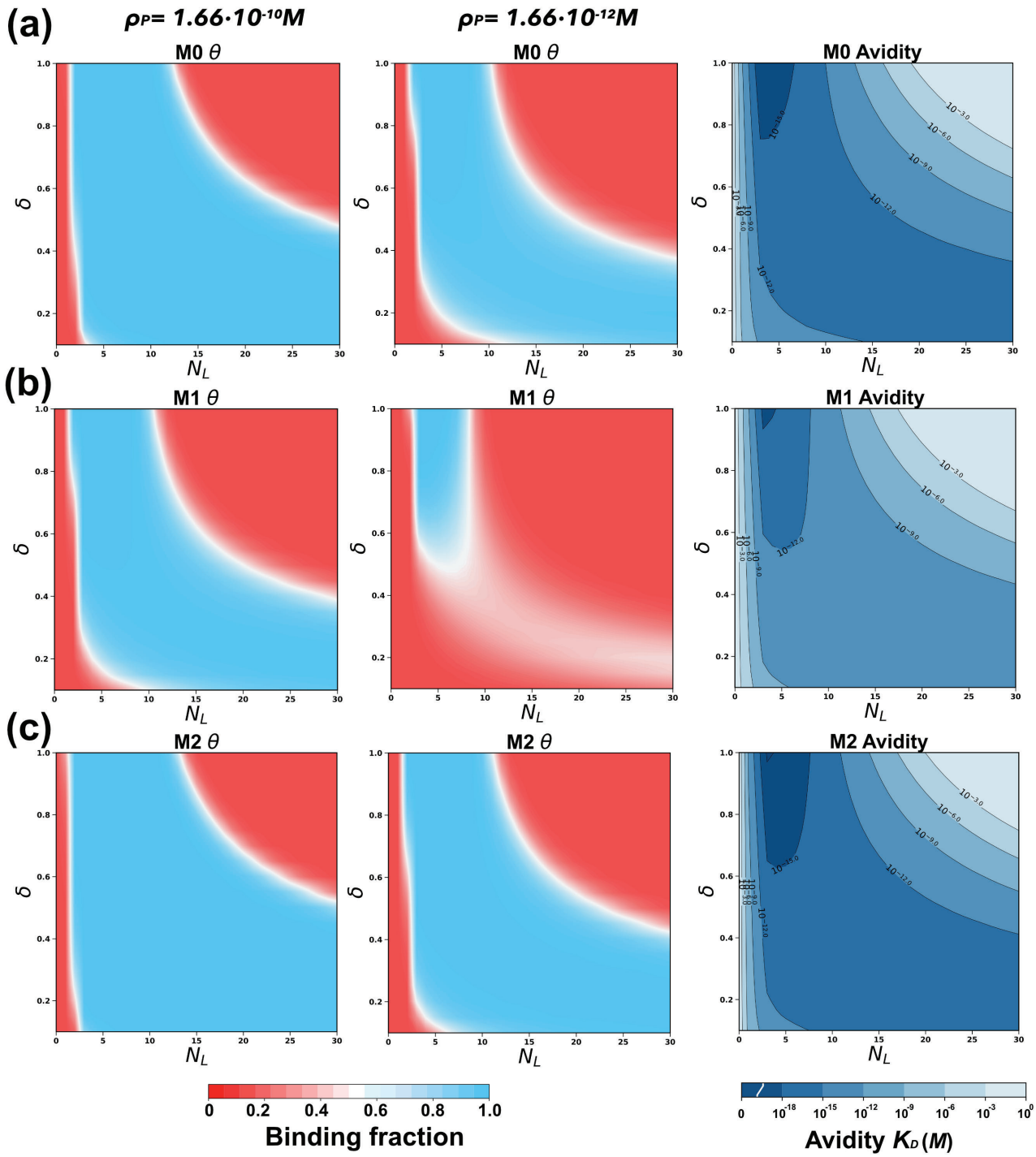

Figure 2.13 The binding fraction and avidity between the mUNO-decorated PEG-PLA nanoparticles and the BMDMs. (a)-(b) Binding fraction and avidity for M0. **(c)-(d)** Binding fraction and avidity for M1. **(e)-(f)** Binding fraction and avidity for M2.

CD206 also serves as a key marker of tumor-associated macrophages (TAMs) and is a promising target in cancer immunotherapy and nanoparticle-based drug delivery strategies[151].

The mUNO peptide (sequence: CSPGAK) is a tumor-homing hexapeptide identified through in vivo phage display for its high affinity and specificity to CD206[152,153]. Unlike classical ligands of CD206, such as mannose or sulfated glycans, mUNO binds to a distinct site between C-type lectin-like domains 1 and 2, engaging the receptor via electrostatic, aromatic, and hydrogen-bond interactions. This unique binding mode allows mUNO to selectively target CD206⁺ M2-polarized macrophages, especially TAMs, without significant off-target accumulation in healthy

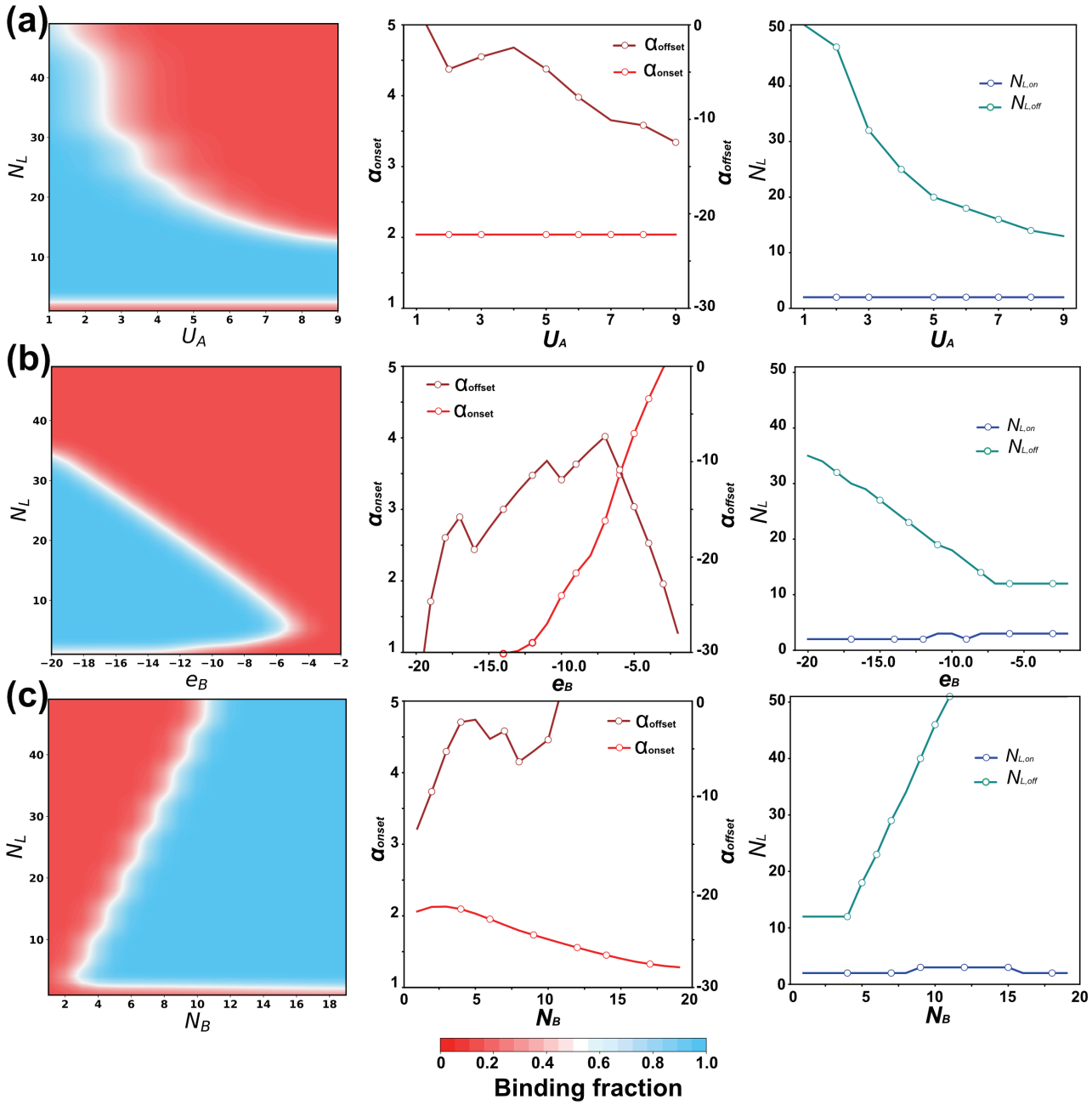

Figure 2.14 Analysis of binding fraction and selectivity as a function of various parameters. (a) Left: Binding fraction as a function of N_L and U_A; Middle: Selectivity parameter as a function of U_A; Right: Onset and offset ligand number value as a function of U_A. **(b)** Left: Binding fraction as a function of N_L and e_B; Middle: Selectivity parameter as a function of e_B; Right: Onset and offset ligand number value as a function of e_B. **(c)** Left: Binding fraction as a function of N_L and N_B; Middle: Selectivity parameter as a function of N_B; Right: Onset and offset ligand number value as a function of N_B.

tissues such as the liver. Fluorescently labeled mUNO (e.g., FAM-mUNO) has demonstrated high selectivity in vitro and various murine tumor models[154]. Importantly, mUNO has been incorporated into nanocarriers such as OximUNO to achieve targeted delivery of therapeutic agents, enhancing drug efficacy in triple-negative breast cancer models while minimizing systemic toxicity[155]. Due to its specificity and functional versatility, mUNO is emerging as a promising ligand for precision targeting in cancer immunotherapy and nanoparticle-based drug delivery.

THP-1-derived macrophages are macrophage-like cells differentiated from the THP-1 cell line, a human monocytic leukemia cell line that is widely used as an in vitro model for human monocytes and macrophages[156,157]. These cells can be polarized into distinct functional phenotypes, including M0 (unpolarized), M1 (pro-inflammatory), and M2 (anti-inflammatory or tissue-repairing) macrophages, making them a valuable tool for modeling macrophage polarization and studying immune responses under controlled laboratory conditions.

We varied the number of mUNO ligands on PEG-PLA nanoparticles from 0 to 30 while keeping the interference parameter fixed at 0.5, aiming to target MDMs. The experimental results after 10 minutes of incubation with different BMDM subtypes are shown in **Fig. 2.12(a)**. For both M0 and M2 macrophages, a binding onset was observed around 7 ligands, with no clear binding offset even at 27 ligands. In contrast, the M1 macrophages displayed a pronounced onset and offset trend, indicating more defined ligand-dependent binding behavior. Since the mUNO receptor is a large receptor, we applied the PAT for large receptors to estimate the binding fraction of the nanoparticles, as illustrated in **Fig. 2.12(b)**. According to the model, the predicted onset of binding begins at around 3 ligands, which does not align with the experimental data. This discrepancy may be attributed to the relatively short incubation time (10 minutes), which might not have been sufficient for nanoparticles to reach equilibrium binding. However, the predicted offset matches the experimental results, supporting the hypothesis that steric hindrance from unbound ligands plays a significant role in limiting further binding at high ligand densities.

Using the PAT for large receptors, we modeled the binding fraction and avidity of mUNO-decorated PEG-PLA nanoparticles to BMDMs as a function of both the number of ligands and the interference parameter, as shown in **Fig. 2.13**. The overall trends closely resemble those observed in **Fig. 2.10**. Specifically, the model predicts that a high interference parameter results in strong binding but within a narrower range of ligand numbers. In contrast, nanoparticles with a low interference parameter allow for effective binding across a broader range of ligand densities. The highest avidity is observed at low ligand numbers combined with a high interference parameter, consistent with the trend previously shown in **Fig. 2.10**.

2.3.2 Selectivity analysis for PAT based on a large receptor

Both the AP2-decorated PEG-PCL and mUNO-decorated PEG-PLA nanoparticles demonstrate that the number of ligands can regulate the onset and offset of binding. To further investigate general principles for optimizing nanoparticle design, we examine how the binding fraction varies with the number of ligands and other critical parameters. These include steric repulsion from unbound ligands, ligand-receptor binding energy, and the number of binding sites available on the receptor. The prediction and selectivity analysis are shown in **Fig.2.14**.

Steric hindrance from unbound ligands limits the ability to modulate the number of ligands for controlling nanoparticle binding. As shown in **Fig. 2.14(a)**, we examine the binding fraction as a function of steric repulsion and ligand number. In the left panel, the range of effective ligand numbers narrows as steric effects increase, indicating that stronger repulsion restricts binding. The middle panel displays the selectivity parameter. Although the differences across α_{max} are not large, a general decrease in α_{min} is observed, suggesting that higher steric penalties from unbound ligands lead to a sharper decline in binding fraction beyond the optimal ligand number. The right panel shows the onset and offset ligand numbers, which align well with model predictions.

Binding energy is another critical parameter influencing nanoparticle-cell interactions. In **Fig. 2.14(b)**, we present the binding fraction as a function of binding energy. For stronger binding energies, a broader range of ligand numbers supports effective binding. In contrast, high

receptor numbers do not lead to strong binding when the binding energy is low. The selectivity parameter for the onset of binding increases as binding energy decreases, consistent with the design principle that multivalent interactions enhance selective binding. The offset selectivity parameter shows some fluctuations but remains within a low range. Overall, the predicted onset and offset ligand numbers are in good agreement with the observed trends.

The number of binding sites on large receptors forms the foundation for supporting multivalent interactions. **Fig. 2.14(c)** presents the predicted binding fraction as a function of the number of binding sites. In the left panel, the effective ligand range shifts toward higher values as the number of binding sites increases. This is because more available sites allow more ligands on the nanoparticle to participate in binding, thereby reducing steric penalties from unbound ligands. As the number of binding sites increases, the onset selectivity parameter decreases, indicating easier initiation of binding. Additionally, the increase in binding sites leads to a higher offset ligand number, as shown in the right panel.

This study deepens our understanding of multivalent interactions between nanoparticles and cellular receptors by dissecting the roles of ligand number, steric hindrance, binding energy, and receptor site availability. One key insight is that the design of nanoparticle-based targeting systems must strike a careful balance: too many ligands can introduce steric penalties that hinder binding, while too few may not provide sufficient avidity. Additionally, the findings underscore the importance of tailoring nanoparticle properties to the biological context-strong binding energy and abundant receptor sites can widen the effective binding window, improving targeting efficiency. Looking forward, this framework provides a quantitative basis for rational nanoparticle design in drug delivery, diagnostics, and immunotherapy.

2.4 Thermodynamic model for rational design of nanoparticle properties

To support the design of precision-targeted nanoparticles, we adopt a multiscale modeling approach to simulate nanoparticle-cell interactions. All-atom simulations are employed to estimate binding affinities and characterize binding scenarios at the molecular level. Based on these interaction profiles, and given specific cell types and nanoparticle properties, we use the PAT to forecast the binding fraction. These predictions are further validated and refined using experimental data, enabling a more accurate and rational design of nanoparticle-based delivery systems.

In this section, we introduced two thermodynamic models within the PAT framework to describe nanoparticle-cell interactions. The first model addresses small ligands, where binding is primarily governed by the receptor density within the nanoparticle's binding volume. The second model focuses on large receptors with multiple binding sites, where the receptor's internal architecture supports multivalent interactions. The parameters, such as the ligand number, size, binding energy, and topology, can be integrated into the PAT to give an inside look at the nanoparticle-cell interaction. Together, these models provide a quantitative foundation for the rational design of nanoparticle properties-including ligand number, size, and affinity-to optimize binding strength and selectivity.

While the thermodynamic approach captures equilibrium behavior and key design principles, it does not account for the dynamic processes underlying nanoparticle binding over time. To address this, the next section introduces a dynamic model that incorporates binding kinetics, enabling the prediction of time-dependent behaviors such as binding rates, accumulation, and saturation. This extension is essential for guiding the design of nanoparticles in time-sensitive biomedical applications.

Chapter 3: Multivalency uptake kinetics

3.1 Multivalency uptake kinetics

3.1.1 From multivalency thermodynamics to multivalency kinetics

Understanding the interaction between nanoparticles and cell surface receptors requires a comprehensive approach that encompasses both the thermodynamics and kinetics of multivalent binding. While thermodynamics offers valuable insights into the overall binding affinity and stability by accounting for the cumulative effects of multiple ligand-receptor interactions, studying the kinetics is equally crucial to reveal how rapidly these interactions form and dissociate, and through which pathways. By integrating thermodynamic principles with kinetic analyses, we can achieve a more accurate prediction and control of nanoparticle behavior in biological systems, ultimately enhancing the effectiveness of targeted delivery strategies.

The study of binding kinetics remains challenging, not only because numerous factors influence it, but also due to our limited quantitative understanding of how these factors impact the process. By drawing an analogy to a single ligand-receptor interaction, we model the binding of a nanoparticle capable of engaging with multiple receptors as a chemical reaction. Consequently, the binding kinetics of nanomedicine, incorporating multivalent effects, can be expressed as follows:

$$P + \nu R \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} PR_{\nu} \qquad (3.1)$$

where PR_{ν} is the nanomedicien/receptor complex.

It is essential for a system to overcome the reaction barrier to enable the reaction to proceed, as shown in **Fig. 1.7(a)**. In designing nanomedicines that exploit the multivalency effect, we exploit thermodynamic models to predict the binding fraction of nanoparticles with target cells-that is, to determine whether interactions are likely to occur. However, it is important to emphasize that even if the calculated binding energy is negative (indicating a thermodynamically favorable interaction), this does not guarantee immediate binding. A relevant analogy is the reaction between pure hydrogen (H_2) and oxygen (O_2) gases, which can coexist for extended periods without reacting. Although the formation of water from these gases is highly exothermic and thermodynamically favorable, the reaction does not proceed spontaneously due to a significant activation energy barrier. Similarly, in the PAT framework, steric hindrance acts as a barrier, functionally analogous to activation energy, that must be overcome for effective binding. Therefore, to fully understand and design multivalent nanomedicines, it is crucial to extend the PAT framework to include kinetic considerations. This integration would enable a more comprehensive description of nanoparticle behavior, encompassing not only whether binding is possible, but also how quickly and efficiently it occurs.

A comprehensive understanding of reaction rates and pathways remains a significant challenge in the study of molecular interactions. This challenge is analogous to the classical brachistochrone problem: given two points A and B in a vertical plane, what is the curve along which a ball will descend from A to B in the shortest time, under gravity and without friction? Nanoparticle–target avidity can be conceptualized as the vertical distance between two points, representing the total binding avidity, which can be estimated using the PAT. In contrast, binding kinetics describe how the 'ball' travels between A and B, reflecting the rate at which binding occurs and dissociates over time. Achieving an understanding of multivalent kinetics is crucial, as it provides a rational framework for controlling the kinetics of molecular interactions.

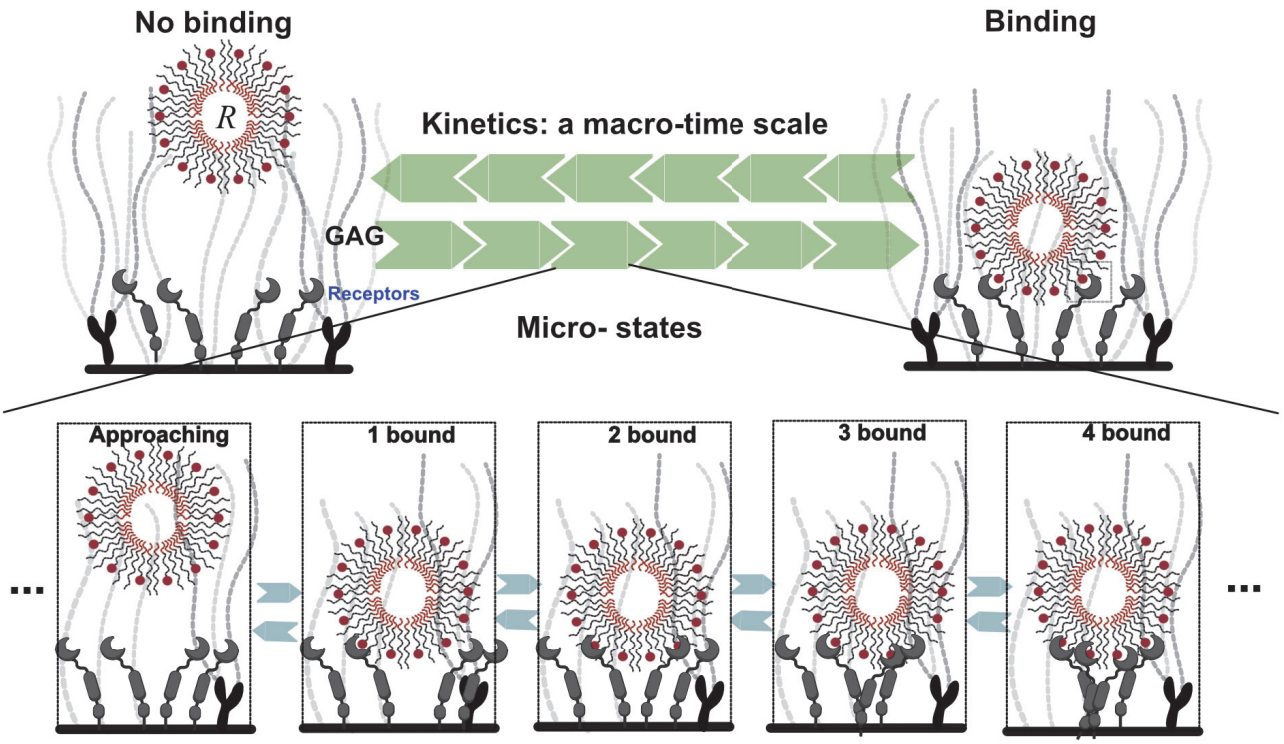

Figure 3.1. The schematic shows the time scale of nanoparticle receptor interaction. Multivalent binding is not a static state characterized by a fixed number of bound ligand-receptor complexes. Instead, it is a dynamic system capable of exploring all the possible binding configurations. Transitions between these states occur on a micro-timescale. From this perspective, the nanoparticle is generally considered to be in a "bound" state, as the fraction of time spent in a completely unbound state is minimal compared to the ensemble of partially or fully bound configurations.

Compared to simple ligand-receptor interactions, binding kinetics in multivalent systems are influenced by additional factors such as nanoparticle size, shape, and the nature of functional ligands. It is even worse to determine how the elements affect the kinetics. Thus, accurately predicting binding kinetics under these complex parameters remains extremely challenging. Further research is essential to deepen our understanding of multivalent interactions, enabling fewer clinical trials and more rational medicinal design.

3.1.2 A mathematical model to quantify the multivalency binding kinetics

To estimate multivalent binding kinetics based on ligand–receptor interactions, various models have been developed to simulate the binding landscape and quantify reaction rates. Hannah West *et. al* introduced a model that accounts for different binding states involving multiple ligands[158]. In this framework, the on-rate and dissociation rate are governed by the affinity of individual ligand-receptor pairs. If we have a single ligand/receptor reaction rate, the model could tell us the overall binding rate, which is calculated as the sum of the binding rates of all possible configurations with varying numbers of bound ligands. As the model considers all the binding states, the differential equations are very complex and are not easy to solve[159,160]. For a multivalence system of small molecules that ignores the steric potential, the model works well. However, when we tune to the nanoparticle-cell binding scenario, it becomes essential to consider the influence of steric potential, as it significantly affects the final avidity between the nanoparticle and the receptor. According to the reaction equation as shown in the previous section, we can write the reaction model as:

$$\frac{\partial \rho_P}{\partial t} = -k_{on}\rho_P\rho_R^{\nu} + k_{off}\rho_{PR_\nu}$$
$$\frac{\partial \rho_R}{\partial t} = -k_{on}\rho_P\rho_R^{\nu} + k_{off}\rho_{PR_\nu} \qquad (3.2)$$
$$\frac{\partial \rho_{PR_\nu}}{\partial t} = k_{on}\rho_P\rho_R^{\nu} - k_{off}\rho_{PR_\nu}$$

The advantage of this equation system lies in its simplicity and its ability to capture the overall binding rate, where both the on-rate and dissociation rates reflect the combined effects of binding avidity and steric potential. When the reaction reaches an equilibrium, we can get the association constant $K_{av} = \frac{\rho_{PR_\nu}}{\rho_P\rho_R^{\nu}}\rho_0^{\nu-1} = \frac{k_{on}}{k_{off}}\rho_0^{\nu-1}$, with $\rho_0 = 1M$ being the standard concentration.

Through this model, we can quantitatively analyze how multivalent interactions influence binding kinetics. As valency increases while keeping the on-rate of individual ligand–receptor interactions constant, the overall reaction rate also increases. In contrast to monovalent systems, multivalent binding exhibits nonlinear behavior because higher valency fundamentally alters the kinetic profile. These interactions introduce cooperative binding effects and steric hindrance, both of which require a more refined and complex kinetic framework to describe the system accurately.

Several techniques are available to estimate the real-time binding kinetics between nanomedicines and cells, including surface plasmon resonance (SPR) and laser scanning confocal microscopy[161,162]. Each of the methods measures the kinetics on a macro-time scale. The interaction between the ligand and receptor happens on the micro-time scale[160]. The interaction schematic is shown in **Fig. 3.1**. From a thermodynamic perspective, the multivalent system naturally tends to explore all accessible states to maximize its overall entropy and minimize its free energy. This means that rather than remaining fixed in a single configuration, the system fluctuates among multiple microstates, each representing a different binding state. The macro-timescale may encompass thousands of distinct micro-binding states, reflecting the inherent capacity of multivalence systems to explore all the possible binding configurations. These states may include an unbound state. If the unbound state plays a minimal role compared to the ensemble of bound states, the association constant is linked to the thermodynamic model known as the PAT. Thus, we can estimate the binding avidity and write the association constant as:

$$K_{av} = \frac{\rho_{PR_\nu}}{\rho_P\rho_R^{\nu}}\rho_0^{\nu-1} = \frac{e^{-E_B/kT}}{\rho_0} = \frac{k_{on}}{k_{off}}\rho_0^{\nu-1} \qquad (3.3)$$

This equation incorporates three approaches for estimating the association constant: estimating the concentrations of components at equilibrium, directly measuring the association and dissociation rates, and calculating it using the PAT model. We advocate for the use of multiple methods to estimate the association constant, both to ensure the reliability of the measured value and to connect thermodynamic and kinetic perspectives for improved nanomedicine design.

3.1.3 A mathematical model to quantify the multivalency uptake kinetics

Once a nanoparticle binds to the cell surface via receptor-ligand interactions, the next critical step in cellular uptake is internalization. This process refers to the active transport of surface-bound materials into the intracellular space, typically through endocytosis. Internalization is

essential for the therapeutic efficacy of many nanomedicine systems, as it determines whether the delivered cargo can reach intracellular targets.

The internalization pathways include phagocytosis, pinocytosis, and receptor-mediated endocytosis, each with distinct regulatory mechanisms and preferences for cargo size, shape, and surface chemistry[163,164]. Phagocytosis and pinocytosis are non-specific uptake mechanisms. In nanoparticle–receptor targeting systems, internalization primarily proceeds via receptor-mediated endocytosis, facilitating specific and efficient uptake into cells. The receptor-mediated endocytosis rate can be expressed as k_e. Thus, we modify the uptake kinetics for the multivalency binding system as:

$$P + \nu R \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} PR_\nu \xrightarrow{k_e} P_E \qquad (3.4)$$

Residence time plays a pivotal role in nanoparticle internalization[94]. The dynamics of internalization are closely linked to the strength and stability of the initial binding event, often characterized by the nanoparticle–receptor residence time. A longer residence time typically provides the cell with sufficient opportunity to initiate endocytic processes, thereby increasing the likelihood of successful internalization. In contrast, weak or transient interactions may result in rapid dissociation before internalization can proceed, leading to reduced uptake efficiency. While the on-rate contributes to the frequency of initial binding events, it is the residence time that plays a more critical role in determining internalization efficiency.

Through the reaction equation, the uptake kinetics can be expressed as the following equation:

$$\begin{aligned}
\frac{\partial \rho_P}{\partial t} &= -k_{on}\rho_P\rho_R^\nu + k_{off}\rho_{PR_\nu} \\
\frac{\partial \rho_R}{\partial t} &= -k_{on}\rho_P\rho_R^\nu + k_{off}\rho_{PR_\nu} \\
\frac{\partial \rho_{PR_\nu}}{\partial t} &= k_{on}\rho_P\rho_R^\nu - k_{off}\rho_{PR_\nu} - k_e\rho_{PR_\nu} \\
\frac{\partial \rho_E}{\partial t} &= k_e\rho_{PR_\nu}
\end{aligned} \qquad (3.5)$$

where ρ_E stands for the endocytosed nanoparticles[165]. The equations provide a quantitative framework for measuring internalization rates, enabling us to evaluate how different nanoparticle formulations influence uptake dynamics. By comparing these rates, we can rationally design nanoparticles with tailored internalization behaviors, optimizing their performance for specific biological or therapeutic applications.

Internalization pushes the multivalence system out of equilibrium. As receptors on the cell membrane are internalized along with the nanoparticles, the number of available binding sites decreases over time, hindering the system from achieving a true equilibrium state. This dynamic behavior complicates the experimental estimation of the association constant, particularly when it depends on equilibrium concentration measurements. Notably, cellular endocytosis can be inhibited by lowering the temperature to 18 °C. However, since the association constant is temperature-dependent and nanomedicines are intended to function at physiological (body) temperature, measurements taken at reduced temperatures may not accurately reflect the *in vivo* binding behavior. Thus, we need to estimate k_{on} and k_{off} from the binding experiment, which could be driven by fitting and simulation of multivalent uptake kinetics.

3.2 Binding kinetics of PMPC nanoparticles

3.2.1 Observation of the uptake kinetics on the membrane

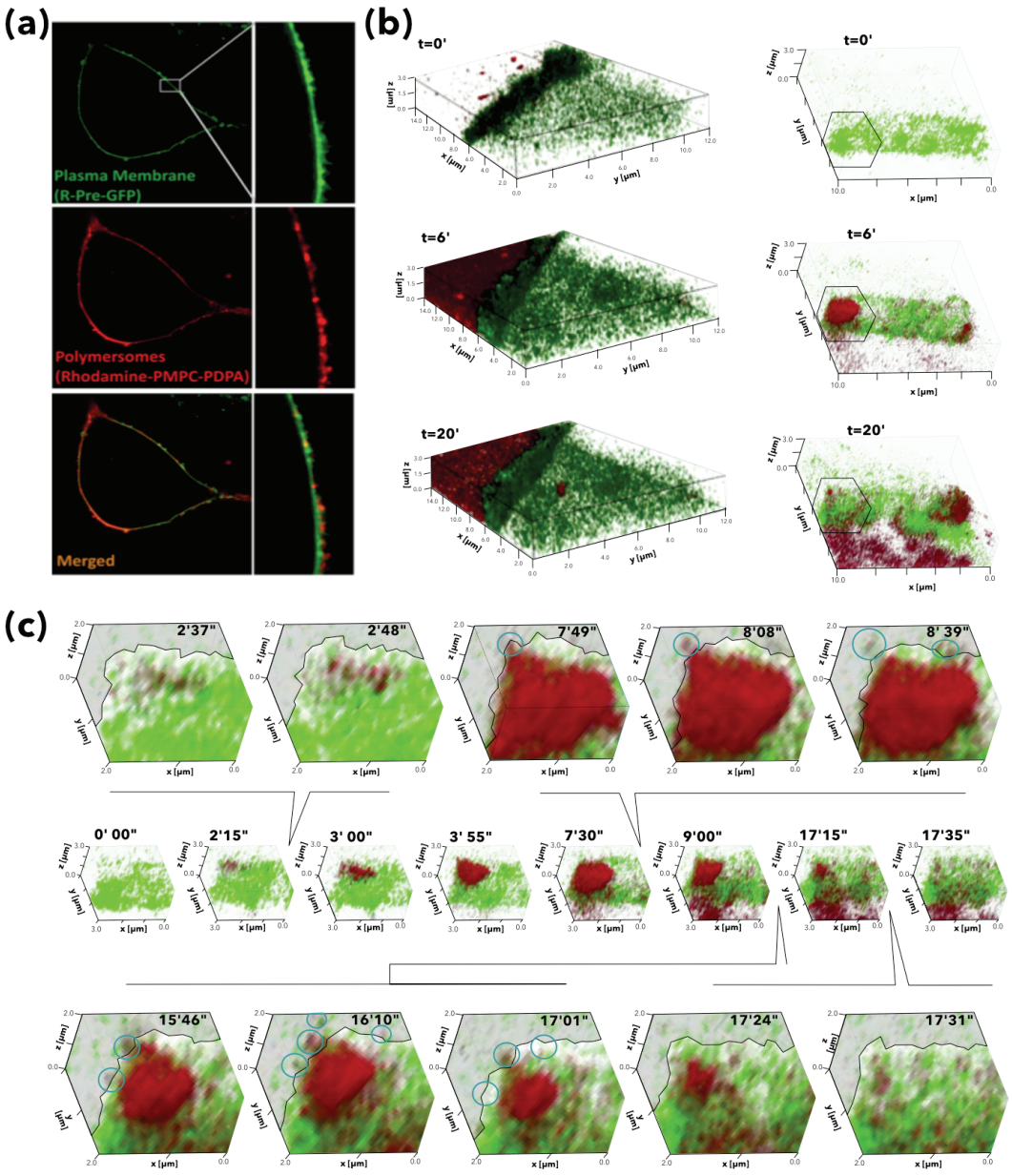

Figure 3.2. Kinetics of endocytosis at the plasma membrane. (a) 2D confocal microscopy pictures showing the binding of the polymersomes (red channel, middle image) to the membrane of one single cell (green channel, top image); the bottom image shows the green and red channels merged. **(b)** The images on the right are magnifications of the membrane. Magnified 3D confocal microscopy snapshots at three different times. The top image illustrates the initial diffusion of PMPC nanoparticle (red) as they approach the plasma membrane (green). The middle image depicts the binding of nanoparticles to the membrane, while the bottom image highlights the onset of initial internalization events. **(c)** A magnified 3D confocal image showcasing the kinetics of nanoparticle uptake occurring within the cube highlighted in (b). At the initial time, the membrane was fully green. The nanoparticles approached the cell membrane and bound to the receptors, as evidenced by the enhanced signal intensity in the red channel. After binding, the nanoparticles were internalized by the cell, resulting in a decrease in red channel intensity.

We employed PMPC nanoparticles to target the A549 cell line, a human lung adenocarcinoma

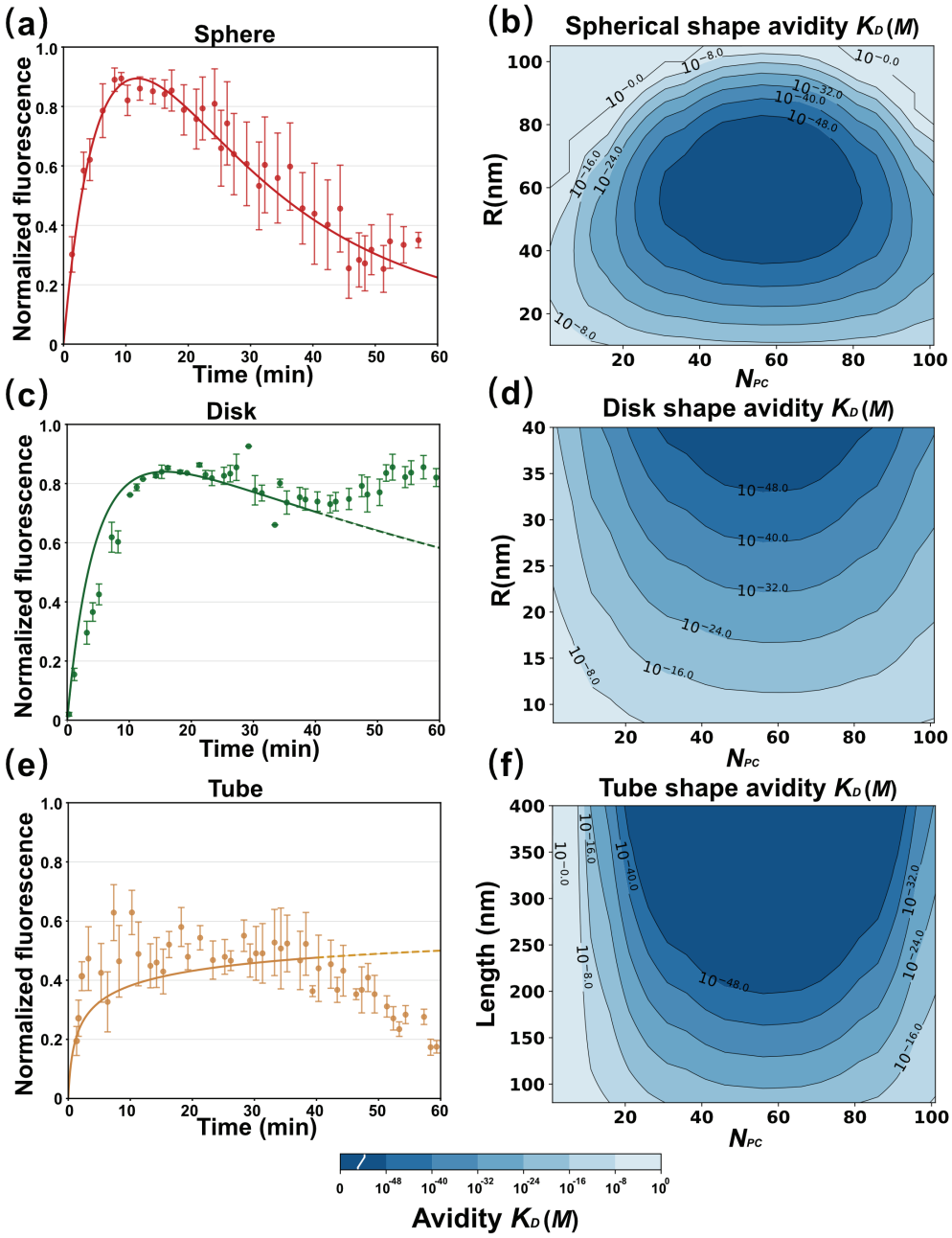

Figure 3.3. Binding kinetics of nanoparticles with different topologies on the cell membrane. (a) Spherical PMPC nanoparticle binding kinetics and fitting results using the uptake kinetics model. **(b)** Avidity prediction of spherical nanoparticles versus PMPC polymerization (N_{PC}) and radius (R). **(c)** Disk binding kinetics and fitting result. **(d)** Avidity prediction of disk nanoparticles versus N_{PC} and R. (e). Tube binding kinetics and fitting result. **(f)** Avidity prediction of tube nanoparticles versus N_{PC} and R.

epithelial cell line derived from alveolar basal epithelial cells. We used fast live cell imaging to monitor both the binding of the nanoparticles to the cell membrane and the subsequent internalization by A549 cells. Cy3-tagged nanoparticles were incubated with A549 cells pre-labeled with a green fluorescent membrane marker (R-Pre fused to GFP) and monitored over time using live-cell confocal microscopy at an acquisition rate of 0.2 frames per second. The resulting data were analyzed using *ImageJ* software. In **Fig. 3.2**, we showed both the 2D confocal microscopy picture and magnified 3D confocal microscopy snapshots at different time points. This binding is a reversible process, mediated by specific interactions between PMPC

Table 3.1: Parameter from the thermodynamics model (PAT) and uptake kinetics model

nanoparticle type	Size R and L [nm]	Particle Molarity [*M*]	Valence	K_{av} [M^{-1}]	k_{on} [$M^{-1}{\cdot}s^{-1}$]	k_{off} [s^{-1}]	Residence time (s)	k_e [s^{-1}]
Psomes	45	$1.43{\cdot}10^{-8}$	2.5	$2.14{\cdot}10^{37}$	$5.38{\cdot}10^{17}$	$2.51{\cdot}10^{-20}$	$3.99{\cdot}10^{19}$	$1.33{\cdot}10^{-3}$
Disks	16	$5.14{\cdot}10^{-7}$	1.1	$8.06{\cdot}10^{18}$	$2.28{\cdot}10^{7}$	$3.20{\cdot}10^{-12}$	$3.12{\cdot}10^{11}$	$1.50{\cdot}10^{-4}$
Tubes	40 and 350	$1.63{\cdot}10^{-9}$	5.5	$1.42{\cdot}10^{52}$	$8.28{\cdot}10^{47}$	$5.84{\cdot}10^{-5}$	$1.71{\cdot}10^{4}$	0

ligands and receptors on the cell surface. The image shows the real-time uptake kinetics of PMPC nanoparticles as they approach and bind to the cell membrane. After binding and remaining on the membrane for a certain period, the nanoparticles are internalized by the cells, leading to a decrease in red channel fluorescence intensity. As discussed earlier, this internalization process drives the PMPC–cell binding system out of equilibrium.

To investigate the effect of nanoparticle architecture, three different PMPC nanoparticle topologies were used (disk, tube, and spherical nanoparticle). The normalized uptake kinetics on the cell membrane are presented in **Fig. 3.3(a), (c),** and **(e)**, with each data point representing the average of 15 individual cells. The observation period was set to approximately 1 hour to capture the full progression of nanoparticle binding and uptake, allowing sufficient time to observe both the initial interaction with the cell membrane and subsequent internalization dynamics. For all nanoparticle morphologies, the fluorescence intensity on the cell membrane exhibited a sharp initial increase, indicating that the binding rate exceeded the rate of endocytosis. The peak membrane-associated fluorescence for spherical nanoparticles occurred at approximately 12 minutes, while for disk-like nanoparticles, it was reached around 15 minutes. After reaching their respective maxima, the fluorescence from spherical nanoparticles declined, suggesting that the rate of endocytosis had surpassed the binding rate. This decline may be attributed to the depletion of available receptors on the cell membrane. Disk-like nanoparticles also showed a gradual decrease in fluorescence up to 40 minutes, followed by a secondary increase, which could be due to receptor recycling. Compared to spherical nanoparticles, disk-like nanoparticles exhibited a slower endocytosis rate. For tube-like nanoparticles, endocytosis appeared to be minimal, as evidenced by a plateau in fluorescence after the initial binding phase.

Given the differences in plasma membrane association kinetics observed for the three nanoparticle morphologies via live-cell imaging, we conducted a series of fluorescence-activated cell sorting (FACS) experiments to obtain more comprehensive and statistically robust data. FACS allowed us to analyze thousands of cells, offering a broader view of nanoparticle uptake across the population and better mimicking in vivo tissue-level interactions. Although FACS lacks the temporal resolution for continuous tracking and is less sensitive at early time points, its high throughput significantly increased the statistical power compared to the limited sampling size of confocal microscopy. To enhance the reliability of our conclusions, we compared the FACS data with the microscopy results and found strong agreement between the two methods. The confocal image of the spherical PMPC nanoparticle uptake is shown in **Fig. 3.4(a)**. FACS data (**Fig. 3.4(b)**) revealed that the fluorescence intensity associated with spherical PMPC nanoparticles and disk-shaped nanoparticles during membrane binding was significantly higher than that of tubular nanoparticles. While live-cell imaging showed a plateau in fluorescence for the tubular morphology, the FACS data indicated a slow, continuous increase in signal over time. The peak membrane-associated fluorescence was observed at approximately 20 minutes for Psomes and around 30 minutes for disks (**Fig. 3.4(b)**). Overall, the FACS measurements corroborated the microscopy observations and provided a more comprehensive

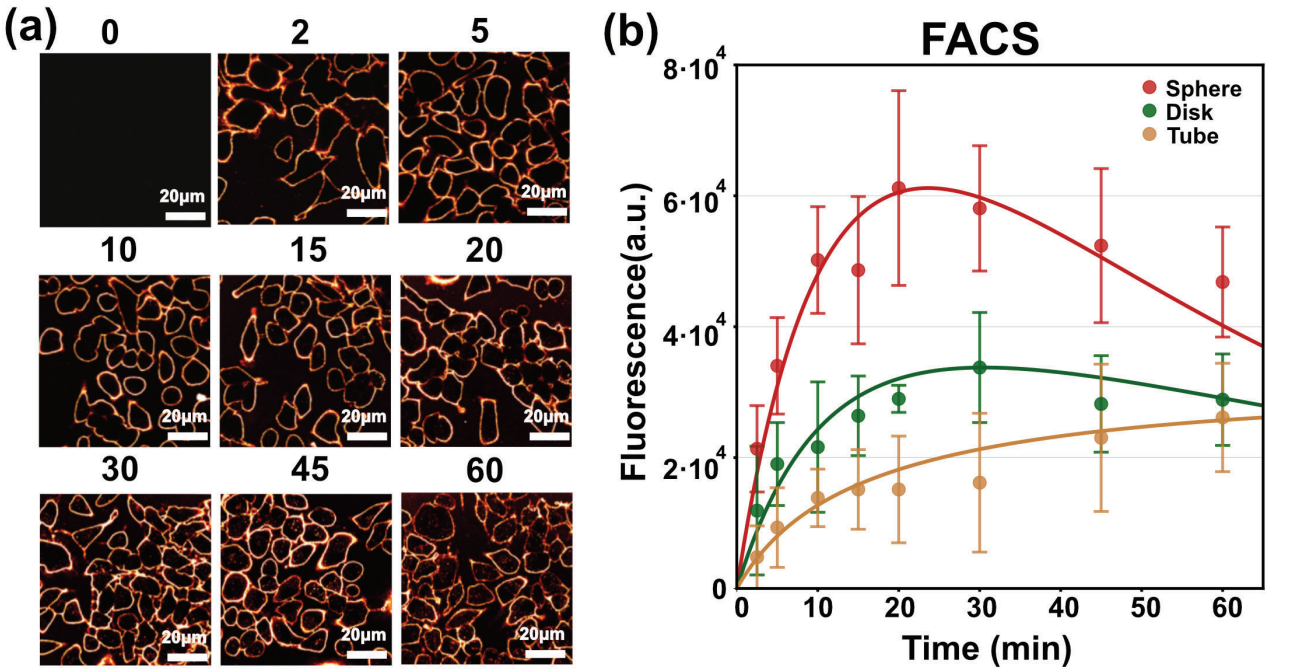

Figure 3.4. FACS validation of the nanoparticle internalization. (a) Connfocal microscopy images at different time points (0min, 2min, 5min, 10min, 15min, 20min, 30min, 45min, and 60min.). **(b)** Corresponding fluorescence measured by FACS for A549 cells after incubation with Cy3-labelled nanoparticles of different morphologies: Psome, disks, and tubular vesicles and the fitting curves using the uptake kinetics model.

view of the binding kinetics and cellular uptake behavior associated with different nanoparticle topologies.

3.2.2 Fitting and analysis of the uptake kinetics

To quantitatively analyze the uptake kinetics, we used Python-based data fitting approaches, employing libraries such as *NumPy*, *SciPy*, and *Matplotlib*. Nonlinear regression was performed using *scipy.optimize.curve_fit* to model the experimental fluorescence intensity data over time. This allowed us to extract key kinetic parameters, including binding and internalization rates, and compare the behavior across different nanoparticle morphologies. The fitting results for different topologies of PMPC nanoparticles binding with A549 cells observed by the microscope are shown in **Fig. 3.2(a)**, **(c)**, and **(e)**. The fitting results for the FACS are shown in **Fig. 3.3(b)**. The fitting results for the parameters are shown in **Table 3.1**.

As shown in the equation, the key parameters are the binding valence, on-rate, dissociation rate, and endocytosis rate. Valence is influenced by both receptor expression on the cell surface and the size of the nanoparticle. When cells exhibit higher receptor density and nanoparticles are bigger, the system tends to display greater multivalency. Valence, in turn, plays a critical role in determining the overall association constant. For a nanoparticle with fixed physicochemical properties, an increase in valence typically leads to a higher association constant, reducing the required dosage concentration. However, the precise impact of valence on binding kinetics remains poorly understood, and further research is necessary to fully elucidate these relationships.

In the binding scenario between PMPC nanoparticles and A549 cells, the cell type remains constant, allowing us to predict binding avidity by varying the properties of the nanoparticles, as demonstrated in the previous section. The predicted binding behaviors for all nanoparticle

topologies are shown in **Fig. 3.2(b)**, **(d)**, and **(f)**, which correspond to the same figures referenced earlier. From the binding kinetics, we can readily extract the on-rate. However, accurately determining the dissociation rate through experimental fitting is challenging. When the dissociation rate falls below a certain magnitude, it exerts a significant influence on the binding kinetics, yet remains difficult to quantify precisely. Our method here is to combine the PAT and binding kinetics to give a more rational estimation of the dissociation rate, as shown by **Eq.(3.3)**.

The resident time is the reverse of the dissociation constant, $\tau = \frac{1}{k_{off}}$. Among all these receptors, the spherical PMPC nanoparticle has the highest residence time. Although the tube-like PMPC nanoparticles exhibit the highest valence, their residence time is relatively shorter compared to the disk-like and spherical nanoparticles. This reduction in residence time is likely due to their larger size, which incurs a greater steric repulsion from the cell-surface glycocalyx.

Among the three nanoparticle morphologies, the tube-like PMPC nanoparticles exhibited the lowest endocytosis rate. Virtually no internalization was observed, as they appeared to remain attached to the cell membrane. In contrast, the spherical nanoparticles showed the highest endocytosis rate, while the disk-like nanoparticles displayed a moderate uptake rate. As discussed in the previous section, nanoparticles with radii between 30 and 100 nanometers are more readily internalized. Disk-like nanoparticles exhibit a significantly lower endocytosis rate than spherical ones, primarily due to their smaller structure. Their reduced rigidity and dynamic morphology can hinder effective membrane wrapping, which is a critical step in the initiation of endocytosis. Additionally, tubular nanoparticles are generally considered favorable for endocytosis when their aspect ratio is approximately 2 to 3. However, in this case, the larger size and elongated shape of the tube-like nanoparticles may have hindered efficient uptake. Nanoparticles with very high aspect ratios often remain on the cell surface or are internalized inefficiently, particularly when their long axis lies parallel to the membrane, limiting membrane wrapping and endocytosis, which is consistent with our observation.

3.3 Binding kinetics of other nanoparticles

3.3.1 Fitting the uptake kinetics between the mUNO-decorated PEG-PLG nanoparticle and the THP1-derived macrophages

In the previous section, we described the binding interactions between mUNO-decorated PEG-PLA nanoparticles and target cells. The binding kinetics and nanoparticle uptake at 10 minutes are presented in **Fig. 2.12**. We show the kinetics for the binding in **Fig. 3.5**. Two sets of experiments were conducted: one using a high nanoparticle-to-cell ratio of $4{\cdot}10^5$ nanoparticles per cell (the incubation molarity is $1.66{\cdot}10^{-10}$ *M*, shown in **Fig.3.5 (a)-(c)**), and another with a lower ratio of $4{\cdot}10^3$ nanoparticles per cell (the incubation molarity is $1.66{\cdot}10^{-12}$ M, shown in **Fig.3.5 (c)-(d)**).

We varied the number of ligands and the incubation concentration to study the binding kinetics. Under high-dose conditions, binding exceeding the nonspecific interaction was observed in formulations with 7, 20, and 27 ligands for M0 cells, 7 ligands for M1 cells, and again 7, 20, and 27 ligands for M2 cells. Under low-dose conditions, binding is enhanced by the mUNO-CD206 interaction in formulations with 6, 8, and 14 ligands for M0 cells, 6, 8, and 14 ligands for M1 cells, and again 8 and 14 ligands for M2 cells. The binding happens within the range of the onset ligand number and the offset ligand number.

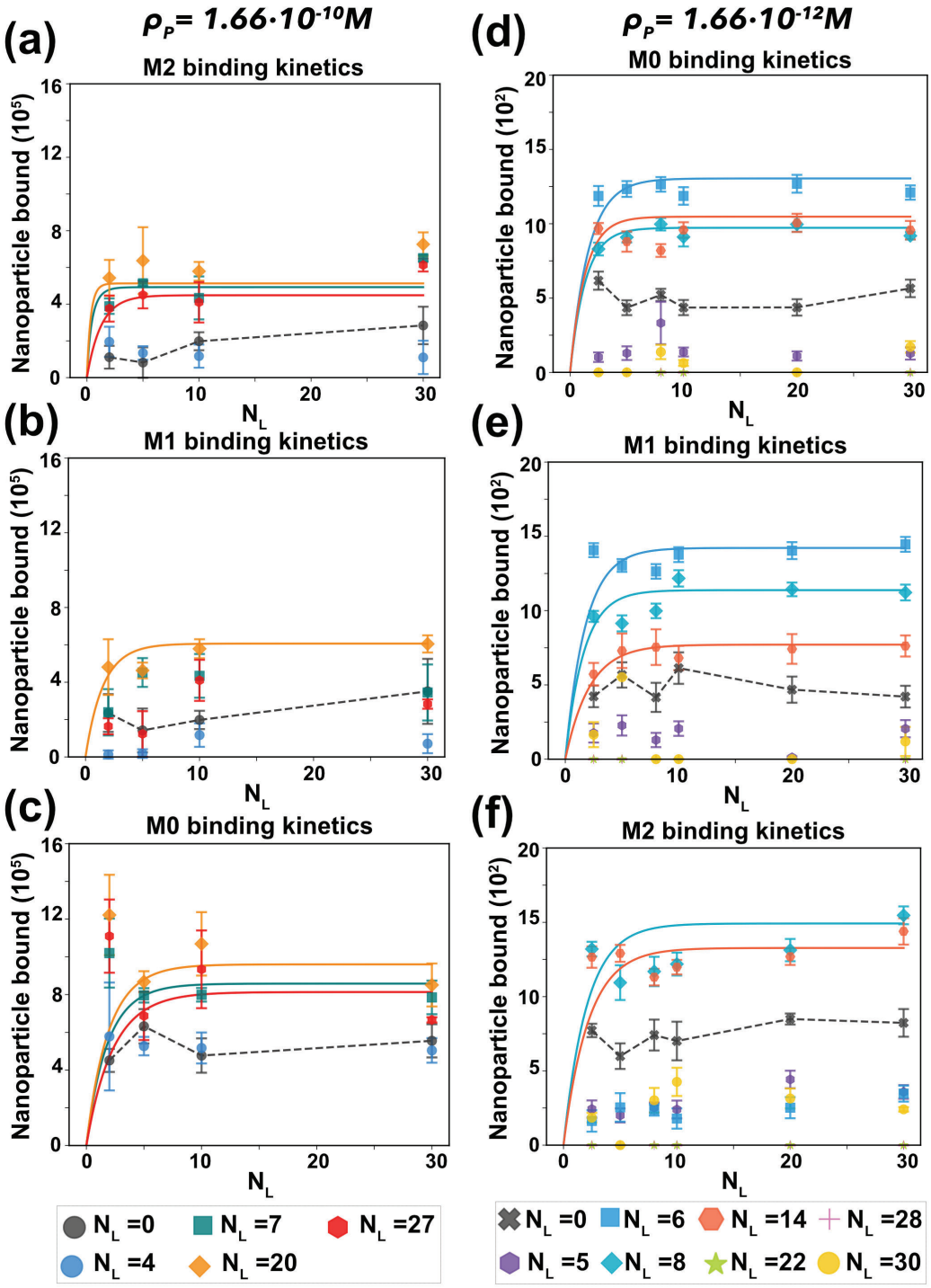

Figure 3.5 The binding kinetics between the mUNO-decorated PEG-PLG nanoparticle and the BMDM cells.

The uptake kinetics remained relatively flat, within the margin of error, suggesting that binding occurred without substantial internalization. The uptake kinetics were modeled using the same formulation as **Eq. (3.2)**, and the fitted kinetic curves are represented by lines in the corresponding colors, shown in **Fig. 3.5**. The fitting parameters are provided in **Table 2**. The interaction between mUNO-decorated nanoparticles and macrophages has a high dissociation rate and a short residence time. This short residence time implies that the nanoparticles did not remain bound long enough to be efficiently internalized via endocytosis. This confirms that internalization does not occur during the interaction, allowing the data to be accurately fitted without accounting for internalization.

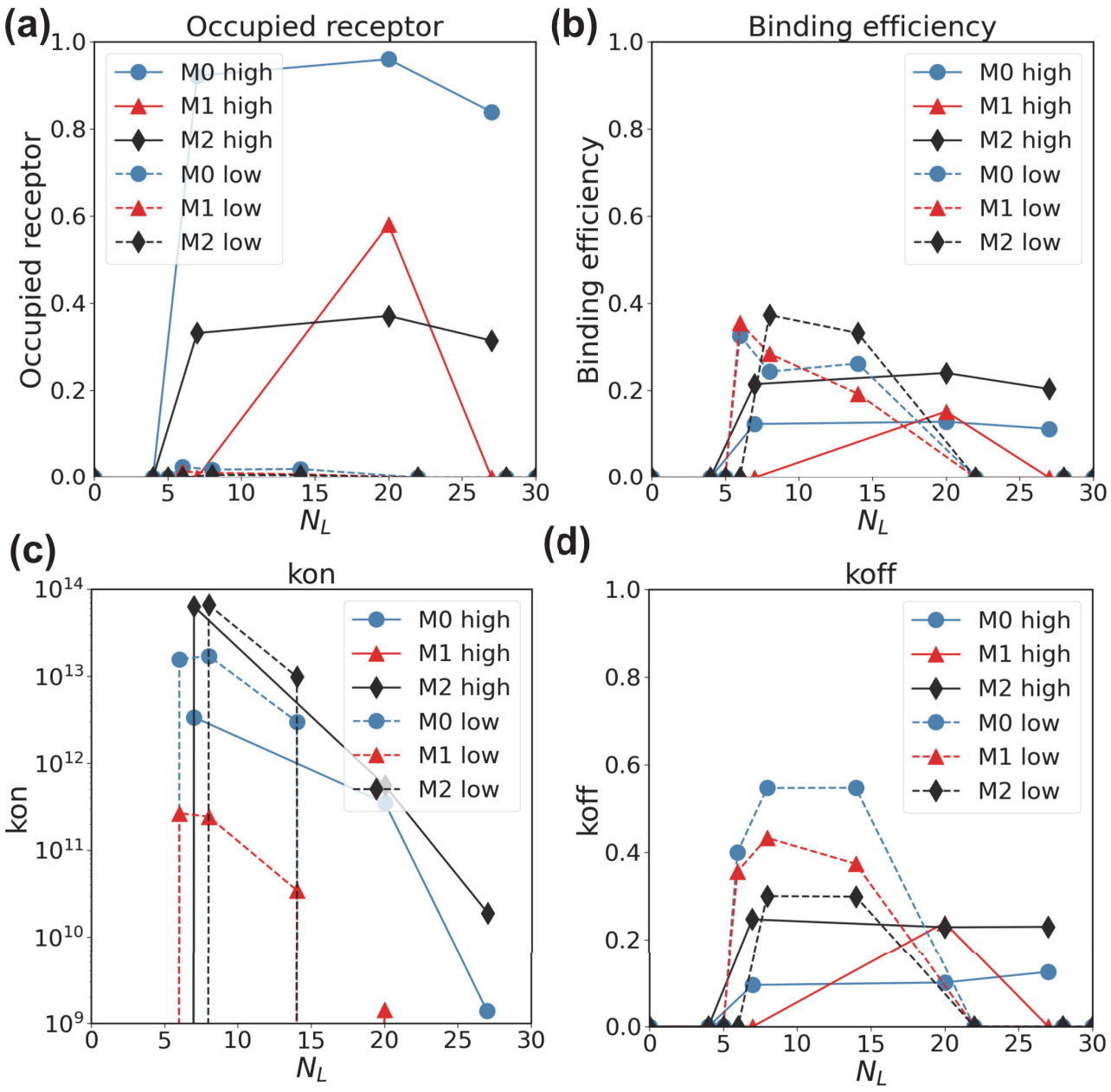

Figure 3.6 Analysis of the fitting results. (a)-(d) The occupied receptor ratio, binding efficiency, association rate (on-rate), and dissociation rate for experiments conducted at high ($1.66 \cdot 10^{-10}$ M) and low ($1.66 \cdot 10^{-10}$ M) incubation concentrations.

The mUNO receptor contains three potential binding sites. However, based on the fitted parameters, the reaction rate (valence) ranges from 0.9 to 1.6. In this binding scenario, the interaction occurs between a single nanoparticle and a single receptor. From the perspective of mass conservation, the valence does not imply that multiple receptors bind simultaneously to the same nanoparticle. Instead, it represents the average number of binding sites engaged on the receptor during the interaction.

As shown in **Table 3.2** and **Fig. 3.6 (c)-(d)**, the on-rate decreases as the number of ligands increases when targeting a single cell type. Among the three cell types, M2 cells exhibit the highest on-rate under the same ligand formulation, which may be attributed to their higher binding avidity. For a given cell type, the dissociation rates remain relatively consistent despite changes in the number of functional ligands.

Table 3.2: Parameters from the PAT and uptake kinetics model for mUNO-decorated nanoparticles

Cell type	Particle Molarity [*M*]	Ligand number	Occupied receptor	Valence	K_{av} (fit) [M^{-1}]	K_{av} (PAT) [M^{-1}]	k_{on} [$M^{-v} \cdot s^{-1}$]	k_{off} [s^{-1}]	Residence time (s)
M0	1.66e-10	7	0.92	1.228	3.5693e+13	3.6347e+13	3.3730e+12	0.09450	10.60
M0	1.66e-10	20	0.96	1.027	3.4954e+11	3.5037e+11	3.4954e+11	0.1000	10.00
M0	1.66e-10	27	0.84	0.956	1.1069e+10	1.1316e+10	1.3781e+09	0.1245	8.03
M1	1.66e-10	20	0.58	0.980	5.9831e+09	6.2331e+09	1.4294e+09	0.2389	4.19
M2	1.66e-10	7	0.33	1.476	2.5595e+14	2.5580e+14	6.3169e+13	0.2468	4.05
M2	1.66e-10	20	0.37	1.267	2.4260e+12	2.4658e+12	5.5483e+11	0.2287	4.37
M2	1.66e-10	27	0.31	1.136	8.2682e+10	7.9638e+10	1.8992e+10	0.2297	4.35
M0	1.66e-12	6	0.24	1.304	3.9034e+13	3.8807e+13	1.5614e+13	0.4000	2.50
M0	1.66e-12	8	0.018	1.312	3.1276e+13	3.1126e+13	1.7102e+13	0.5468	1.83
M0	1.66e-12	14	0.020	1.236	5.4738e+12	4.7572e+12	2.9959e+12	0.5473	1.83
M1	1.66e-12	6	0.013	1.170	7.4577e+11	6.9038e+11	2.6609e+11	0.3568	2.81
M1	1.66e-12	8	0.011	1.172	5.6178e+11	5.5373e+11	2.4330e+11	0.4331	2.31
M1	1.66e-12	14	7.37e-03	1.118	9.2653e+10	8.4632e+10	3.4638e+10	0.3739	2.67
M2	1.66e-12	8	5.76e-03	1.461	2.1993e+14	2.1905e+14	6.5978e+13	0.3000	3.33
M2	1.66e-12	14	5.12e-03	1.386	3.2779e+13	3.3480e+13	9.8304e+12	0.2999	3.34

In **Fig. 3.6**, we present the occupied receptor ratio, binding efficiency, on-rate, and dissociation rate for experiments conducted at different incubation concentrations. As predicted by the PAT model, the binding fractions approach 1. At high incubation concentrations, the occupied receptor ratio on M0 cells approaches 1, indicating that nanoparticles occupy most receptors. In contrast, for the other two cell types, the occupied receptor ratios range between 0.4 and 0.6. At low incubation concentrations, the occupied receptor ratios are below 0.05 across all cell types. This is because the number of nanoparticles per cell is significantly lower than the number of available receptors. Interestingly, the ratio of bound nanoparticles to the total number of incubated nanoparticles is higher under low incubation conditions than at high concentrations. This suggests that optimizing the incubation concentration could improve targeting efficiency.

Compared with the predicted binding avidity and the experimental binding outcomes, we observe a strong correlation between high ligand numbers and successful binding. When the number of ligands is fewer than five, the experimental results show poor binding performance. This may be due to the limited probability of lower-valency nanoparticles reaching and effectively engaging with the receptor binding sites. As a result, despite potentially high predicted avidity, these formulations do not perform well kinetically.

Upon lowering the incubation concentration, effective binding events shift toward lower ligand numbers in M1 cells, with more successful bindings observed in formulations containing approximately 6 ligands. This suggests that at higher concentrations, competitive effects between nanoparticles may hinder binding efficiency. The improved performance at lower concentrations highlights the influence of nanoparticle competition on binding kinetics, particularly in M1 cells.

3.3.2 A comprehensive consideration of the nanoparticle cell interaction

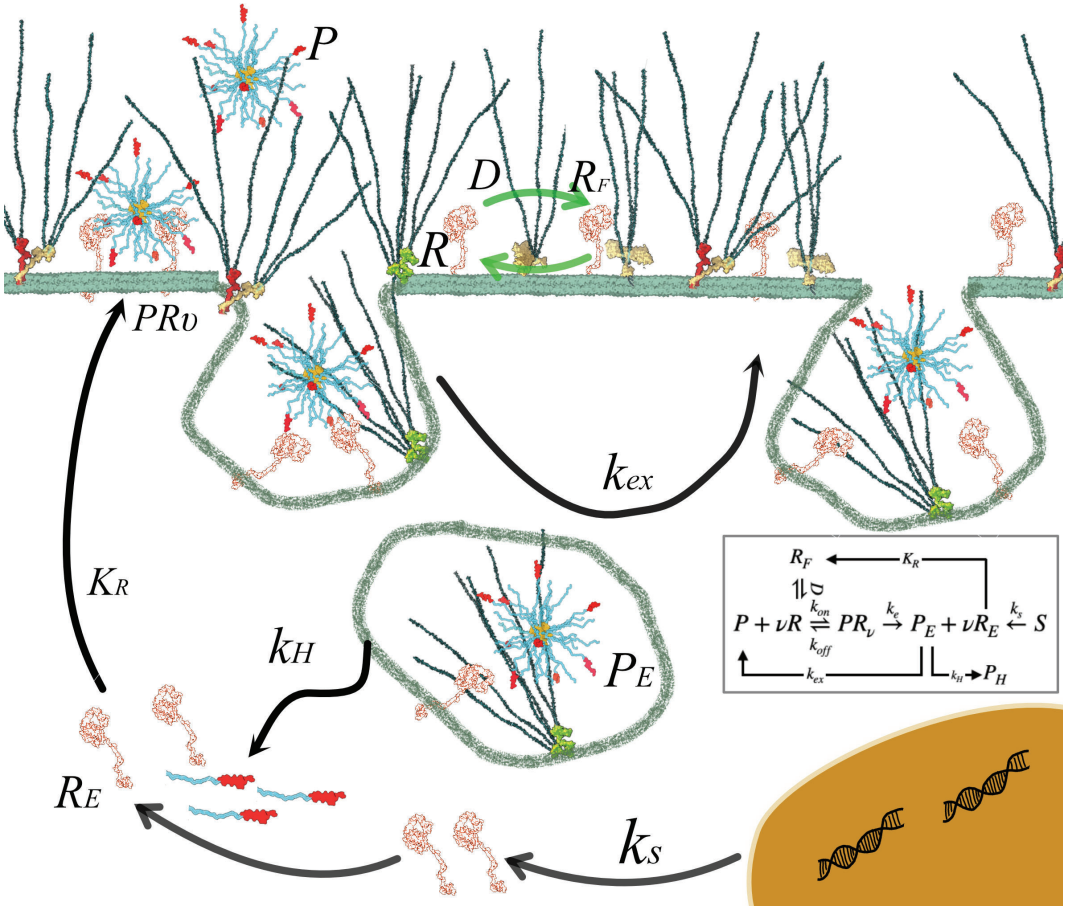

Figure 3.7 The schematic for the anoparticle-cell interactions. The insertion shows the process as a chemical reaction. The nanoparticle first overcomes steric hindrance and binds to receptors on the cell surface, forming a nanoparticle–receptor complex. This complex is then internalized via endocytosis, entering the cell within an early endosome, a membrane-bound compartment marked by Rab5 that functions as the primary sorting station. From the early endosome, nanoparticles may follow several intracellular routes: they can be recycled back to the plasma membrane via Rab11-positive recycling endosomes, transported to Rab7-positive late endosomes, or, in some cases, escape into the cytoplasm. As endosomes mature, they acidify and eventually fuse with lysosomes, where the nanoparticle is degraded by hydrolytic enzymes in the low pH environment. Receptors on the cell membrane are dynamic and mobile. They can diffuse across the membrane, engage in repeated binding events with nanoparticles, and undergo multiple rounds of internalization. Once internalized, receptors may be recycled back to the cell surface following exocytosis or nanoparticle degradation. Additionally, new receptors can be synthesized through gene expression to maintain receptor availability on the membrane.

Through the mathematical model discussed before, we can simulate the nanoparticle's journey from initial binding to endocytosis. However, several post-endocytic mechanisms, such as exocytosis and receptor degradation, are not accounted for in this model. To develop a more comprehensive understanding of nanoparticle-cell interactions, including membrane-level binding and intracellular trafficking, we represent the overall process as shown in the schematic **Fig. 3.7**.

The binding step can be experimentally estimated by labeling the cell membrane with fluorescent markers. As previously discussed, nanoparticles must first overcome steric barriers to bind with cell surface receptors-a process that can be predicted using PAT. However, as reported

by Zhaoping Xie *et al.*, receptor clustering also plays a critical role in enhancing multivalent binding[91]. This phenomenon is particularly common in certain cell types, such as T cells, where clustered receptors significantly increase binding valency. To account for this effect, we introduce a receptor diffusion parameter, denoted as D in the schematic, which reflects the dynamic clustering behavior. However, in some other cell types, the receptor diffusion rate is relatively slow and can therefore be considered negligible. Whether to introduce the diffusion parameter or not depends on the real receptor diffusion rate.

We have discussed the uptake kinetics model, which includes both the binding of nanoparticles to the cell membrane and their subsequent endocytosis. Once internalized, nanoparticles are encapsulated within early endosomes. In the case of PMPC nanoparticles, they often escape into the cytoplasm before the early endosome matures. However, nanoparticles may also follow alternative intracellular pathways. One possibility is exocytosis, in which nanoparticles are recycled back to the cell membrane through the formation of exosomes. This exocytic process is represented by the rate constant k_{ex} in the schematic. Alternatively, as endosomes mature, they acidify and fuse with lysosomes, where nanoparticles are degraded by hydrolytic enzymes in the acidic environment. This degradation process is characterized by the hydrolysis rate k_h.

Endocytic receptors are recycled back to the cell membrane after internalization, allowing them to participate in nanoparticle uptake repeatedly. The recycling rates vary among different cell types[166-170]. This recycling process should be incorporated into kinetic models, as it helps explain non-linear or unexpected increases in binding behavior.

In addition to recycling, receptors on the cell surface can also be newly synthesized within the cell through gene expression, a process that begins in the nucleus and proceeds through transcription, translation, and trafficking to the plasma membrane. Sometimes, cells upregulate receptor synthesis, resulting in higher receptor density on the membrane. This increased expression enhances nanoparticle binding and facilitates more efficient internalization.

In summary, the entire nanoparticle uptake process includes membrane binding, receptor diffusion, nanoparticle endocytosis, intracellular trafficking, exocytosis, nanoparticle degradation, receptor recycling, and receptor synthesis. By accounting for all these steps, a more comprehensive mathematical model can be developed to more accurately capture the complex intracellular journey of nanoparticles:

$$
\begin{aligned}
\frac{\partial \rho_P}{\partial t} &= -k_{on}\rho_P\rho_R^{\nu} + k_{off}\rho_{PR_\nu} + k_{ex}\rho_{P_E} \\
\frac{\partial \rho_{R_F}}{\partial t} &= K_R\rho_{R_E} - D\rho_{R_F} + D\rho_R \\
\frac{\partial \rho_{PR_\nu}}{\partial t} &= k_{on}\rho_P\rho_R^{\nu} - k_{off}\rho_{PR_\nu} - k_e\rho_{PR_\nu} \\
\frac{\partial \rho_R}{\partial t} &= -k_{on}\rho_P\rho_R^{\nu} + k_{off}\rho_{PR_\nu} - D\rho_{R_F} + D\rho_R \qquad (3.6) \\
\frac{\partial \rho_{P_E}}{\partial t} &= k_e\rho_{PR_\nu} - k_{ex}\rho_{P_E} - k_H\rho_{P_E} \\
\frac{\partial \rho_{R_E}}{\partial t} &= -K_R\rho_{R_E} + k_e\rho_{PR_\nu} + f(s) \\
\frac{\partial \rho_{P_H}}{\partial t} &= k_H\rho_{P_E}
\end{aligned}
$$

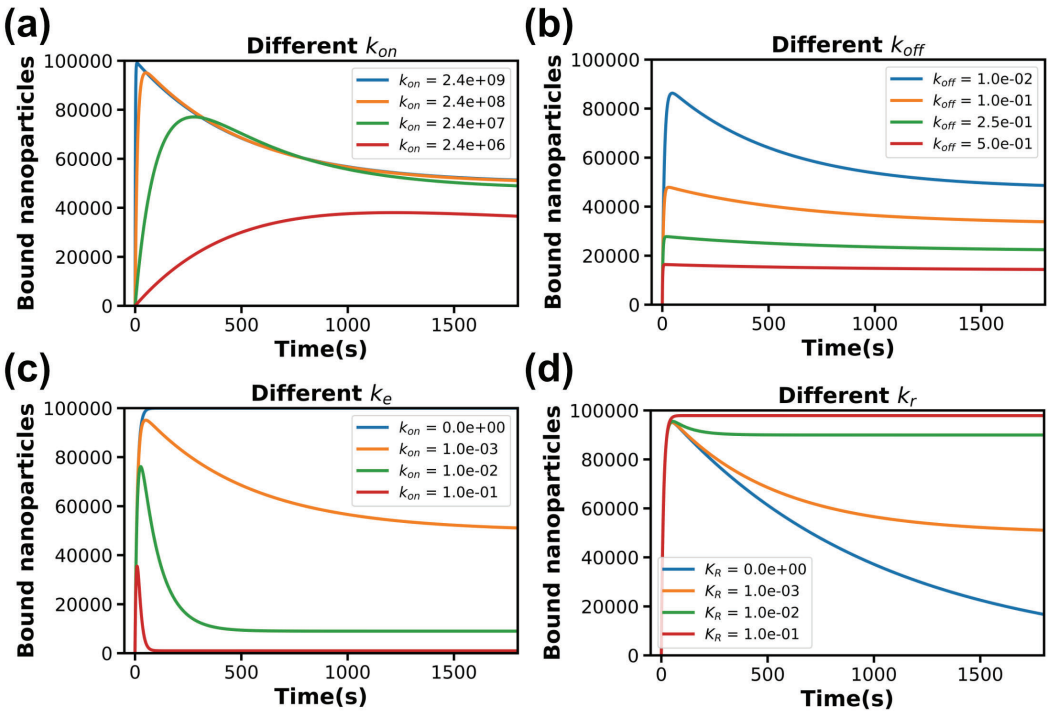

Figure 3.8 The binding kinetics with different parameters. (a)-(d) The bound nanoparticle with different k_{on}, k_{off}, k_e and K_R, respectively.

where $f(s)$ is the receptor synthesis function. These complex equations require additional datasets to provide a more precise understanding of nanoparticle-cell interactions. Intracellular processes may follow multiple pathways, and by comparing the values of key parameters, we can determine which routes contribute most significantly to the overall intracellular behavior.

3.3.3 Theoretical analysis.

We varied different parameters to investigate how distinct receptor properties influence the kinetics. In **Fig. 3.8**, we present the number of bound nanoparticles under varying on-rates, dissociation rates, endocytosis rates, and receptor recycling rates. When adjusting one parameter, all other parameters were held constant.

As the on-rate increases, a sharp rise in the number of bound nanoparticles is observed. Specifically, in **Fig. 3.8(a)**, the bound nanoparticle count reaches a maximum at very high on-rates, indicating that nearly all receptors are occupied. In contrast, at low on-rates, the number of bound nanoparticles does not reach saturation. This phenomenon highlights the critical role of the on-rate in determining binding efficiency.

We varied the dissociation rate and present the results in **Fig. 3.8(b)**. It is evident that higher dissociation rates lead to a lower maximum number of bound nanoparticles compared to lower dissociation rates. As previously discussed, a lower dissociation rate corresponds to a longer residence time and improved drug efficacy. These findings underscore the importance of designing drugs with low dissociation rates to enhance therapeutic performance.

A higher endocytosis rate is desirable for drugs designed for intracellular delivery. With an increased endocytosis rate, nanoparticles are less likely to remain on the cell membrane. Instead, they are more rapidly internalized, leading to a reduction in the number of membrane-bound nanoparticles, as shown in **Fig. 3.8(c)**. In the absence of endocytosis, the number of bound nanoparticles is primarily governed by surface binding interactions.

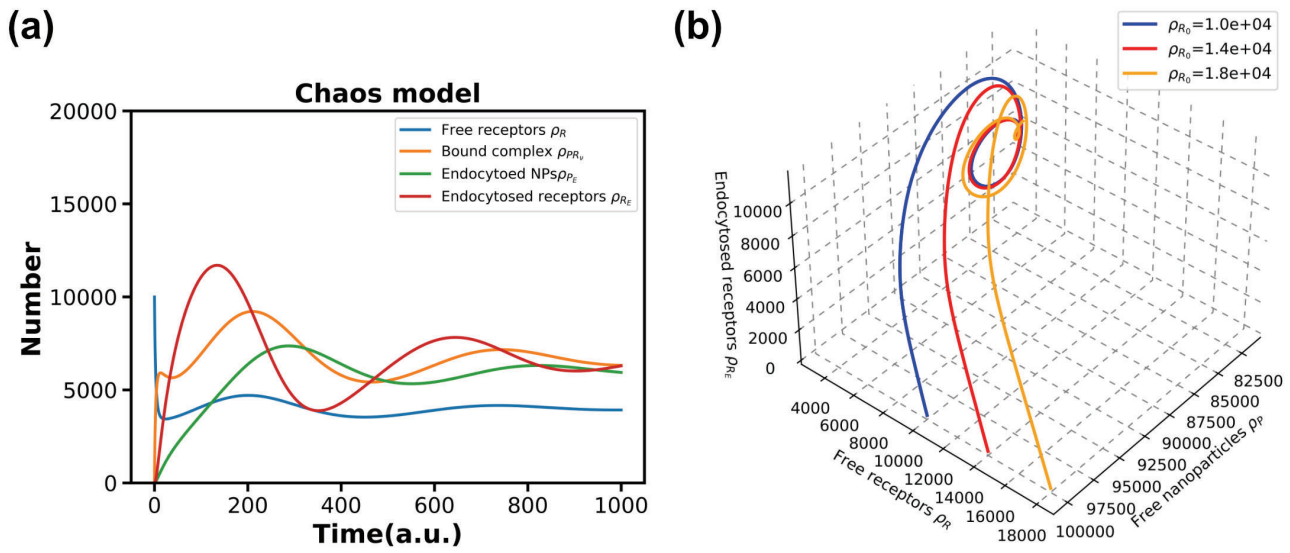

Figure 3.9 The simulation of the chaos model. (a) Predictions for the concentrations of free receptors, bound complexes, endocytosed nanoparticles, and endocytosed receptors. **(b)** Phase plots showing the dynamics of free receptors, free nanoparticles, and endocytosed receptors under varying initial free receptor levels.

Different receptor types exhibit varying recycling rates. In **Fig. 3.8(d)**, we present the number of bound nanoparticles under different receptor recycling rates. At higher recycling rates, the number of bound nanoparticles remains elevated, indicating that recycled receptors are reused efficiently to facilitate multiple rounds of nanoparticle uptake.

By analyzing these parameters, we gain a clearer understanding of how each one influences the uptake kinetics. This enables us to identify the key factors that can be optimized to enhance nanoparticle uptake. Overall, the model provides valuable guidance for the rational design of nanoparticles and more effective drug formulations.

3.3.4 A chaos model in the uptake kinetics

As nanoparticles bind to receptors on the cell membrane or are internalized via endocytosis, these interactions are often linked to receptor activation or downstream signaling pathways such as transcription, which can lead to increased receptor expression. At the same time, the ongoing endocytosis reduces receptor density on the cell surface. In experiments, this dynamic interplay often results in oscillations in surface receptor density. To explain this phenomenon, we developed a model to describe receptor oscillation:

$$
\begin{aligned}
\frac{\partial \rho_P}{\partial t} &= -k_{on}\rho_P\rho_R^{\nu} + k_{off}\rho_{PR_\nu} + k_{ex}\rho_{P_E} \\
\frac{\partial \rho_R}{\partial t} &= -k_{on}\rho_P\rho_R^{\nu} + k_{off}\rho_{PR_\nu} + k_{ex}\rho_{P_E} + K_R\rho_{R_E} \\
\frac{\partial \rho_{PR_\nu}}{\partial t} &= k_{on}\rho_P\rho_R^{\nu} - k_{off}\rho_{PR_\nu} - k_e\rho_{PR_\nu} \\
\frac{\partial \rho_{P_E}}{\partial t} &= k_e\rho_{PR_\nu} - k_{ex}\rho_{P_E} - k_H\rho_{P_E} \\
\frac{\partial \rho_{R_E}}{\partial t} &= k_e\rho_{PR_\nu} - K_R\rho_{R_E} + k_s\frac{1-(\rho_R+\rho_{P_E})}{\rho_{R_0}}\rho_{PR_\nu}
\end{aligned}
\tag{3.7}
$$

here, we ignore the receptor diffusion on the cell membrane and define the receptor synthesis function as: $f(s) = k_s \frac{1-(\rho_R+\rho_{P_E})}{\rho_{R_0}} \rho_{PR_\nu}$, with k_s being the receptor synthesis rate. The bound nanoparticle facilitated the synthesis of the receptor. However, the free receptor on the membrane and endocytosis would hinder the fabrication of the receptor to avoid overexpression of the receptors.

In **Fig. 3.9**, we present simulation results from the kinetic model. As shown in **Fig. 3.9(a)**, the density of the bound complexes exhibits oscillatory behavior over time. This dynamic closely aligns with the receptor density oscillations observed experimentally. In **Fig. 3.9(b)**, we display the corresponding phase plots under different initial concentrations. These results demonstrate that the oscillatory behavior is sensitive to the initial receptor concentration, where variations in starting conditions lead to significant differences in the amplitude and pattern of the oscillations. This emphasizes the importance of controlling the initial concentration in the chaos system.

We use the Lyapunov coefficient of the model to validate the model. The calculated Lyapunov coefficients exceed 1, indicating that the system exhibits chaotic dynamics. In dynamical systems theory, a Lyapunov exponent quantifies the rate of separation of infinitesimally close trajectories. When the largest Lyapunov exponent is greater than zero (or specifically greater than 1 in normalized conditions), it implies that small perturbations in the initial conditions lead to exponentially diverging outcomes over time. This sensitivity to initial conditions is a hallmark of chaotic systems, meaning that while the system is deterministic, its behavior becomes unpredictable in the long term. In our model, this suggests that the receptor–nanoparticle interaction dynamics are highly sensitive to initial conditions and parameter variations, which may explain the oscillatory or irregular behavior observed experimentally.

3.4 Perspective for integrating the PAT and kinetics model

We introduced an approach to integrate the PAT framework with a kinetics model. This integration allows us to quantify key kinetic parameters such as the on-rate, dissociation rate, and endocytosis rate. In the multivalent kinetics model, valency plays a critical role-it not only influences the reaction rate but also determines how many receptors or binding sites are occupied during multivalent interactions. Importantly, valency significantly enhances the overall binding avidity, making it a crucial factor in the design of efficient nanoparticle-based drug delivery systems.

By fitting the uptake kinetics using the integrated model, we are able to explain experimental observations of nanoparticle behavior. For instance, the tube-like PMPC nanoparticles got stuck on the cell membrane due to their prolonged residence time and low endocytosis rate. These quantified parameters offer deeper insight into the mechanisms underlying the observed uptake kinetics.

By quantifying key parameters, we can observe how changes in nanoparticle design influence kinetic behaviors. One of our primary goals is to enable the prediction of kinetic outcomes during the nanoparticle design phase. Through this modeling framework, we aim to uncover the underlying mechanisms of nanoparticle–cell interactions, thereby deepening our understanding of the factors that govern cellular uptake kinetics. This insight could ultimately guide the rational design of more effective nanomedicines.

We also aim to extend our model to capture more complex biological systems. Some receptors are capable of initiating intracellular signaling, thereby regulating cellular behavior. To account for dynamic receptor responses, we introduced a chaos-based model to explain receptor

oscillations. Our framework has the potential to mimic more intricate biological phenomena and provide deeper insights into nanoparticle-cell interactions.

Chapter 4: PBPK optimizing dosing strategy

4.1 PBPK model for non-specific interactions

4.1.1 Nanoparticles with varying methionine (MET) polymerization

Methionine (MET), a natural amino acid, offers several advantages as a building block in polymeric nanoparticles, particularly when incorporated into PEG-b-PMET copolymers[171-174]. Its inherent biocompatibility and biodegradability ensure that the resulting materials degrade into non-toxic products, making them suitable for biomedical applications. A key benefit of MET is its redox-responsiveness, owing to the thioether group in its side chain, which can be selectively oxidized in reactive oxygen species (ROS)-rich environments, enabling controlled and stimuli-responsive drug release. Additionally, MET's hydrophobicity can be finely tuned by adjusting the degree of polymerization or oxidation state, allowing precise control over nanoparticle self-assembly. This tunability supports the formation of diverse nanoparticle morphologies, including spherical micelles, worm-like micelles, and vesicles, which significantly influence circulation time, cellular uptake, and biodistribution. Furthermore, the sulfur-containing side chains of MET offer chemical modifiability, allowing for further functionalization to achieve targeted delivery or enhanced responsiveness. Collectively, these properties make MET a highly versatile and valuable component in the design of a smart drug delivery system.

Chlorin e6 (Ce6) is a second-generation photosensitizer widely used in photodynamic therapy (PDT) due to its strong absorption in the near-infrared (NIR) region, which allows for deeper tissue penetration and efficient generation of ROS upon light activation[175,176]. Derived from chlorophyll, Ce6 exhibits excellent photophysical properties, including a high singlet oxygen quantum yield and photostability[177]. Its amphiphilic nature enables it to interact with various nanocarriers, improving its solubility, stability, and cellular uptake. When encapsulated within nanoparticles, Ce6 can achieve enhanced tumor accumulation via the EPR effect, leading to more effective and localized ROS generation under light irradiation. These features make Ce6 a promising therapeutic agent for cancer treatment, especially when integrated into stimuli-responsive nanocarrier systems that enable controlled delivery and release in response to specific biological cues.

Nanoparticle morphology plays a crucial role in nanomedicine, particularly in influencing biological responses. Here, we synthesized nanoparticles with different topologies, spherical micelles, worm-like micelles, and vesicles, by varying the degree of MET polymerization[178]. More exactly, PEG_{5k}-$PMET_{5/40/120}$ self-assembles into spherical micelles, worm-like micelles, and vesicles, respectively. All nanoparticles were loaded with the photosensitizer chlorin e6 (Ce6). The resulting topologies are illustrated in **Fig. 4.1**. In **Fig. 4.1(a)**, we show the synthetic route of PEG_{5k}-$PMET_{5/40120}$ and its self-assembly into spherical micelles, worm-like micelles, and vesicles, controlled by the molecular weight of the PMET block. We took the TEM image of PEG_{5k}-$PMET_{5/40120}$ nanoparticles in both the solid phase and liquid phase. The TEM image of Ce6@PEG_{5k}-$PMET_{5/40120}$ nanoparticle in solid phase is shown in **Fig. 4.1(d)**. The hydrodynamic diameters of micelles and vesicles were measured using dynamic light scattering (DLS) (**Fig. 4.1(e)**). Stability studies demonstrated that the particle sizes of the formulations remained relatively stable over 72 hours in aqueous solution, PBS, and 10% serum, indicating good colloidal stability and supporting the reliability of subsequent biological experiments. Fluorescence spectra (**Fig. 1(f)**) and UV-visible absorption spectra (**Fig. 4.1(g)**) were also collected to characterize both PEG_{5k}-$PMET_{5/40120}$ and Ce6-loaded PEG_{5k}-$PMET_{5/40120}$. The results confirmed that the blank carrier exhibited no overlapping fluorescence with the Ce6-loaded

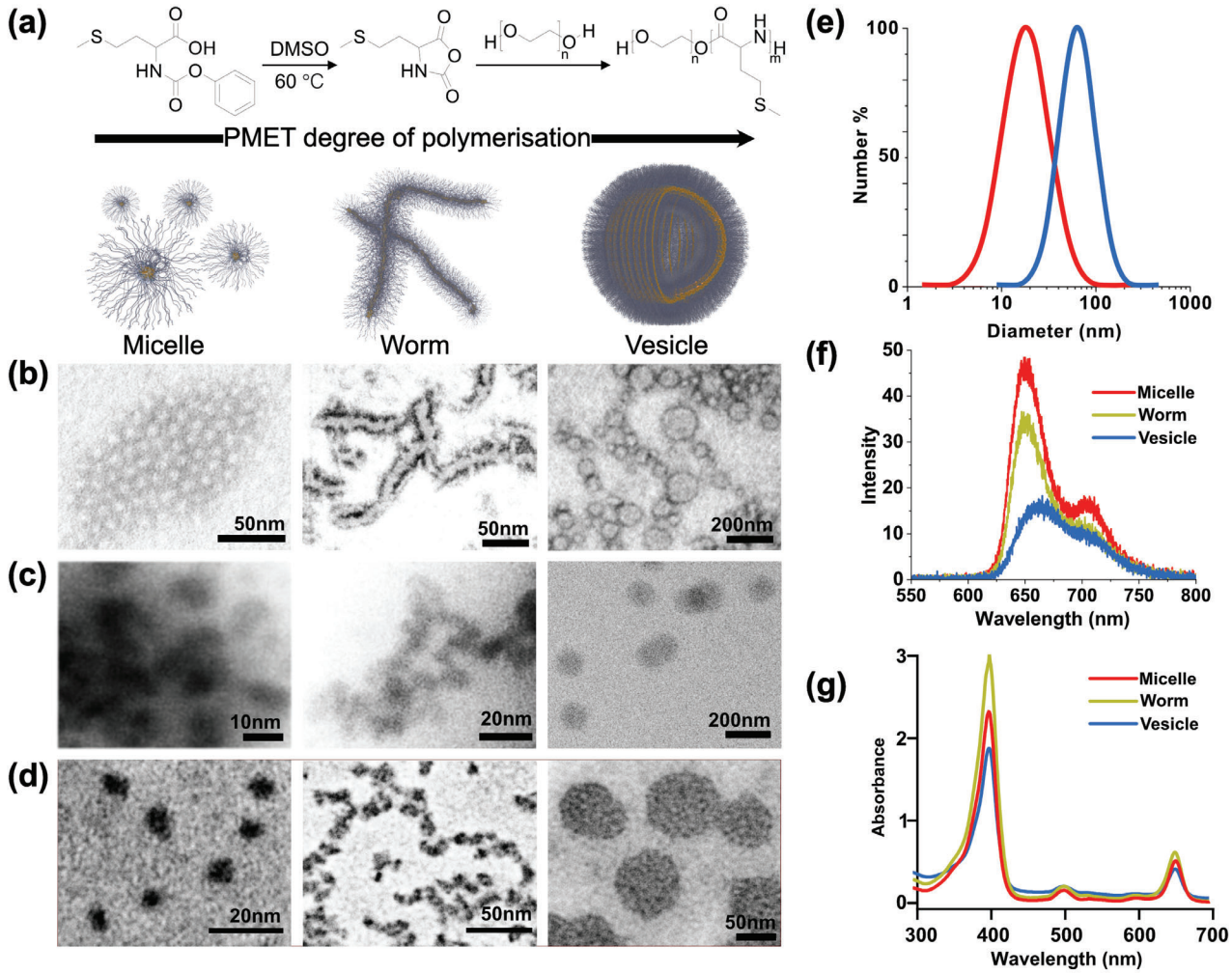

Figure 4.1. Synthesis and Characterization of Ce6@PEG$_{5k}$-PMET$_{5/40/120}$ nanoparticles. (a) Synthetic route of PEG$_{5k}$-PMET$_{5/40120}$ and its self-assembly into spherical micelles, worm-like micelles, and vesicles, controlled by the molecular weight of the PMET block. **(b)** TEM images of PEG$_{5k}$-PMET$_{5/40120}$ nanoparticles in the solid state. (c) TEM images of PEG$_{5k}$-PMET$_{5/40120}$ nanoparticles in the liquid phase. **(d)** TEM images of Ce6@PEG$_{5k}$-PMET$_{5/40120}$ nanoparticles in the solid state. **(e)** DLS size distribution of micelles and vesicles. **(f)** Fluorescence spectra of loaded Ce6@PEG$_{5k}$-PMET$_{5/40/120}$. **(g)** Absorption spectra of loaded Ce6@PEG$_{5k}$-PMET$_{5/40/120}$.

nanoparticles, validating the feasibility of using fluorescence-based methods in biological assays.

4.1.2 Pharmacokinetics of Ce6@nanoparticles in tumor-bearing mice

HepG2 tumor-bearing mice were established by subcutaneously inoculating 8·10^6 HepG2 cells into 4–6-week-old nude mice. Once the tumors reached approximately 100 mm^3 in BALB/c mice, Ce6-loaded nanoparticles were intravenously injected at a Ce6 dose of 5 mg/kg. Following administration, fluorescence signals were monitored using a multimodal *in vivo* imaging system (AniView100, Guangzhou Biolight Biotechnology, Guangzhou, China). Mice were euthanized at 3 h, 6 h, 24 h, and 48 h post-injection, and major organs, including the heart, liver, spleen, lung, kidney, and tumor tissue, were collected for *ex vivo* imaging.

Following injection, fluorescence rapidly spread throughout the body, with nanoparticles gradually accumulating at tumor sites (**Fig. 4.2(A)–(F)**). At 3 hours post-injection, spherical micelles were predominantly found in the kidneys, liver, and tumor, while worm-like micelles

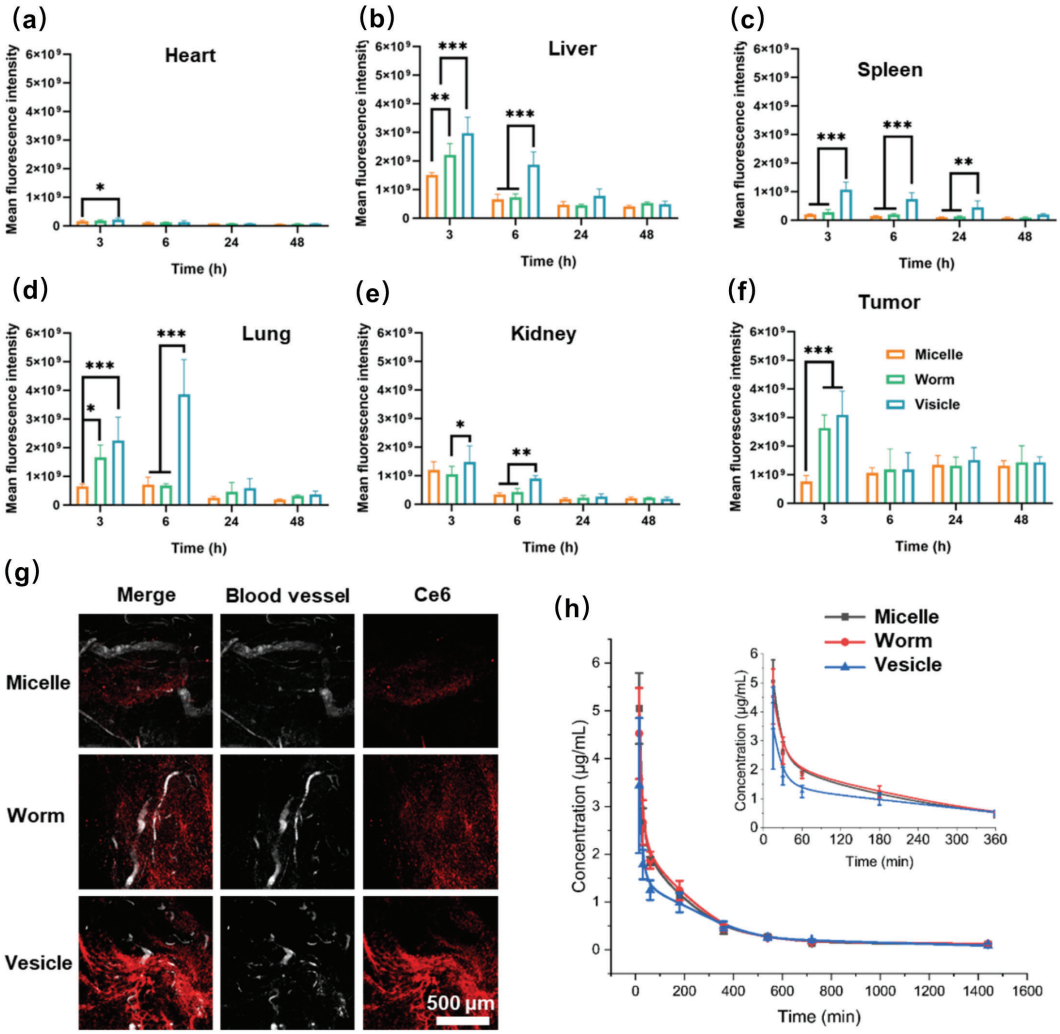

Figure 4.2 Distribution of nanoparticles in major organs and tumor sites and pharmacokinetics. (a-f) Average fluorescence statistics of tumor and major organs (heart, liver, spleen, lung, kidney) *ex vivo*. **(g)** Ex vivo CLSM images for harvested tumors at an equivalent Ce6 dose of 5 mg/kg for 3 hours. Tumor vessels were labeled by FITC-dextran (white), and the red signal was from Ce6. **(h)** In vivo plasma Ce6 concentration after intravenous injection of nanoparticles (the concentration of Ce6 at 0.5 mg/mL).

were primarily distributed in the liver, lungs, and tumor. Vesicles were present in all major organs except the heart. By 6 hours, spherical micelles remained concentrated in the liver and tumor, while their presence in the kidneys declined. The overall organ distribution of worm-like micelles and vesicles decreased. At 24 hours, both spherical micelles and worms showed predominant accumulation at the tumor site with minimal presence in other organs. Vesicles were also localized mainly in the tumor, although some residual distribution was noted in the liver and lungs. By 48 hours, all three nanoparticle types exhibited strong tumor accumulation with minimal distribution in non-target organs. Notably, while spherical micelles and worm-like micelles showed distinct organ distribution patterns early on, their profiles converged over time, likely due to structural changes in the worm-like micelles caused by biological factors.

To visualize tumor vasculature, blood vessels were labeled using FITC-dextran. After 3 hours, blood vessels were labeled with FITC-conjugated dextran (50 mg/kg, 2000 kDa, Cat#FD2000S, Sigma-Aldrich), injected into the tail vein 15 minutes before tumor excision. Tumors were then

harvested, washed with PBS, and placed in 35-mm glass-bottom dishes for confocal imaging. Confocal laser scanning microscopy of tumor tissue confirmed the in vivo distribution patterns, with Ce6 fluorescence indicating nanoparticle localization (**Fig. 4.2(g)**).

The molar concentration of Ce6 in blood over time was measured. Normal BALB/c mice (18–22 g) were randomly divided into three groups, with five mice in each group. Ce6-loaded nanoparticles (Ce6 dose: 5 mg/kg) were administered via tail vein injection. Blood samples (20 μL) were collected from the fudus venous plexus at predetermined time points: 15 min, 30 min, 1 h, 3 h, 6 h, 9 h, 12 h, and 24 h post-injection. Each blood sample was collected into an EP tube pre-filled with 80 μL of ultrapure water and stored at –20 °C. Blank blood samples were also collected as controls. The collected blood samples were freeze-dried for further analysis. To each dried sample, 100 μL of DMSO was added, followed by ultrasonication and incubation overnight at 4 °C. After vortexing for 1 minute, the samples were centrifuged at 14,000 rpm for 10 minutes. The resulting supernatant was transferred to a black 96-well plate for fluorescence measurement. A standard curve was prepared by spiking Ce6 into blank plasma and processing it in the same manner. The fluorescence intensity of Ce6 in the supernatant was measured using a multifunctional microplate reader (Ex=400 nm, Em=660 nm). Worm-like micelles demonstrated the longest circulation half-life (563.29 min), followed by vesicles (427.88 min) and spherical micelles (313.41 min), as shown in **Fig. 4.2(h)**. This highlights the influence of nanoparticle morphology on circulation time and biodistribution.

4.1.3 PBPK model of Ce6@nanoparticles in tumor-bearing mice

As discussed before, Ce6@nanopaticles have no specific interaction with the receptors on the cell surface and mainly exploit the EPR effect to achieve enhanced tumor accumulation. To better understand the biodistribution, clearance, and overall efficiency of Ce6@nanoparticles, we employed a first-order differential equation to model the opposing processes of influx and efflux, forming the basis of a PBPK model grounded in systemic blood circulation[126-128,179,180]. In the PBPK model, each organ is treated as an individual compartment. In this study, the body was divided into seven compartments: plasma, tumor, kidney, spleen, liver, lung, and heart. The overall compartmental schematic is presented in **Fig. 4.3(a)**, while the detailed distribution within the tumor is illustrated in **Fig. 4.3(b)**. Plasma serves as the central transport medium, delivering nanoparticles to various organs. The PBPK model used in this study is a generalized framework based on systemic blood circulation dynamics. The mass balance in the plasma of the nanoparticle is described as:

$$V_P \frac{\partial \rho_P}{\partial t} = \sum (-V_P k_{P-i} \rho_P + V_i k_{i-P} \rho_i) \qquad (4.1)$$

The equation is the same as described in **Eq.(1.13)**. As discussed previously, the mass of nanoparticles in organs with fenestrations but without active uptake can be expressed as (same with **Eq.(1.14)**):

$$V_i \frac{\partial \rho_i}{\partial t} = V_P k_{P-i} \rho_P - V_i k_{i-P} \rho_i \qquad (4.2)$$

The glomerular filtration rate (GFR) is a key physiological parameter that quantifies how efficiently the kidneys filter blood, typically measured in milliliters per minute. In the context of nanoparticle clearance, GFR determines the extent to which small nanoparticles can be directly removed from the plasma via renal filtration. This filtration process significantly influences the nanoparticles' circulation time and overall biodistribution. Consequently, PBPK models incorporate GFR to estimate renal clearance rates, particularly for nanoparticles small enough to pass through the glomerular barrier without substantial binding to plasma proteins or reabsorption in the renal tubules. Here, we can write the function as:

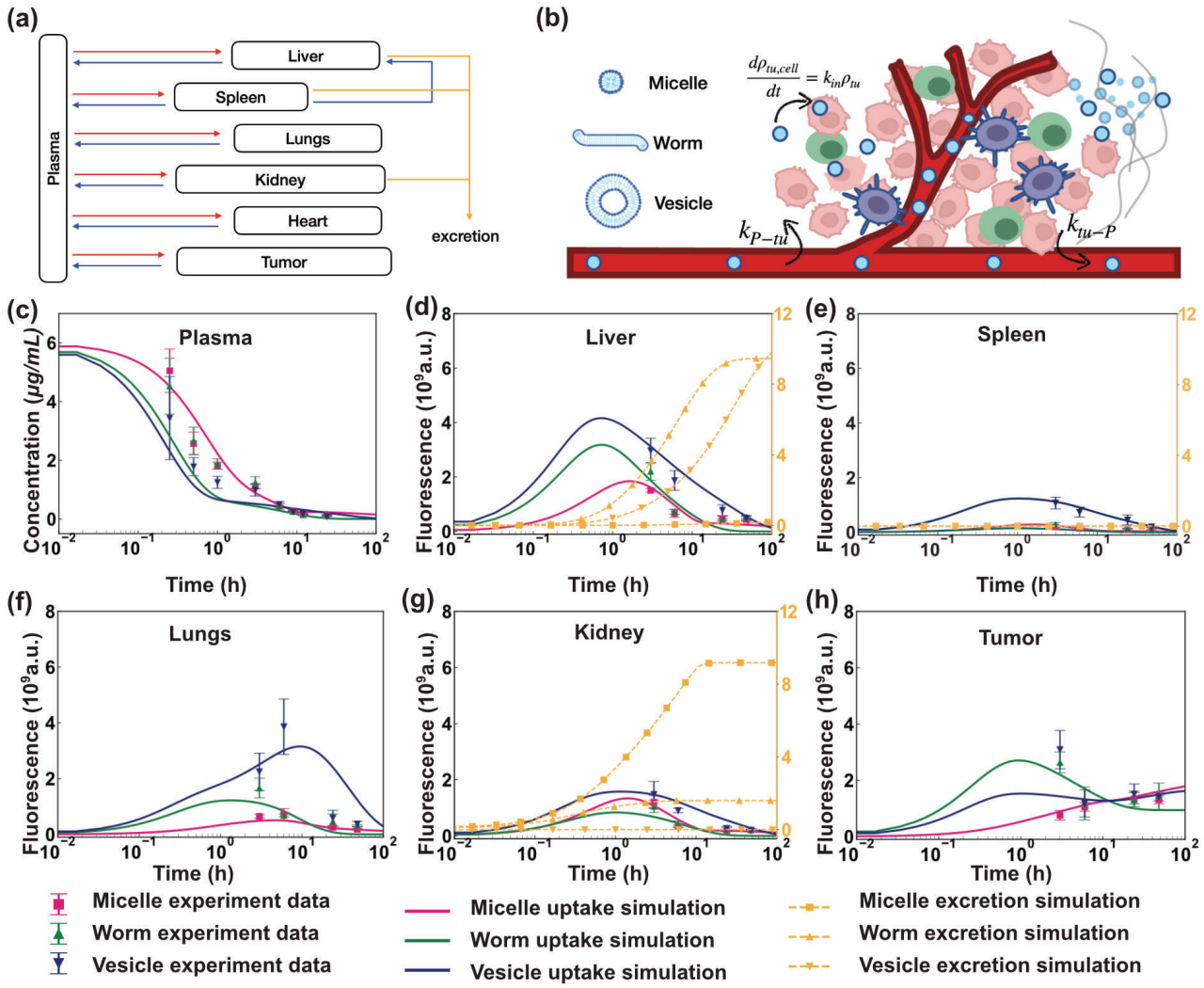

Figure 4.3. PBPK Model Based on Blood Circulation. (a) Schematic diagrams of the PBPK model illustrate systemic blood circulation following tail vein injection of Ce6@nanoparticles. **(b)** Schematic diagrams of Ce6@nanoparticles pharmacokinetics within the tumor. **(c)-(h)** The fitting results for Ce6@nanoparticles fluorescence changes in plasma, fluorescence uptake in the tumor, lungs, liver, spleen, and kidney, and fluorescence clearance in the liver, spleen, and kidney.

$$\frac{dU}{dt} = k_{GFR} V_P \rho_P \qquad (4.3)$$

with k_{GFR} being the GFR and U being the nanoparticle mass in the urine.

The reticuloendothelial system (RES) also contributes to the clearance of Ce6@nanoparticles, primarily through phagocytic activity in the liver and spleen. Thus, we write the concentration in the liver and spleen as:

$$V_i \frac{d\rho_i}{dt} = k_{P-i}\rho_P V_P - (k_{i-P} + k_{cl,i})\rho_i V_i \qquad (4.4)$$

where $k_{cl,i}$ refers to the clearance rate in the liver or spleen.

Ce6@NPs are capable of penetrating tumor cells and exerting therapeutic effects. The adsorption efficiency of Ce6@nanoparticles within tumor tissue is a critical determinant of therapeutic efficacy. The accumulation of Ce6@nanoparticles in the tumor is described by the following equation:

$$\frac{d\rho_{tu,cell}}{dt} = k_{pen}\rho_{tu} \qquad (4.5)$$

with k_{pen} being the penetration rate.

As previously described, Ce6@nanoparticles concentration data in plasma and fluorescence data from various organs were collected. However, these datasets were in different units. To address this, we treated parameters such as initial Ce6@nanoparticles fluorescence, influx rate, efflux rate, glomerular filtration rate, and clearance rate as adjustable variables. Simulations were performed using varying parameter values. Both the simulation and experimental plasma concentration data were normalized into dimensionless ratio forms for comparison. The Pearson product-moment correlation coefficient was used to assess the accuracy and reliability of the model. Particle swarm optimization and some other optimization methods were employed to explore the parameter space and obtain parameter values. The fitting results using the PBPK model are shown in **Fig. 4.3 (c)-(h)**.

The PBPK model allows us to infer the primary clearance pathways of Ce6@nanoparticles. Micelles, which had exhibited the longest circulation time, were primarily eliminated through renal excretion, as shown in **Fig. 4.3(g)**. In contrast, larger Ce6@nanoparticles (vesicles) were more readily sequestered and cleared by the RES, as illustrated in **Fig. 4.3(d)**. Worm-like Ce6@nanoparticles exhibited intermediate clearance characteristics, influenced by both their relatively small size and unique elongated shape. Some of the smaller worms still underwent renal clearance, while the larger fraction was mainly eliminated via hepatic filtration. Both the liver and spleen contributed to RES-mediated clearance, but the liver played a dominant role. Anatomically, the liver is larger and receives a greater share of cardiac output than the spleen. Additionally, it filters not only arterial blood but also blood from the portal vein, allowing it to process a larger volume of circulating Ce6@nanoparticles. These anatomical and physiological factors accounted for the liver's higher Ce6@nanoparticles clearance efficiency compared to the spleen.

Among all Ce6@nanoparticle topologies, vesicles showed the highest tendency to accumulate in non-target organs. As illustrated in **Fig. 4.3(d)** and **(f)**, vesicles predominantly accumulated in the liver and lungs, which was largely attributed to their larger size. Additionally, their distribution in the spleen and kidneys was also higher compared to micelles and worm-like micelles. In contrast, micelles, due to their small size, were less likely to be sequestered by the RES, resulting in the lowest off-target organ distribution. Worms exhibited intermediate behavior, with a moderate level of distribution in non-target organs relative to micelles and vesicles.

All Ce6@nanoparticle topologies exhibited similar levels of tumor accumulation after 12 hours, as shown in Fig. 4.3(h). Vesicles and worm-like Ce6@nanoparticle showed higher initial distribution in the tumor compared to spherical micelles, indicating a stronger EPR effect. However, by 6 hours, a decline in tumor accumulation was observed, suggesting that Ce6@nanoparticles began to efflux from the tumor site. In contrast, micelles displayed a lower initial tumor distribution, but their accumulation gradually increased over time. According to parameters derived from the PBPK model, vesicles and worms exhibited higher influx rates into the tumor than micelles. Nevertheless, their tumor penetration rates were lower, which may have limited their deep tissue delivery compared to micelles.

4.1.4 Validation of the PBPK model for Ce6@nanoparticles

After determining the parameters for each type of nanoparticle, we performed a sensitivity analysis[126]. Sensitivity analysis can be categorized into two types: local sensitivity analysis and global sensitivity analysis. The sensitivity was calculated using the following equation:

$$SC = \frac{\left(AUC'_{0-\infty} - AUC_{0-\infty}\right)/AUC_{0-\infty}}{(Par' - Par)/Par} \qquad (4.6)$$

where SC is the sensitivity; $AUC_{0-\infty}$ and $AUC'_{0-\infty}$ are the area under the curve before and under perturbed, respectively; Par' and Par are the perturbed value and reference value of the parameter under investigation, respectively. Through this function, we validate the sensitivity of the parameters in different ranks, as shown in **Fig. 4.4**.

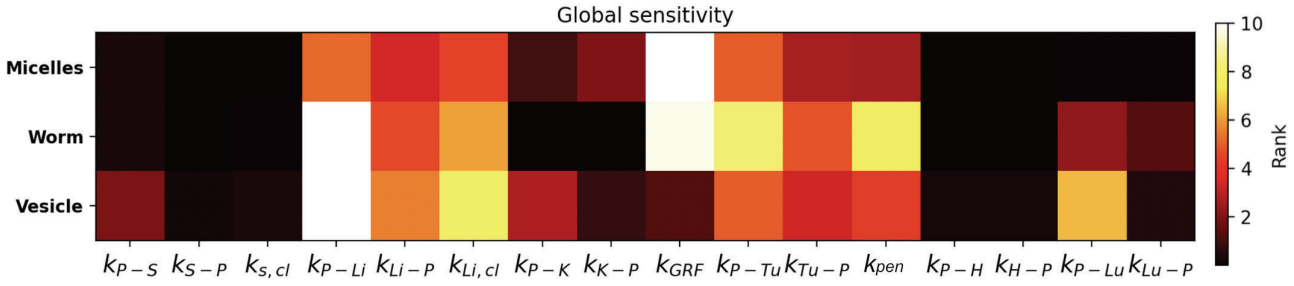

Figure 4.4 Parameter rank based on global sensitivity in the PBPK model. The parameter with the highest sensitivity in the model is ranked 10, while the one with the lowest sensitivity is ranked 0.

4.2 PBPK model based on PAT

4.2.1 Pharmacokinetics of PMPC nanoparticles

The PMPC nanoparticles we used here have a radius of 30 nm and a polymerization of 25. To determine the pharmacokinetics of PMPC nanoparticles in mice, the animals were sacrificed at predetermined time points (1 min, 1 h, 2 h, 4 h, 6 h, 24 h, and 7 days) in accordance with ethical guidelines. Blood was collected via cardiac puncture or retro-orbital bleeding into anticoagulant-coated tubes and centrifuged to obtain plasma. Major organs, including the liver, spleen, kidneys, lungs, and heart, were quickly excised, rinsed with cold PBS to remove residual blood, blotted dry, and weighed. Each organ was then homogenized in cold PBS or lysis buffer using a mechanical homogenizer or bead mill. For fluorescent nanoparticles, DMSO or another suitable solvent was added to the homogenate, followed by vortexing and ultrasonication to extract the fluorescent payload. The mixture was centrifuged, and the supernatant was analyzed using a fluorescence plate reader. For non-fluorescent compounds, the payload was extracted using organic solvents such as methanol or acetonitrile, and the supernatant was analyzed by HPLC or LC-MS/MS. Quantitative data were obtained using calibration curves and normalized to tissue weight, enabling accurate assessment of nanoparticle biodistribution, clearance, and accumulation across different organs.

The fluorescence signals of the nanoparticles in both plasma and various organs were collected using a consistent method, enabling accurate normalization of fluorescence intensity across all samples. The normalized plasma concentration of PMPC nanoparticles is shown in **Fig. 4.5(b)-(f)**. A rapid decline in plasma fluorescence was observed immediately after injection, indicating fast initial clearance from circulation. Since PMPC nanoparticles are designed to actively target specific cell types, this rapid decrease in blood concentration suggests efficient uptake by the targeted cells, which will be discussed in detail later.

The majority of PMPC nanoparticles were primarily distributed in the liver and gastrointestinal (GI) tract at the initial time point (10 minutes). High levels of PMPC nanoparticle accumulation were consistently observed across all time points. In the GI tract, nanoparticle levels increased steadily before 6 hours, followed by a decline between 6 and 24 hours. However, a notable accumulation was observed again at 7 days, suggesting delayed retention or redistribution. The spleen also showed relatively high PMPC nanoparticle accumulation across all time points. Due to its smaller anatomical size compared to the liver, the normalized fluorescence ratio remained

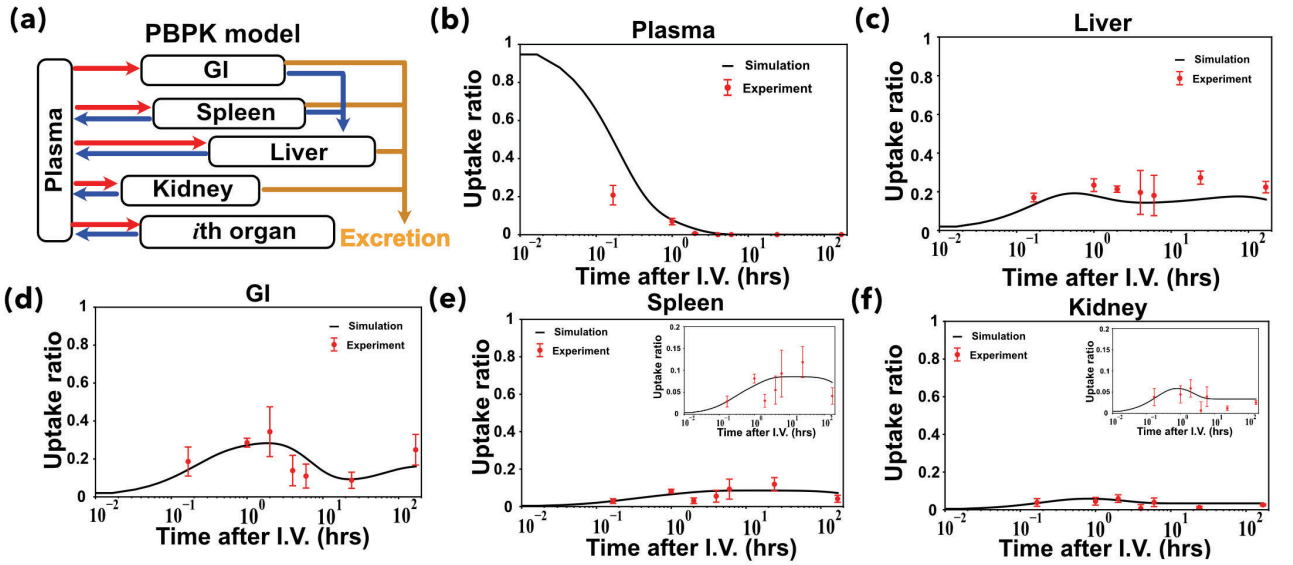

Figure 4.5 PBPK model based on the blood circulation. (a) The schematic of the PBPK model based on the blood circulation. **(b)-(f)** The normalized pharmacokinetics of PMPC nanoparticle in plasma, liver, GI, spleen, and the kidney.

much lower than that observed in the GI tract and liver. The accumulation of PMPC nanoparticles in the kidneys was relatively high at the initial time points and decreased after 6 hours.

4.2.2 PBPK model of PMPC nanoparticles

As discussed before, the PMPC nanoparticle can actively target SRB1, CD36, and CD81. The PMPC nanoparticles have specific interactions with different cell types. We developed a PBPK model that incorporates active targeting for PMPC nanoparticles. This model accounts for cellular uptake, enabling detailed simulations of the nanoparticles' biodistribution. It allows us to predict nanoparticle concentrations not only within target tissues but also at the subcellular level, which is critical for evaluating therapeutic efficacy. To capture the dynamics of active targeting, we integrated the PAT and multivalent uptake kinetics into the PBPK framework. This enhanced model reflects the active uptake behavior, providing insights for optimizing PMPC nanoparticle design, including parameters such as nanoparticle radius and the degree of PC chain polymerization. These optimizations improve targeting efficiency and therapeutic outcomes. Additionally, by incorporating the PAT approach, we can estimate the association constant, which aids in determining optimal dosing strategies.

Given the specific cell types, we can estimate the binding fraction between PMPC nanoparticles and cells based on the PAT model. To ensure accuracy, all cell types that the nanoparticles can potentially interact with should be considered. We collected data on cell types and their corresponding receptor expression levels in different organs using experimental measurements and the Mouse Protein Atlas. In the PBPK model, organs predominantly composed of a single cell type were simplified by representing them with only that major cell type. This approach enables a more manageable yet biologically representative simulation of nanoparticle-cell interactions across various tissues.

The uptake kinetics of PMPC nanoparticles follow a multivalent interaction mechanism. At the single-cell level, we modeled this behavior using multivalent uptake kinetics to accurately represent the dynamics of nanoparticle-cell binding. Unlike in vitro incubation experiments, where only one cell type is typically considered, in vivo systems require accounting for multiple cell types across different organs, making the modeling significantly more complex. Moreover,

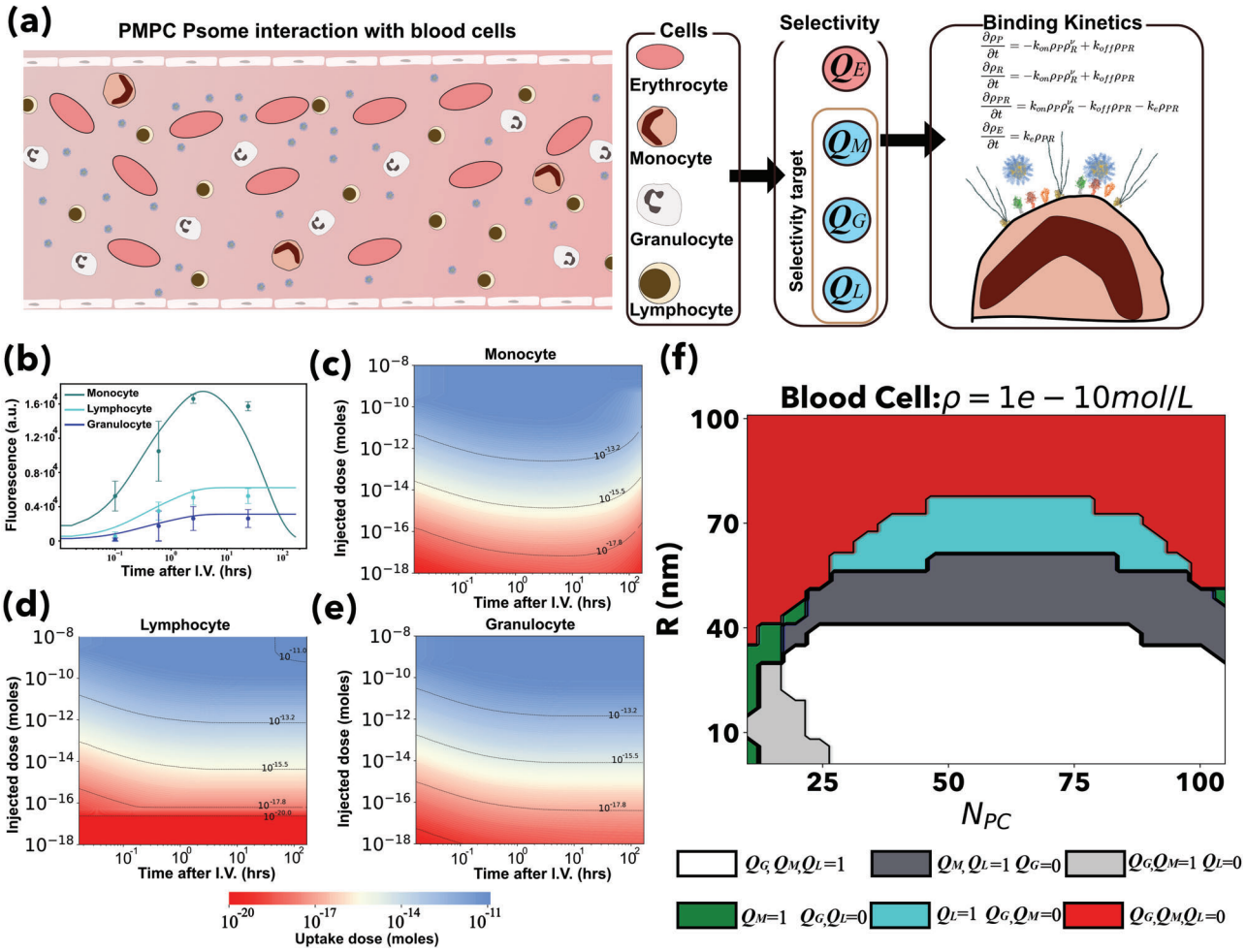

Figure 4.6. Blood cell uptake kinetics. (a) Scemetics of PMPC Psome selectivity derived from PAT and the blood cell uptake mimicked by the multivalent uptake kinetics model. Considering the thermodynamic and kinetic elements, we predict the behavior of PMPC Psome in the blood versus different parameters. **(b)** Uptake kinetics of blood cells from *in vivo* experiments with fitting results using the PBPK model. **(c)** Prediction of monocyte uptake kinetics versus injected dose. **(d)** Prediction of lymphocyte uptake kinetics versus injected dose. **(e)** Prediction of granulocyte uptake kinetics versus injected dose. **(f)** Discrimination among blood cells at an incubation concentration of 10^{-10} mol/L.

the concentration dynamics are considerably more complex due to simultaneous processes such as biodistribution, cellular uptake, and systemic clearance.

We integrated the PAT and multivalence uptake kinetics into the PBPK model. The schematic of the PBPK model for PMPC nanoparticle is shown in **Fig. 4.5(a)**. The model is also based on the blood circulation. The nanoparticle concentration in plasma is influenced not only by cellular uptake but also by biodistribution across organs, further complicating the system dynamics. Considering the cell uptake, we can write the mass of PMPC nanoparticles in the blood as:

$$V_P \frac{\partial \rho_P}{\partial t} = \sum_i \left(-V_P k_{P-i} \rho_P + V_i k_{i-P} \rho_i\right) + \sum_j f(\rho_P, C_{M_{\phi,j}}) \qquad (4.7)$$

where $f(\rho_P, C_{M_{\phi,j}})$ is the multivalence uptake function, which is the same as **Eq.(3.5)**, for the jth cell in the plasma. The organ uptake can be written as:

$$V_i \frac{\partial \rho_i}{\partial t} = V_P k_{P-i} \rho_P - V_i k_{i-P} \rho_i + \sum_j f(\rho_i, C_j) \qquad (4.8)$$

where $f(\rho_i, C_j)$ is the multivalence uptake function for the jth cell in the ith oragn. To explain the accumulation of PMPC nanoparticles in the GI tract after 24 hours, we hypothesize that certain

cells that internalize the nanoparticles may migrate and accumulate in the GI. Based on this, the concentration of PMPC nanoparticles in the GI can be described as follows:

$$V_{GI}\frac{\partial \rho_{GI}}{\partial t} = V_P k_{P-GI}\rho_P - V_{GI}k_{i-P}\rho_{GI} + \sum_j f(\rho_{GI}, C_j) + \sum_k g(C_k) \qquad (4.9)$$

The fitting results are shown in **Fig. 4.5(b)-(f)**. The influx rates of PMPC nanoparticles into the liver and GI tract are significantly higher than those into the spleen and kidneys, which explains the initially high distribution observed in the liver and GI tract. Due to the smaller anatomical size of the spleen and kidneys, their influx rates are lower compared to those of the liver and GI tract. Nevertheless, the PBPK model still indicated a relatively high accumulation of PMPC nanoparticles in the spleen and kidneys.

The PMPC nanoparticle targets receptors such as SR-B1, CD36, and CD81, which are expressed on specific cell types, particularly macrophages. In other words, the PMPC nanoparticle exhibits selective targeting of certain cells, as validated by the PAT model. Once internalized by these target cells, the PMPC nanoparticles are subsequently degraded.

The PMPC nanoparticles used in this study have a radius of 30 nm and a polymerization degree of 25. Due to their size, the GFR is very low, indicating that these nanoparticles are unlikely to be excreted through the urine. Instead, the primary clearance pathways involve the liver, spleen, and GI tract.

4.2.3 Selectivity targeting among different cell groups

The PBPK model for PMPC nanoparticles incorporates both the PAT framework and multivalent uptake kinetics, enabling prediction of nanoparticle interactions with various cell types. In this context, we focus on the interactions between PMPC nanoparticles and blood cells to demonstrate their targeting selectivity. This analysis also provides insights for optimizing dosing strategies and guiding the design of PMPC nanoparticles.

Following the injection of PMPC nanoparticles, we observed that they selectively targeted three blood cell types: monocytes, lymphocytes, and granulocytes. To validate this specificity, we examined the expression levels of key receptors-SRB1, CD36, and CD81-on these cells. For monocytes, the fluorescence intensities corresponding to SRB1, CD36, and CD81 were 5000, 2000, and 18,000, respectively. In comparison, lymphocytes showed values of 1000, 300, and 1800, while granulocytes exhibited intensities of 4000, 2000, and 1800. As shown in **Fig. 4.6(a)**, we evaluated the binding avidity of PMPC nanoparticles across various blood cell types. Only monocytes, lymphocytes, and granulocytes exhibited strong binding at the injected dose, consistent with the experimental data. Other blood cell types did not show significant binding avidity, indicating minimal interaction with the nanoparticles. After defining the binding fraction, we applied multivalent uptake kinetics to simulate nanoparticle uptake specifically for these three cell types. In the PBPK model, we treated other organs and tissues using the same approach: defining the binding affinities for relevant cell types and applying multivalent uptake kinetics, as shown in **Eq. (4.8)**.

In **Fig. 4.6(b)**, we present the uptake kinetics and PBPK model fitting results for monocytes, lymphocytes, and granulocytes. The observed decrease in monocyte-associated fluorescence over time reflects their migration to other organs. Monocytes exhibited the highest nanoparticle uptake and uptake rate, which can be attributed to their significantly higher receptor expression compared to lymphocytes and granulocytes. Although granulocytes had slightly higher receptor expression than lymphocytes, lymphocytes captured more PMPC nanoparticles. This suggests a higher effective binding rate in lymphocytes. One possible explanation is that the dense

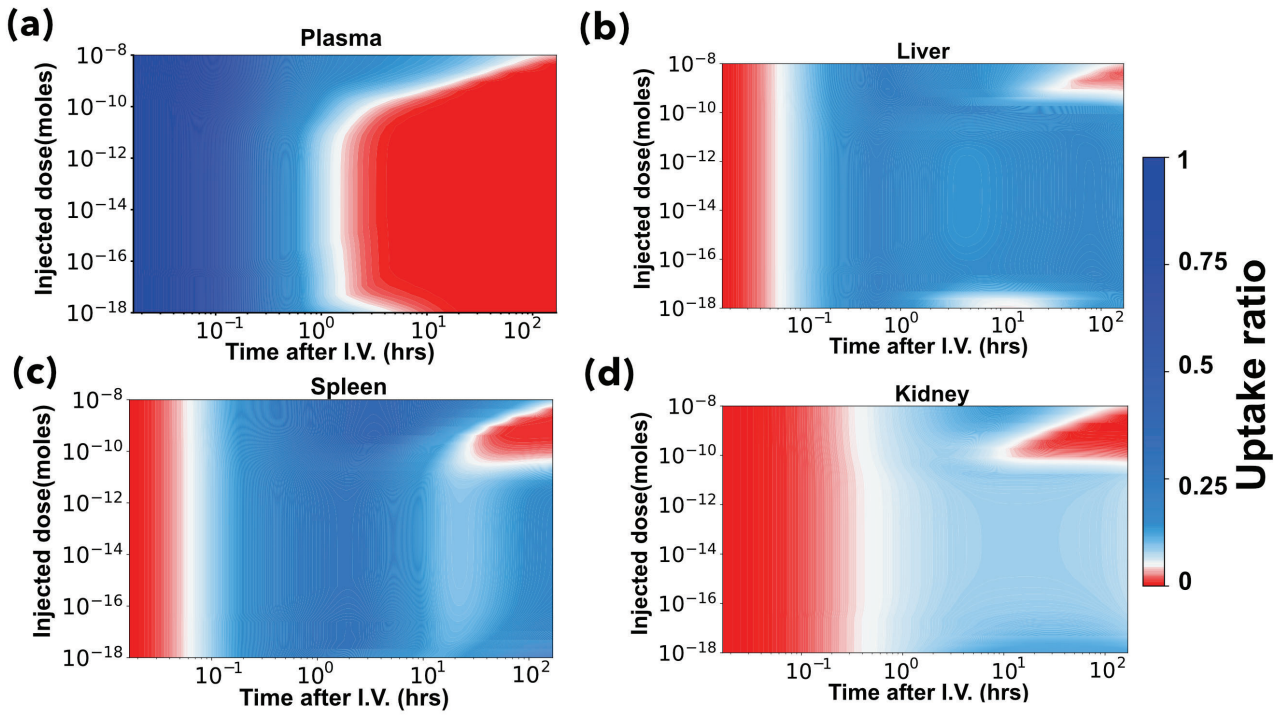

Figure 4.7 Prediction of pharmacokinetics in various organs versus injected dose. **(a)** PMPC nanoparticle concentration in plasma versus the injected dose predicted by the PBPK model. **(b)-(d)** PMPC nanoparticle pharmacokinetics in the liver, spleen, and kidney versus the injected dose predicted by the PBPK model.

glycocalyx on granulocytes may have hindered nanoparticle binding, resulting in a lower binding efficiency.

Altering the polymersome radius or polymer chain length affects key binding processes, such as polymersome diffusion and ligand–receptor interactions, thereby influencing the binding on-rate. However, the relationship between these kinetic parameters and the physicochemical properties of PMPC polymersomes has not been fully explored. In this study, we assume that polymersome concentration does not affect the on-rate or dissociation rate. Based on this assumption and the PBPK model, we were able to predict the uptake kinetics of different blood cell types across a range of injected doses. Specifically, we examined the uptake behavior of monocytes, lymphocytes, and granulocytes at doses ranging from 10^{-18} M to 10^{-8} M. As shown, no significant binding occurred between PMPC nanoparticles and lymphocytes at very low doses, indicating that lymphocytes exhibit the lowest binding avidity among the three cell types. These findings suggest that adjusting the injected dose can enhance targeting selectivity and improve specificity toward high-affinity cells. Optimizing the dose, particularly by lowering the injected amount while maintaining therapeutic efficacy, can lead to a safer and more precise administration strategy.

Based the cell type, we could optimizing the nanoparticle parameter to make it more selectivity. As shown in **Fig. 4.6(f)**, we show the PMPC nanoparticle selectivity to blood cells versus the radius and PMPC chain polymerization with an injected dose of 10^{-10} M. For monocytes, an increase in PMPC nanoparticle radius leads to stronger steric repulsion, which outweighs the enhanced attractive interactions resulting from the increased number of PC ligands. As shown in the figure, there is a significant overlap between monocyte binding and granulocyte binding.

The steric potential for a dense glycocalyx also hinders effective binding to the granulocyte. In contrast, lymphocytes benefit more from the increased ligand density, as the associated steric hindrance is less significant, allowing the attractive forces to dominate. When increasing the

polymerization degree of the PC ligands, the binding energy between the ligands and their receptors initially strengthens but eventually diminishes. Once the polymerization exceeds a certain threshold, the steric hindrance from the extended polymer chains surpasses the binding energy, limiting receptor interactions to the outermost layer of the PC ligands.

4.2.4 Predict the distribution in different organs.

As we derive the parameters for the PBPK model, we explore how nanoparticle distribution changes in response to variations in key design parameters. One particularly critical factor for administration is the concentration of the PMPC nanoparticles. As shown in **Fig. 4.7**, we present heat maps showing distribution changes in the plasma, liver, spleen, and kidney when the administration dose varies from 10^{-12} M to 10^{-8} M. We use the normalized ratio to present the nanoparticle concentration changes. This would lead to a better optimization strategy for dosing.

In the context of personalized medicine, where therapy is tailored to individual patient characteristics such as genetic markers, receptor expression, or cellular phenotypes, our model provides a framework for predicting patient-specific interactions between PMPC nanoparicles and target cells. By focusing on both nanoparticle properties and cellular phenotypes, the model helps identify whether effective nanoparticle-cell interactions are likely to occur in a given individual.

4.3 Designing precise nanomedicine based on the multivalency effect

In the design of precision nanomedicine, it is essential to tailor nanoparticle properties to achieve optimal targeting and therapeutic performance. Here, we introduced a PBPK model integrating the PAT and cell uptake kinetics.

The process begins with fabricating the physicochemical properties of the nanoparticles with guidelines from the PAT. Before conducting in vivo experiments, we utilize data from the Human Protein Atlas, or experimental validation of specific cell types, to construct a pharmacokinetic model that incorporates these relevant cell types. Using the PAT, we estimate the binding fraction at the cellular level and iteratively adjust nanoparticle parameters to ensure selective interaction with the target cells only.

Then, we collected pharmacokinetic data and used a PBPK model to fit these data. In parallel, we gathered cell-level binding data, enabling quantification of uptake kinetics at the cellular scale. By integrating the PBPK model with PAT, we achieved more accurate estimates of key parameters such as the on-rate and dissociation rate. Simulation results were compared with experimental data, and the model was refined iteratively. Guided by insights from the PBPK analysis and established design principles, we then optimized the nanoparticle formulation to improve targeting efficiency, biodistribution, and overall therapeutic efficacy.

In future work aimed at improving the PBPK model, we plan to incorporate fluid dynamics to simulate the transport of nanoparticles within the bloodstream. This will allow us to investigate how nanoparticle parameters, such as size, shape, and surface charge, influence their diffusion, circulation, and accumulation in target tissues. By integrating these physical transport mechanisms, we aim to enhance the predictive power of the PBPK model and better inform the rational design of nanomedicines.

Chapter 5: Discussion and future work

5.1 Discussion

5.1.1 The case of PMPC nanoparticle

In this thesis, we have shown the thermodynamics model-PAT. We use the partition function to mimic the interaction between the nanoparticle and the cell. Given the cell phenotype and the nanoparticle properties, we can predict the binding fraction between the nanoparticle and the cell. Through the model, we can also drive the binding avidity, which could help us to determine the on-rate and dissociation rate more accurately.

In the case of PMPC nanoparticles, we explore how changes in nanoparticle shape influence binding behavior. Depending on the topology, the activity of nanoparticles and receptors within the binding volume (the interaction valence) varies. To improve prediction accuracy, we adjust model parameters according to nanoparticle size and geometry. PMPC nanoparticles are designed to target three different receptors simultaneously, a strategy referred to as multiplexing. We analyze the selectivity parameter as a function of both the nanoparticle properties and the characteristics of the three receptors. This parameter provides insight into how sensitively the binding fraction responds to changes in these variables. To better visualize the onset of binding across the three receptor types, we generate a 4D plot along with onset surfaces for nanoparticles with fixed properties. Through this approach, we present a framework to analyze and visualize the mechanisms and efficiency of multiplexed targeting.

We then integrated the PMPC-based PAT model into the uptake kinetics analysis. Uptake data for nanoparticles with different topologies were collected and fitted using a mathematical kinetics model. From this fitting, we obtained key parameters such as the on-rate for binding and the endocytosis rate for each nanoparticle topology. In this context, the reaction valence is linked to the number of receptors within the binding volume. Due to limitations in the model and experimental deviations, it was not possible to accurately determine the dissociation rate when it falls below $10^{-3}\,s^{-1}$. However, by integrating the PAT with the kinetics model, we could estimate the dissociation constant and derive the residence time. All nanoparticle topologies demonstrated long residence times. Among the different shapes, spherical PMPC nanoparticles exhibited the highest endocytosis rate. Tube-like nanoparticles had the lowest endocytosis rate, while disk-like nanoparticles also showed a low rate, likely due to their small size, which demands higher membrane energy for cellular wrapping. The combination of long residence time and low endocytosis rate explains why tube-like nanoparticles remain on the cell membrane longer, as observed in experiments. Overall, the integrated kinetics and PAT model provides a deeper understanding of how nanoparticle topology influences cellular uptake dynamics.

Finally, we demonstrate how to simulate the journey of PMPC nanoparticles using a PBPK model. This model integrates PAT and cellular-level uptake kinetics to capture the behavior of nanoparticles within the body. By incorporating specific cell phenotypes, the PBPK model integrated with PAT predicts binding fractions across different cell types, enabling us to explore how selectivity can be achieved by tuning nanoparticle parameters. We fit the in vivo experimental data using this PBPK framework. With information from cellular uptake kinetics and PAT predictions, we can simulate the uptake behavior of blood cells under various injection doses. Importantly, we demonstrate that selectivity can be modulated by adjusting the injection dose, leading to reduced off-target binding and minimized potential side effects. In addition, we model the ADME of nanoparticles at the organ level across different dosing strategies. Together,

this modeling-experimental pipeline provides a powerful tool for optimizing therapeutic design, improving targeting efficiency, and enhancing treatment efficacy.

5.1.2 The case of nanoparticle targeting large receptors

When targeting large receptors such as LRP1 and CD206, we observed that nanoparticle formulations with a high number of ligands did not exhibit strong binding. To explain this, we introduced a steric penalty arising from unbound ligands. In the case of large receptors, the nanoparticle inserts into the glycocalyx and adjusts its position to interact effectively. Ligands on the receptor-facing side bind to the receptor, while ligands on the opposite side may interact with components on the cell surface, creating steric hindrance. Additionally, compression-induced steric penalties become significant when targeting large receptors. To account for these effects, we developed a model that incorporates both osmotic and compression steric penalties. By including these factors, we refined and optimized the PAT model, making it more suitable for scenarios involving large receptor targeting.

We predicted the binding fraction as a function of ligand number and the interference parameter. The steric hindrance caused by unbound ligands is related to the interference parameter. For formulations with a lower interference parameter, the range of ligand numbers that results in a high binding fraction is broader compared to those with higher interference parameters.

We analyzed the selectivity parameter as a function of ligand number. To quantify this, we defined the onset and offset ligand numbers, along with their corresponding selectivity values, α_{onset} and α_{offset}. As the ligand number increases, the binding fraction first rises and then declines, resulting in positive values for both α_{onset} and α_{offset}. This analysis allows us to identify the ligand number range in which selectivity can be achieved. Furthermore, we examined how selectivity is influenced by several key factors: the steric penalty from a single unbound ligand, the ligand–receptor binding energy, and the number of available binding sites. For receptors such as LRP1 and CD206, these parameters vary significantly, leading to differences in binding performance and selectivity behavior.

We integrated the PAT model for large receptors with the uptake kinetics model to quantify the on-rate and dissociation rate of the binding kinetics between mUNO-decorated nanoparticles and THP-1-derived macrophages across different formulations. The dissociation constants in these binding systems were relatively high, indicating short residence times and consequently low endocytosis rates by the cells. Through this integrated modeling approach, we were able to clarify the observed experimental phenomena, highlighting the importance of coupling the kinetics model with PAT. This strategy enables the rational design of nanoparticles and supports the optimization of formulations for enhanced and effective cellular binding.

5.1.3 Improving the kinetics model

We present a comprehensive kinetics model that incorporates binding, dissociation, endocytosis, receptor diffusion, receptor recycling, exocytosis, and receptor synthesis to more accurately mimic the real uptake kinetics between nanoparticles and cells. To capture the observed oscillatory behavior, we assume that the receptor synthesis rate is regulated by feedback from bound nanoparticles, free receptors remaining on the cell surface, and internalized nanoparticles. This feedback may originate either proximally or distally relative to the receptor. Using this model, we successfully simulated the oscillatory dynamics and validated the system through the Lyapunov coefficients. Notably, the model is highly sensitive to initial

conditions, highlighting its potential to explain a broader range of uptake phenomena by adjusting parameters or starting values.

5.1.4 PBPK model for nanoparticles with nonspecific interaction

We applied the PBPK model to fit the ADME profiles of PEG-PMET nanoparticles with different topologies. We found that micelles were primarily cleared via renal excretion, while worm-like nanoparticles and vesicles were mainly eliminated by the liver and spleen. Among all topologies, micelles exhibited the highest tumor penetration rate. After parameter estimation, we performed sensitivity analysis on the PBPK model to identify factors that influenced biodistribution and clearance mostly. These insights guided us in determining the most suitable nanoparticle topologies for efficient drug delivery.

5.2 Future work

5.2.1 Validation, refinement, and optimization

We gather experimental data and use *in silico* methods to replicate and simulate these experiments. In silico approaches can be employed to generate research hypotheses that are later validated through *in vitro* or *in vivo* experimentation. The resulting experimental data are then used to refine and enhance the accuracy and complexity of the in silico models, thereby increasing their ability to generate realistic and predictive results.
In this work, the *in silico* modeling includes the PAT framework, an uptake kinetics model, and a PBPK model. Our goal is to develop a streamlined workflow for designing precision nanomedicines, a tool that can assist researchers in interpreting experimental data and gaining deeper mechanistic insights. The tool is developed as a Python package and is maintained on our group's GitHub repository (*https://github.com/Molecular-Bionics-Labs/mbpk*). The models are governed by fundamental physical principles. Based on the experimental results, we identify physical mechanisms that may have been overlooked or previously unconsidered (for example, the steric from the unbound ligands) and incorporate them into the models to improve their predictive capabilities. In short, we use *in silico* methods to interpret experimental results, while the experimental data, in turn, refine and optimize the in silico models.

5.2.2 Towards personalized nanomedicine

Through our model, we aim to achieve more precise measurements of cellular phenotypes, particularly focusing on receptor expression and glycocalyx composition. We plan to collect comprehensive phenotype data from various human tissues to capture the diversity of cell profiles across the body. Using this phenotype information, we can pre-design nanoparticles to enhance their target selectivity and minimize off-target interactions, thereby reducing toxicity. We will then test different nanoparticle formulations to evaluate their targeting kinetics and identify the optimal formulation that ensures efficient and specific cellular uptake. Beyond in vitro validation, we plan to integrate individual gene expression data into the model, enabling personalized predictions of nanoparticle ADME through the PBPK model for humans. This will support the development of personalized nanomedicine tailored to individual patients.

5.2.3 Integrating machine learning (ML) into design nanomedicine

In recent years, machine learning (ML) has emerged as a transformative tool in the design of nanomedicine[181]. By leveraging large experimental datasets and extracting property-function relationships, ML has been applied to rationalize nanoparticle formulation, predict biodistribution, and assess cellular uptake efficiency[182]. Integrating ML into nanomedicine research offers a more efficient pathway toward the development of next-generation therapeutic nanoparticles.

Using the thermodynamic model, we can predict nanoparticle binding properties across different cell types. However, this model overlooks certain factors, such as particle surface potential. By applying ML, we can integrate all nanoparticle properties with their corresponding binding efficiencies and identify the key parameters governing interactions. These parameters can then be compared with the PAT to uncover previously neglected information and reveal underlying physical mechanisms.

Physics-Informed neural networks (PINNs) are a class of machine learning models that incorporate physical laws, typically expressed as differential equations, directly into the training process[183-185]. Unlike traditional data-driven approaches, PINNs embed governing equations (e.g., conservation laws, ODEs, PDEs) into the loss function, ensuring that predictions remain consistent with known physics. Since both the kinetics and PBPK models in this study are based on ODEs, PINNs can provide more accurate solutions for predicting outcomes. They can also assist in parameter identification for kinetics models, uncover intrinsic PBPK structures directly from data, and integrate time-dependent parameters to enhance pharmacokinetic modeling[186-188]. In short, PINNs represent a powerful tool for improving the accuracy and interpretability of our kinetics models.

5.2.4 Fractional model and chaotic model

Pharmacokinetics are closely intertwined with complex physiological processes. Traditional pharmacokinetic models often assume that drug clearance and diffusion follow simple exponential (first-order) kinetics. In reality, however, many biological systems display non-exponential, anomalous behavior in drug disposition, driven by long-tailed distributions, memory effects, and tissue trapping[189,190]. These deviations highlight the need to incorporate non-linear elements into pharmacokinetic models.

A fractional pharmacokinetic model applies fractional calculus, derivatives of non-integer order, rather than conventional integer-order differential equations, to describe drug ADME. By introducing memory and heterogeneity through fractional derivatives, these models capture complex behaviors that traditional approaches cannot. For a single component, fractional elimination can be expressed as:

$$D_t^{\alpha} C(t) = -kC(t) \qquad (5.1)$$

with D_t^{α} being the Caputo derivative and $C(t)$ being the component condition. The solution would be:

$$C(t) = C_0 E_{\alpha}(-kt^{\alpha}) \qquad (5.2)$$

With C_0 being the initial condition and E_{α} being the Mittag–Leffler function (generalizes the exponential)[191]. By introducing a fractional derivative, we can better mimic nonlinear biological processes. For instance, integrating fractional derivatives with Fick's law allows the modeling of long-tailed distributions, thereby providing more accurate predictions[192].

Pharmacokinetics can display chaotic behavior arising from the intrinsic complexity of biological systems. A key contributor to this complexity is the network of biological clocks. These clocks are synchronized by periodic environmental cues, most prominently the light-dark cycle[193,194]. The suprachiasmatic nucleus (SCN) in the brain serves as the central pacemaker, responding to light-

dark signals and regulating systemic variables such as body temperature, hormone secretion, and activity levels. To account for these dynamics, PBPK models can incorporate periodic functions that capture circadian influences on drug disposition. Certain diseases can disrupt circadian rhythms, impairing the synchronization between the central clock and peripheral tissues. This desynchrony can, in turn, lead to chaotic behavior in drug ADME[195]. Organ interactions further amplify chaotic dynamics. Each organ (liver, kidney, gut, brain, etc.) possesses its own circadian clock that regulates local drug transporters and metabolizing enzymes[196-198]. When peripheral clocks drift out of sync with one another or with the SCN, their outputs, such as blood flow, enzyme activity, and clearance rates, become misaligned, leading to nonlinear and potentially chaotic pharmacokinetics. Moreover, drug distribution in one organ can influence others through systemic signaling. For instance, liver metabolism generates metabolites or alters hormone levels that circulate throughout the body and, in turn, modulate kidney clearance, gut absorption, or brain exposure[199-201]. This would lead to the quadratic items, such $\rho_{Li}(t)\rho_{GI}(t)$, into the PBPK system. This inter-organ communication reinforces the dynamic, and sometimes chaotic, nature of pharmacokinetics. The sparse identification of nonlinear dynamics (SINDy) can help determine PBPK equations that include quadratic elements[202,203]. As a data-driven method, SINDy identifies the governing equations of complex dynamical systems by selecting the most relevant terms from a large set of candidate functions. Applied to pharmacokinetics, it can reveal the underlying physical mechanisms and control loops governing drug distribution.

5.2.5 Fluid dynamics in understanding the nanomedicine therapy.

At the tissue level, nanoparticles distribute among cells through the interstitial fluid. Since organ geometries differ significantly, as noted earlier, this distribution is shaped by both fluid dynamics and interactions between cell types[204]. Building models that integrate fluid dynamics with cell–cell interactions while accounting for organ geometry provides a powerful tool to predict nanoparticle fate. Such models can be designed to mimic physical properties, aiding in the understanding of pathology and the rational design of nanomedicines. For example, incorporating biological processes such as transcytosis into these models demonstrates the potential of combining fluid dynamics with biological mechanisms to capture nanoparticle behavior more accurately[205]. We will focus on the RES. To this end, organ-specific geometries will be constructed to mimic the unique architecture of the liver, kidney, and spleen. Key geometric properties, such as fenestration size, tissue surface-to-volume ratio, and macrophage localization, will be incorporated to enable more precise predictions of the PBPK model[206].

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